Contents

1 Lecture 1

1.1 Vector spaces

Let F be a field. Informally, a field is a set of numbers equipped with the operations $+, -, \times, \div$ that satisfy the usual arithmetic rules. A precise mathematical definition is not required in this course; for a rigorous treatment and an introduction to the beautiful Galois theory of fields, see [1, Chapters 13-14]. In this course, we will primarily focus on the cases where $F = \mathbb{R}$, the field of real numbers, or $F = \mathbb{C}$, the field of complex numbers.

What is a vector? Mathematicians do not answer the question directly. Instead we answer the question what operations can be done on vectors. As long as these operations can be done, they are vectors.

Definition 1.1. A set V is called a **vector space** (**linear space**) over F if it is equipped with addition

$$V \times V \to V, \quad (x, y) \mapsto x + y$$

and scalar multiplication

$$F \times V \to V, \quad (a, x) \mapsto ax$$

satisfying the following axioms: for any $a, b \in F$, $x, y, z \in V$

(I) $x + y = y + x$, commutativity

(II) $(x + y) + z = x + (y + z)$, associativity

(III) There exists $0 \in V$ such that $x + 0 = x$, additive identity

(IV) For any $x \in V$, there exists $-x \in V$ such that $x + (-x) = 0$, additive inverse

(V) $a(bx) = (ab)x$, multiplicative associativity

(VI) $1x = x$, multiplicative identity

(VII) $a(x + y) = ax + ay$, distributive property 1

(VIII) $(a + b)x = ax + bx$. distributive property 2

An element $x \in V$ is called a **vector** and an element $a \in F$ is called a **scalar**.

Remark 1.2. The axioms above seems a lot, but they are just saying that the usual algebraic rules work for vector addition and scalar multiplication. This can be seen even more clearly in Proposition 1.3. It should be noted that the multiplication of two vectors is <u>not</u> defined within the structure of a vector space.

Proposition 1.3. *The following are direct consequences of the axioms*

(i) The element $0 \in V$ *is unique.*

(ii) *For any* $x \in V$, $0x = 0$.

(iii) *For any* $a \in F$, $a0 = 0$.

(iv) *If* $a \in F$, $x \in V$ *such that* $ax = 0$, *then either* $a = 0$ *(in* F*) or* $x = 0$ *(in* V*).*

(v) *For any* $x \in V$, *the element* $-x$ *is unique.*

(vi) *For any* $x \in V$, $(-1)x = -x$.

(vii) *If* $x + z = y + z$, *then* $x = y$.

Proof. (i)-(vi) are left as exercise. We now proof (vii). We have

$$x = x+0 = x+(z+(-z)) = (x+z)+(-z) = (y+z)+(-z) = y+(z+(-z)) = y+0 = y$$

where we used (III), (IV), (II), the assumption, (II), (IV), (III) in each equality. $\square$

Example 1.4.

$$F^n = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} : x_i \in F, i = 1, \dots, n \right\}$$

equipped with addition

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{pmatrix}$$

and scalar multiplication

$$c \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} cx_1 \\ cx_2 \\ \vdots \\ cx_n \end{pmatrix}.$$

The zero vector is $\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ and the additive inverse of $\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ is $\begin{pmatrix} -x_1 \\ -x_2 \\ \vdots \\ -x_n \end{pmatrix}$.

Example 1.5. An $m \times n$ **matrix** A over F is an array of elements in F with m rows and n columns:

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} = (a_{ij})_{1 \le i \le m, 1 \le j \le n}$$

where $a_{ij} \in F$ for $1 \leq i \leq m$, $1 \leq j \leq n$. If the range of i, j are clear from context, we omit the range and write (a_{ij}). The set of all $m \times n$ matrices over F is denoted by $M_{m\times n}(F)$. It is equipped with the addition

$$(a_{ij}) + (b_{ij}) = (a_{ij} + b_{ij})$$

and scalar multiplication

$$c(a_{ij}) = (ca_{ij}).$$

The zero vector is $0 = 0_{m\times n}$ an $m \times n$ matrix whose entries are all 0 and the additive inverse is$-(a_{ij}) = (-a_{ij})$. Then $M_{m\times n}(F)$ is a vector space. As a vector space, $M_{m\times n}(F)$ is simply $F^{m\times n}$, since a matrix can be flattened into a column vector with $m \times n$ components.

Given a matrix A its **transpose** is the matrix A^t given by transforming its rows into columns. For example

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^t = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}.$$

It is a convention in math to write a vector in F^n as a column vector as we did in Example 1.4. But writing $\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ in the notes all the time takes a lot of space, so for the rest of the notes we make use of the transpose notation and write $(x_1, \ldots, x_n)^t$.

Example 1.6. Let $k \in \mathbb{N}$ be a natural number and $C^k(\mathbb{R})$ be the set of functions $f : \mathbb{R} \to \mathbb{R}$ whose derivatives up to order k are continuous. We also write $C(\mathbb{R}) = C^0(\mathbb{R})$ for the set of continuous function. We equip $C^k(\mathbb{R})$ with the usual addition $(f+g)(x) = f(x)+g(x)$ and scalar multiplication $(cf)(x) = cf(x)$ for $c \in \mathbb{R}$. Then $C^k(\mathbb{R})$ is a vector space over $\mathbb{R}$.

Example 1.7. The space $\mathcal{P}_n(F)$ of polynomials of degree at most n consists of polynomials of the form

$$p(x) = a_n x^n + \cdots + a_1 x + a_0$$

where $a_i \in F$ for $0 \leq i \leq n$. Under the addition

$$(a_n x^n + \cdots + a_1 x + a_0) + (b_n x^n + \cdots + b_1 x + b_0) = (a_n + b_n)x^n + \cdots + (a_1 + b_1)x + (a_0 + b_0)$$

and scalar multiplication

$$c(a_n x^n + \cdots + a_1 x + a_0) = ca_n x^n + \cdots + ca_1 x + ca_0$$

$\mathcal{P}_n(F)$ is a vector space over F.

1.2 Subspaces

Definition 1.8. A subset $W \subset V$ of a vector space V is called a **subspace** if

(i) $0 \in W$

(ii) For any $x, y \in W$, we have $x + y \in W$

(iii) For any $a \in F$, $x \in W$, we have $ax \in W$

Lemma 1.9. *A subspace is a vector space.*

Proof. We first show that if $x \in W$ then $-x \in W$. By Proposition 1.3 (v) and definition of a subspace, $-x = (-1)x \in W$.

By the definition of a subspace, addition and scalar multiplication are well-defined on W. The axioms (I)-(VIII) are satisfied on W since they are satisfied for any vectors in V. $\square$

Example 1.10. A line through the origin in $\mathbb{R}^3$ is given by $\{tv : t \in \mathbb{R}\}$ where $v = (v_1, v_2, v_3)^t \neq 0$ is the direction of the line.

A plane through the origin in $\mathbb{R}^3$ is given by $\{tv + sw : t, s \in \mathbb{R}\}$ where v, w are two vectors in $\mathbb{R}^3$ not parallel to each other.

One can check that lines and planes are subspaces of $\mathbb{R}^3$. In fact $\{0\}$, lines, planes, $\mathbb{R}^3$ are the only subspaces of $\mathbb{R}^3$.

Example 1.11. A matrix A is called a **square** matrix if the number of columns is the same as the number of rows i.e. $A \in M_{n\times n}(F)$. A square matrix is **symmetric** if $A = A^t$. It is anti-symmetric if $A^t = -A$. The set of symmetric matrices and anti-symmetric matrices are subspaces of $M_{n\times n}(F)$ can be seen as follows: 0 matrix is both symmetric and anti-symmetric and for any $A, B \in M_{n\times n}(F)$ we have $(A+B)^t = A^t + B^t$ and $(cA)^t = cA^t$. Thus if A, B are symmetric, then $(A + B)^t = A^t + B^t = A + B$ and $(cA)^t = cA^t = cA$. The anti-symmetric case is similar.

Example 1.12. $C^k(\mathbb{R})$ is a subspace of $C^{k-1}(\mathbb{R})$ for any $k \in \mathbb{N}$. $\mathcal{P}_n(\mathbb{R})$ is a subspace of $C^k(\mathbb{R})$ for any $k \in \mathbb{N}$.

Solutions to a linear differential equation form a subspace.

Example 1.13. Let $V = \{f \in C^2(\mathbb{R}) : f'' + f = 0\}$ with the addition and scalar multiplication from $C^2(\mathbb{R})$. Then V is a subspace of $C^2(\mathbb{R})$ since $0 \in V$ and if $f, g \in V$ and $c \in \mathbb{R}$, we have $(f + g)'' + f + g = 0$ and $(cf)'' + cf = 0$. In fact, we know from ODE that $f(x) = c_1 \cos x + c_2 \sin x$ for $c_1, c_2 \in \mathbb{R}$ from which one can also check that V is a subspace. The point here is that we do not need the explicit formula for solutions to conclude that the set of all solutions form a subspace.

1.3 Span

We want to build the vector space out of the smallest amount of vectors. Addition and scalar multiplication are the only operations we have in a vector space. If we have $x, y \in V$, then $x + y$ and ax for $a \in F$ can be built out of them. Given a set of vectors $v_1, \ldots, v_n$ what are all vectors that can be built from them? This is the idea of linear combination and span.

Definition 1.14. Let V be a vector space over F. A **linear combination** of $v_1, \ldots, v_n \in V$ is a vector of the form

$$a_1v_1 + \cdots + a_nv_n$$

where $a_1, \ldots, a_n \in F$. The set of all linear combinations of $v_1, \ldots, v_n$ is denoted by $\operatorname{span}(v_1, \ldots, v_n)$. We use the convention that $\operatorname{span} \emptyset = \{0\}$.

Proposition 1.15. $\operatorname{span}(v_1, \ldots, v_n)$ *is a subspace of* V.

Proof. By Proposition 1.3 (ii), we have $0 = 0v_1 + \cdots + 0v_n \in \operatorname{span}(v_1, \ldots, v_n)$. If $x = a_1v_1 + \cdots + a_nv_n$ and $y = b_1v_1 + \cdots + b_nv_n$ are two linear combinations, then $x + y = (a_1 + b_1)v_1 + \cdots + (a_n + b_n)v_n$ and $cx = (ca_1)v_1 + \cdots + (ca_n)v_n$ are both linear combinations of $v_1, \ldots, v_n$. Thus $\operatorname{span}(v_1, \ldots, v_n)$ is a subspace. □

Example 1.16. Let $v_1 = (1, 0, 0)^t$. Then $\operatorname{span}(v_1) = \{(a_1, 0, 0)^t : a_1 \in \mathbb{R}\}$ is the line through the origin in the direction of v_1.

Let $v_2 = (0, 1, 0)^t$, then $\operatorname{span}(v_1, v_2) = \{(a_1, a_2, 0)^t : a_1, a_2 \in \mathbb{R}\}$ is the plane determined by 0, v and v_2.

Let $v_3 = (1, 1, 0)^t$, then $\operatorname{span}(v_1, v_2, v_3) = \{(a_1 + a_3, a_2 + a_3, 0)^t : a_1, a_2, a_3 \in \mathbb{R}\}$. This is the same as $\operatorname{span}(v_1, v_2)$ since it is just the set of points with $x_3 = 0$.

Let $v_4 = (0, 0, 1)^t$, then $\operatorname{span}(v_1, v_2, x) = \{(a_1, a_2, a_3)^t : a_1, a_2, a_3 \in \mathbb{R}\} = \mathbb{R}^3$.

Definition 1.17. $v_1, \ldots, v_n$ is a **spanning set** of V if $\operatorname{span}(v_1, \ldots, v_n) = V$.

If V has a finite spanning set, then V is called a **finite dimensional** vector space.

1.4 Linear dependence and independence

We see in Example 2.3 that v_3 is redundant in building vectors i.e. adding v_3 to v_1, v_2 does not change their span. We want to get rid of the redundant vectors to find the smallest set of vectors that span a vector space. The following concepts are important.

Definition 1.18. $v_1, \ldots, v_n$ is **linearly dependent** if there exist $a_1, \ldots, a_n \in F$ not all zero such that $a_1v_1 + \cdots + a_nv_n = 0$.

$v_1, \ldots, v_n$ is **linearly independent** if or it is not linearly dependent, in other words, for any $a_1, \ldots, a_n \in F$ such that $a_1v_1 + \cdots + a_nv_n = 0$ we have $a_1 = \cdots = a_n = 0$.

Example 1.19. $v_1 = (1, 0, 0)^t$, $v_2 = (0, 1, 0)^t$ are linearly independent since for any $a_1, a_2 \in F$ we have $0 = a_1v_1 + a_2v_2 = (a_1, a_2, 0)^t$ implies $a_1 = a_2 = 0$.

Example 1.20. $v_1 = (1, 0, 0)^t$, $v_2 = (0, 1, 0)^t$, $v_3 = (1, 1, 0)^t$ are linearly dependent since $v_1 + v_2 - v_3 = 0$.

The previous examples shows that a linearly dependent set of vectors is not small enough i.e. it has vectors that is redundant and that linearly independent vectors are not redundant. Thus our goal of finding a smallest set of vectors that builds the whole vector space is to find a basis:

Definition 1.21. $v_1, \ldots, v_n$ is a **basis** of V if $\operatorname{span}(v_1, \ldots, v_n) = V$ and $v_1, \ldots, v_n$ are linearly independent.

References

[1] Dummit, David Steven, and Richard M. Foote. Abstract algebra. Vol. 3. Hoboken: Wiley, 2004.

2 Lecture 2

2.1 Span

We want to build a vector space out of the smallest amount of vectors. Addition and scalar multiplication are the only operations we have in a vector space. If we have $x, y \in V$, then $ax + by$ for $a, b \in F$ can be built out of them. Given a set of vectors $v_1, \ldots, v_n$ what are all vectors that can be built from them? This is the idea of linear combination and span.

Definition 2.1. Let V be a vector space over F. A **linear combination** of $v_1, \ldots, v_n \in V$ is a vector of the form

$$a_1 v_1 + \cdots + a_n v_n$$

where $a_1, \ldots, a_n \in F$. The set of all linear combinations of $v_1, \ldots, v_n$ is denoted by $\operatorname{span}(v_1, \ldots, v_n)$. We use the convention that $\operatorname{span} \emptyset = \{0\}$.

Proposition 2.2. $\operatorname{span}(v_1, \ldots, v_n)$ *is a subspace of* V*.*

Proof. By Proposition 1.3 (ii), we have $0 = 0v_1 + \cdots + 0v_n \in \operatorname{span}(v_1, \ldots, v_n)$. If $x = a_1 v_1 + \cdots + a_n v_n$ and $y = b_1 v_1 + \cdots + b_n v_n$ are two linear combinations, then $x + y = (a_1 + b_1)v_1 + \cdots + (a_n + b_n)v_n$ and $cx = (ca_1)v_1 + \cdots + (ca_n)v_n$ are both linear combinations of $v_1, \ldots, v_n$. Thus $\operatorname{span}(v_1, \ldots, v_n)$ is a subspace. $\square$

Example 2.3. Let $v_1 = (1, 0, 0)^t$. Then $\operatorname{span}(v_1) = \{(a_1, 0, 0)^t : a_1 \in \mathbb{R}\}$ is the line through the origin in the direction of v_1.

Let $v_2 = (0, 1, 0)^t$, then $\operatorname{span}(v_1, v_2) = \{(a_1, a_2, 0)^t : a_1, a_2 \in \mathbb{R}\}$ is the plane determined by 0, v_1 and v_2.

Let $v_3 = (1, 1, 0)^t$, then $\operatorname{span}(v_1, v_2, v_3) = \{(a_1 + a_3, a_2 + a_3, 0)^t : a_1, a_2, a_3 \in \mathbb{R}\}$. This is the same as $\operatorname{span}(v_1, v_2)$ since it is just the set of points with $x_3 = 0$.

Let $v_4 = (0, 0, 1)^t$, then $\operatorname{span}(v_1, v_2, v_4) = \{(a_1, a_2, a_4)^t : a_1, a_2, a_4 \in \mathbb{R}\} = \mathbb{R}^3$ and $\operatorname{span}(v_1, v_2, v_3, v_4) = \{(a_1 + a_3, a_2 + a_3, a_4)^t : a_1, a_2, a_3 \in \mathbb{R}\} = V$.

Definition 2.4. $v_1, \ldots, v_n$ is a **spanning set** of V if $\operatorname{span}(v_1, \ldots, v_n) = V$.

If V has a spanning set, then V is called a **finite dimensional** vector space.

A vector space V is **infinite dimensional** if it is not finite dimensional.

2.2 Linear dependence and independence

We see in Example 2.3 that v_3 is redundant in building vectors i.e. adding v_3 to v_1, v_2 does not change their span. We want to get rid of the redundant vectors to find the smallest set of vectors that span a vector space. The following concepts are important.

Definition 2.5. $v_1, \ldots, v_n$ is **linearly dependent** if there exist $a_1, \ldots, a_n \in F$ not all zero such that $a_1 v_1 + \cdots + a_n v_n = 0$.

$v_1, \ldots, v_n$ is **linearly independent** if or it is not linearly dependent, in other words, for any $a_1, \ldots, a_n \in F$ such that $a_1 v_1 + \cdots + a_n v_n = 0$ we have $a_1 = \cdots = a_n = 0$.

Example 2.6. $v_1 = (1,0,0)^t$, $v_2 = (0,1,0)^t$ are linearly independent since for any $a_1, a_2 \in F$ we have $0 = a_1v_1 + a_2v_2 = (a_1, a_2, 0)^t$ implies $a_1 = a_2 = 0$.

Example 2.7. $v_1 = (1,0,0)^t$, $v_2 = (0,1,0)^t$, $v_3 = (1,1,0)^t$ are linearly dependent since $v_1 + v_2 - v_3 = 0$.

We see in Example 2.7 that v_3 is a linear combination of v_1, v_2. In the following lemma, we see that this is always the case for linearly dependent vectors.

Lemma 2.8. *(i) $v_1, \dots, v_n$ is linearly dependent if and only if there exists $1 \le k \le n$ such that $v_k \in \operatorname{span}(v_1, \dots, v_{k-1})$.*

(ii) $v_1, \dots, v_n$ is linearly independent if and only if for any $1 \le k \le n$ we have $v_k \notin \operatorname{span}(v_1, \dots, v_{k-1})$.

Remark 2.9. If $k = 1$, we use the convention that $(v_1, \dots, v_{k-1})$ is the empty set and $\operatorname{span} \emptyset = \{0\}$.

Proof. Note that statement (ii) is logically the negation of statement (i), we only need to show (i).

"$\Longrightarrow$" Since $v_1, \dots, v_n$ is linearly dependent, there exist $a_1, \dots, a_n \in F$ not all zero such that $a_1v_1 + \cdots + a_nv_n = 0$. Let k be the largest index such that $a_k \neq 0$. Then we have $a_1v_1 + \cdots + a_kv_k = 0$. Thus if $k > 1$, then $v_k = -\frac{1}{a_k}(a_1v_1 + \cdots + a_{k-1}v_{k-1})$. If $k = 1$, then $a_1v_1 = 0$ and hence $v_1 = 0$ by Proposition 1.3 (iv).

"$\Longleftarrow$" If $v_k \in \operatorname{span}(v_1, \dots, v_{k-1})$, then there exists $a_1, \dots, a_{k-1} \in F$ such that $v_k = a_1v_1 + \cdots + a_{k-1}v_{k-1}$. In other words, $a_1v_1 + \cdots + a_{k-1}v_{k-1} - v_k = 0$. This shows $v_1, \dots, v_n$ are linearly dependent. $\square$

2.3 Basis

The previous examples shows that a linearly dependent set of vectors is not small enough i.e. it has vectors that are redundant. On the other hand, linearly independent vectors are not redundant. Thus our goal of finding a smallest set of vectors that builds the whole vector space is to find a basis:

Definition 2.10. $v_1, \dots, v_n$ is a **basis** of V if $\operatorname{span}(v_1, \dots, v_n) = V$ and $v_1, \dots, v_n$ are linearly independent.

We can reduce a spanning set to a basis by removing the "redundant" vectors as in the following lemma.

Lemma 2.11. *Every spanning set contains a basis.*

Proof. Let $v_1, \dots, v_n$ be a spanning set i.e. $\operatorname{span}(v_1, \dots, v_n) = V$. We remove vectors in $v_1, \dots, v_n$ through the following process.

Step 1: if $v_1 = 0$, then delete v_1. If $v_1 \neq 0$, leave it unchanged.

Step k: if $v_k \in \operatorname{span}(v_1, \dots, v_{k-1})$ then delete v_k. Otherwise leave v_k unchanged.

Stop the process after step n. We relabel the remaining vectors as $v_1, \ldots, v_m$ preserving the original order. Since each time we discard a vector that is already in the span of previous vectors, we have $\operatorname{span}(v_1, \ldots, v_m) = V$. By construction we also have $v_k \notin \operatorname{span}(v_1, \ldots, v_{k-1})$ for any $1 \leq k \leq m$. By Lemma 2.8, we have $v_1, \ldots, v_m$ is linearly independent. Thus $v_1, \ldots, v_m$ is a basis of V. $\square$

We need the following Lemma for proving the main theorem of this lecture, Theorem 2.13.

Lemma 2.12. *Let $v_1, \ldots, v_n$ be a basis of V. If $v_1 = a_1 w_1 + a_2 v_2 + \cdots + a_n v_n$ where $a_i \in F$, $1 \leq i \leq n$ and $a_1 \neq 0$, then $w_1, v_2, \ldots, v_n$ is also a basis.*

Proof. Since $v_1, \ldots, v_n$ is a basis, for any $x \in V$, we have $x = x_1 v_1 + \cdots + x_n v_n$ for some $x_i \in F$, $1 \leq i \leq n$. Since $v_1 = a_1 w_1 + a_2 v_2 + \cdots + a_n v_n$, we have $x = x_1(a_1 w_1 + a_2 v_2 + \cdots + a_n v_n) + x_2 v_2 + \cdots + x_n v_n = a_1 x_1 w + (a_2 x_1 + x_2) v_2 + \cdots + (a_n x_1 + x_n) v_n$. Thus $\operatorname{span}(w_1, v_2, \ldots, v_n) = V$.

To show that $w_1, v_2, \ldots, v_n$ is linearly independent, let $b_1 w_1 + b_2 v_2 + \cdots + b_n v_n = 0$. Then since $w_1 = \frac{1}{a_1}(v_1 - a_2 v_2 - \cdots - a_n v_n)$, we have $b_1 \frac{1}{a_1}(v_1 - a_2 v_2 - \cdots - a_n v_n) + b_2 v_2 + \cdots + b_n v_n = 0$ i.e. $\frac{b_1}{a_1} v_1 + (b_2 - \frac{b_1}{a_1} a_2) v_2 + \cdots + (b_n - \frac{b_1}{a_1} a_n) v_n = 0$. Since $v_1, \ldots, v_n$ is a basis, we have $\frac{b_1}{a_1} = b_2 - \frac{b_1}{a_1} a_2 = \cdots = b_n - \frac{b_1}{a_1} a_n = 0$. Hence $b_1 = \cdots = b_n = 0$. $\square$

Theorem 2.13 (Replacement theorem)**.** *Let $v_1, \ldots, v_n$ be a basis for V and $w_1, \ldots, w_m$ be linearly independent. Then reordering $v_1, \ldots, v_n$ if necessary, for each $1 \leq k \leq m$, we have $\{w_1, \ldots, w_k, v_{k+1}, \ldots, v_n\}$ is a basis of V. In particular $n \geq m$.*

Proof. We prove the theorem inductively on k.

We first prove that the conclusion is true for $k = 1$. Since $v_1, \ldots, v_n$ is a basis, $w_1 = a_1 v_1 + \cdots + a_n v_n$. Since $w_1, \ldots, w_m$ are linearly independent, $w_1 \neq 0$. Thus there is $1 \leq j \leq n$ such that $a_j \neq 0$. We reorder v_i such that $a_1 \neq 0$. We replace v_1 by w_1. Then $v_1 = \frac{1}{a_1}(w_1 - a_2 v_2 - \cdots - a_n v_n)$. By Lemma 2.12, $w_1, v_2, \ldots, v_n$ is a basis of V.

Suppose the conclusion is true for $k - 1$, we are going to show that the conclusion holds for k. By assumption, we have $w_1, \ldots, w_{k-1}, v_k, \ldots, v_n$ is a basis. Then there exist $a_1, \ldots, a_n \in F$ such that $w_k = a_1 w_1 + \cdots + a_{k-1} w_{k-1} + a_k v_k + \cdots + a_n v_n$. We claim that there exists $k \leq j \leq n$ such that $a_j \neq 0$. If this is not the case, then $a_k = \cdots = a_n = 0$ and we have $w_k = a_1 w_1 + \cdots + a_{k-1} w_{k-1}$. This contradicts the fact that $w_1, \ldots, w_m$ is linearly independent.

Now we reorder v_i such that $a_k \neq 0$ and we replace v_k by w_k. Then $v_k = \frac{1}{a_k}(w_k - a_1 w_1 - \cdots - a_{k-1} w_{k-1} - \cdots - a_n v_n)$. Hence by Lemma 2.12 again, $w_1, \ldots, w_k, v_{k+1}, \ldots, v_n$ is a basis of V. This finishes the induction.

To prove the last statement, we assume for contradiction that $n < m$. Then by the Theorem we just proved, $w_1, \ldots, w_n$ is a basis. Hence $w_{n+1} = a_1 w_1 + \cdots + a_n w_n$ for some $a_1, \ldots, a_n \in F$. This contradicts linear independence of $w_1, \ldots, w_m$ (Lemma 2.8). $\square$

Corollary 2.14. *Let V be a finite dimensional vector space. Then any basis of V has the same number of elements.*

Proof. Let $v_1, \ldots, v_n$ and $w_1, \ldots, w_m$ be two bases of V. Then using $v_1, \ldots, v_n$ is a basis and $w_1, \ldots, w_m$ is linearly independent, we have by Theorem 2.13, $n \le m$. On the other hand, using $w_1, \ldots, w_m$ is a basis and $v_1, \ldots, v_n$ is linearly independent we have, by Theorem 2.13 again, $m \le n$. Hence $m = n$. $\square$

Definition 2.15. Let V be a finite dimensional vector space. The number of elements in a basis is called the **dimension** of V, denoted by $\dim V$. We sometimes need to emphasize the field of the vector space and write $\dim_F V$.

Corollary 2.16. *Let V be a finite dimensional vector space. Then any set of linearly independent vectors can be extended to a basis of V.*

Proof. By Lemma 2.11, there is a basis $v_1, \ldots, v_n$ of V. Let $w_1, \ldots, w_m$ be linearly independent. By Theorem 2.13, we can reorder the v_i's so that $w_1, \ldots, w_m, v_{m+1}, \ldots, v_n$ is a basis. This finishes the proof. $\square$

Corollary 2.17. *Let V be a finite dimensional vector space and $W \subset V$ be a subspace. Then W is finite dimensional and* $\dim W \le \dim V$. *If* $\dim W = \dim V$ *then* $W = V$.

Proof. Suppose $\dim V = n \ge 0$. We apply the following procedure. If $W = \{0\}$, then $\dim W = 0 \le n$ and we are done. If $W \ne \{0\}$, choose $v_1 \in W$ such that $v_1 \ne 0$. If $W = \operatorname{span}(v_1)$ then since v_1 is linearly independent, $\dim W = 1 \le \dim V$. If $W \ne \operatorname{span}(v_1)$, we choose $v_2 \notin \operatorname{span}(v_1)$. Then v_1, v_2 is linearly independent by Lemma 2.8. We continue this process to obtain linearly independent vectors $v_1, \ldots, v_m \in W$. By Theorem 2.13, $m \le n$ and thus this process must stop at a stage where $m \le n$ and $v_1, \ldots, v_m$ is linearly independent and for every vector $v \in W$, $v_1, \ldots, v_m, v$ is linearly dependent. Hence by Lemma 2.8, $v \in \operatorname{span}(v_1, \ldots, v_m)$. Since $v \in W$ is arbitrary, $W = \operatorname{span}(v_1, \ldots, v_m)$. Hence W is finite dimensional and $\dim W = m \le n = \dim V$.

If $\dim W = \dim V = n$, then a basis of W consists of n vectors. By Theorem 2.12, any linearly independent n vectors in V is a basis. Thus a basis of W is also a basis of V hence $V = W$. $\square$

Example 2.18. The standard basis of F^n is

$$v_1 = (1, 0, \ldots, 0)^t,\ v_2 = (0, 1, \ldots, 0)^t, \ldots, v_n = (0, 0, \ldots, 1)^t.$$

Therefore, $\dim F^n = n$.

Example 2.19. $V = \{(a_1, a_2, a_3)^t \in F^3 : a_1 + a_2 + a_3 = 0\}$. It is an exercise to show that V is a subspace of F^3. To find a basis, we write $a_1 = -a_2 - a_3$. Set $a_2 = 1$, $a_3 = 0$ we get the first vector $(-1, 1, 0)^t$. Set $a_2 = 0$, $a_3 = 1$ we get $(-1, 0, 1)^t$. One can check that they are linearly independent and $(-a_2 - a_3, a_2, a_3)^t = a_2(-1, 1, 0)^t + a_3(-1, 0, 1)^t$. Thus $(-1, 1, 0)^t, (-1, 0, 1)^t$ is a basis and $\dim V = 2$. This is a baby version of finding all solutions to a system of linear equation. We will learn how to systematically do it later.

Example 2.20. Let $E^{ij} \in M_{m\times n}(F)$ be the matrix whose (i,j) entry is 1 and other entries are 0. Then $\{E^{ij}\}_{1\le i\le m, 1\le j\le n}$ is a basis of $M_{m\times n}(F)$ and hence $\dim M_{m\times n}(F) = mn$. For example, $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ is a basis of $M_{2\times 2}(F)$.

Example 2.21. The standard basis of $\mathcal{P}_n(F)$ is

$$p_0(x) = 1,\ p_1(x) = x, \ldots, p_n(x) = x^n.$$

Therefore, $\dim \mathcal{P}_n(F) = n + 1$.

Example 2.22. $V = \{f \in C^2(\mathbb{R}) : f'' + f = 0\} = \{c_1 \cos x + c_2 \sin x : c_1, c_2 \in \mathbb{R}\}$. $\cos x$ and $\sin x$ is a basis of V (homework). Therefore $\dim V = 2$.

2.4 Intersection, sum and direct sum of subspaces

In this section, we introduce several constructions of new vector spaces out of old spaces.

Proposition 2.23. *Let V be a vector space over F and $V_1, \ldots, V_m \subset V$ be subspaces, then $V_1 \cap \cdots \cap V_m$ is a subspace.*

Proof. We check the definition of a subspace. Since V_i is a subspace for any $1 \le i \le m$, we have $0 \in V_i$, if $x, y \in V_i$ then $x + y \in V_i$ and if $a \in F$, $x \in V_i$, we have $ax \in V_i$. Then we have $0 \in V_1 \cap \cdots \cap V_m$. If $x, y \in V_1 \cap \cdots \cap V_m$, then $x, y \in V_i$ for any i and hence $x + y \in V_i$ for any i i.e. $x + y \in V_1 \cap \cdots \cap V_m$. Similarly if $a \in F$, then $ax \in V_i$ for any i i.e. $ax \in V_1 \cap \cdots \cap V_m$. □

Example 2.24. Let $V_1 = \operatorname{span}(v_1, v_2) = \{(a_1, a_2, 0)^t : a_1, a_2 \in \mathbb{R}\}$ and $V_2 = \operatorname{span}(v_3, v_4) = \{(a_3, a_3, a_4)^t : a_3, a_4 \in \mathbb{R}\}$. Then $V_1 \cap V_2 = \{(a_3, a_3, 0)^t : a_3 \in \mathbb{R}\} = \operatorname{span}(v_3)$.

Definition 2.25. Let V be a vector space over F and $V_1, \ldots, V_m \subset V$ be subspaces. The **sum** of $V_1, \ldots, V_m$ is denoted by $V_1 + \cdots + V_m = \{v_1 + \cdots + v_m : v_i \in V_i, 1 \le i \le m\}$.

Proposition 2.26. *$V_1 + \cdots + V_m$ is the smallest subspace of V containing all of $V_1, \ldots, V_m$ in the following sense*

(i) $V_1 + \cdots + V_m$ is a subspace.

(ii) If U is a subspace of V containing $V_1, \ldots, V_m$, then U contains $V_1 + \cdots + V_m$

Proof. (i) This is proved by directly checking the definition.

(ii) Let $v_i \in V_i$ for $1 \le i \le m$. Then $v_i \in U$. Since U is a subspace, $v_1 + \cdots + v_m \in U$. Thus $V_1 + \cdots + V_m \subset U$. □

Example 2.27. Let v_1-v_4 be as in Example 2.3 and V_1, V_2 be as in Example 2.24. Then we have $\operatorname{span}(v_1) + \operatorname{span}(v_2) = V_1$ and $V_1 + V_2 = F^3$. If $V_3 = \operatorname{span}(v_4)$, then $V_1 + V_3 = F^3$.

Theorem 2.28. $\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2)$.

3 Lecture 3

3.1 Sum, direct sum, quotient

Theorem 3.1. $\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2)$

Proof. Let $u_1, \ldots, u_k$ be a basis of $V_1 \cap V_2$. By Corollary 2.16, we can extend $u_1, \ldots, u_k$ to $u_1, \ldots, u_k, v_1, \ldots, v_j$ a basis of V_1 and to $u_1, \ldots, u_k, w_1, \ldots, w_l$ to be a basis of V_2. We claim that $B = \{u_1, \ldots, u_k, v_1, \ldots, v_j, w_1, \ldots, w_l\}$ is a basis of $V_1 + V_2$.

Let $x \in V_1$ and $y \in V_2$. Then $x = \lambda_1 u_1 + \cdots + \lambda_k u_k + b_1 v_1 + \cdots + b_j v_j$ and $y = \mu_1 u_1 + \cdots + \mu_k u_k + c_1 w_1 + \cdots + c_l w_l$ thus $x + y = (\lambda_1 + \mu_1)u_1 + \cdots + (\lambda_k + \mu_k)u_k + b_1 v_1 + \cdots + b_j v_j + c_1 w_1 + \cdots + c_l w_l$. Hence we see that $u_1, \ldots, u_k, v_1, \ldots, v_j, w_1, \ldots, w_l$ span $V_1 + V_2$.

To show that B is linearly independent, we suppose $a_1 u_1 + \cdots + a_k u_k + b_1 v_1 + \cdots + b_j v_j + c_1 w_1 + \cdots + c_l w_l = 0$. Then $b_1 v_1 + \cdots + b_j v_j = -a_1 u_1 - \cdots - a_k u_k - c_1 w_1 - \cdots - c_l v_l \in V_2$. Since $b_1 v_1 + \cdots + b_j v_j \in V_1$, we have $b_1 v_1 + \cdots + b_j v_j \in V_1 \cap V_2$. Thus there exists $\lambda_1, \ldots, \lambda_k$ such that $\lambda_1 u_1 + \cdots + \lambda_k u_k + b_1 v_1 + \cdots + b_j v_j = 0$. Since $u_1, \ldots, u_k, v_1, \ldots, v_j$ is a basis of V_1 we have $b_1 = \cdots = b_j = 0$. Similarly we can show that $c_1 = \cdots = c_l = 0$. Thus we have $a_1 u_1 + \cdots + a_k u_k = 0$. Since $u_1, \ldots, u_k$ is a basis of $V_1 \cap V_2$, we have $a_1 = \cdots = a_k = 0$.

In particular, $\dim(V_1 + V_2) = k + j + l = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2)$. $\square$

Remark 3.2. In this proof, we start with a basis of $V_1 \cap V_2$ and extend it to be a basis of V_1 and V_2 respectively. One <u>cannot</u> do it the other way around by directly taking the intersection of bases of V_1 and V_2. The intersection is not necessarily a basis of $V_1 \cap V_2$ as can be seen in the following example. Let $V_1 = \operatorname{span}(v_1, v_2)$ and $V_2 = \operatorname{span}(v_3, v_4)$. We have $\{v_1, v_2\} \cap \{v_3, v_4\} = \emptyset$. However, $V_1 \cap V_2 = \operatorname{span}(v_3)$.

Definition 3.3. We say that $V_1 + \cdots + V_m$ is a **direct sum** if any $x \in V_1 + \cdots + V_m$ can be written as $v_1 + \cdots + v_m$ *uniquely* for $v_i \in V_i$, $1 \le i \le m$. In this case, we write $V_1 \oplus \cdots \oplus V_m$ for $V_1 + \cdots + V_m$.

We want to introduce the summation notation here which will be used in this notes.

Notation 3.4. If $x_1, \ldots, x_n$ are things that can be added together e.g. numbers, vectors, functions, matrices etc, then we write

$$\sum_{i=1}^{n} x_i$$

for

$$x_1 + \cdots + x_n.$$

The symbol reads "summation of x_i for i from 1 to n".

The following Proposition shows that a direct sum is a sum of subspaces which has "no overlap".

Proposition 3.5. *Let V be a finite dimensional vector space. The following are equivalent:*

(i) The only way to write $0 = v_1 + \cdots + v_m$ where $v_i \in V_i$ is by taking $v_i = 0$ for all $1 \le i \le m$.

(ii) $V_1 + \cdots + V_m$ is a direct sum.

(iii) $V_i \cap \sum_{j:j\neq i} V_j = \{0\}$ for all $1 \le i \le m$.

(iv) $\dim(V_1 + \cdots + V_m) = \dim V_1 + \cdots + \dim V_m$.

Proof. (i) $\implies$ (ii) Suppose $x = \sum_{i=1}^m v_j = \sum_{j=1}^m w_j$ where $v_j, w_j \in V_i$ for any j. Then $0 = \sum_{j=1}^m (v_j - w_j)$. By (i) we have $v_j = w_j$ for all $1 \le j \le m$ and hence the decomposition is unique.

(ii) $\implies$ (iii) Let $v \in V_i \cap \sum_{j:j\neq i} V_j$. Then $v \in V_i$ and $v = \sum_{j:j\neq i} v_j$ for $v_j \in V_j$, $j \neq i$. Then $0 = v_1 + \cdots + v_{i-1} - v + v_{j+1} + \cdots + v_m$. By (ii), we have $v = 0$.

(iii) $\implies$ (iv) For any $1 \le j \le m$, let $\{w_i^{(j)}\}_{1\le i\le k_j}$ be a basis of V_j. We claim that $\{w_i^{(j)}\}_{1\le i\le k_j, 1\le j\le m}$ is a basis of $V_1 + \cdots + V_m$. Suppose $\sum_{j=1}^m \sum_{i=1}^{k_j} a_i^{(j)} w_i^{(j)} = 0$ for $a_i^{(j)} \in F$. Then for $1 \le j \le m$, $\sum_{i=1}^{k_j} a_i^{(j)} w_i^{(j)} = -\sum_{l\neq j} \sum_{i=1}^{k_l} a_i^{(j)} w_i^{(l)} \in \sum_{l:l\neq j} V_l$. Since $\sum_{i=1}^{k_j} a_i^{(j)} w_i^{(j)} \in V_j$, we have $\sum_{i=1}^{k_j} a_i^{(j)} w_i^{(j)} \in V_j \cap \sum_{l:l\neq j} V_l = \{0\}$. Thus $\sum_{i=1}^{k_j} a_i^{(j)} w_i^{(j)} = 0$. Since $\{w_i^{(j)}\}_{1\le i\le k_j}$ is a basis of V_j, we have $a_i^{(j)} = 0$ for all $1 \le i \le k_j$. Since this is true for all $1 \le j \le m$, we have $\{w_i^{(j)}\}_{1\le i\le k_j, 1\le j\le m}$ is linearly independent. Since any $v \in V_1 + \cdots + V_m$ can be written as $v = v_1 + \cdots + v_m$ and $v_j \in V_j$, we have $v_j = \sum_{i=1}^{k_j} a_i^{(j)} w_i^{(j)}$. Hence $v = \sum_{j=1}^m \sum_{i=1}^{k_j} a_i^{(j)} w_i^{(j)}$. Therefore, $V_1 + \cdots + V_m = \text{span}\{w_i^{(j)}\}_{1\le i\le k_j, 1\le j\le m}$.

(iv) $\implies$ (i) For any $1 \le j \le m$, let $\{w_i^{(j)}\}_{1\le i\le k_j}$ be a basis of V_j. As in previous step we can show that $V_1 + \cdots + V_m = \text{span}\{w_i^{(j)}\}_{1\le i\le k_j, 1\le j\le m}$. Since $\dim(V_1 + \cdots + V_m) = \dim V_1 + \cdots + \dim V_m = k_1 + \cdots + k_m$, we must have $\{w_i^{(j)}\}_{1\le i\le k_j, 1\le j\le m}$ is linearly independent since otherwise the spanning set $\{w_i^{(j)}\}_{1\le i\le k_j, 1\le j\le m}$ strictly contains a basis (by linear dependence and Lemma 2.11) which has elements $< \dim(V_1 + \cdots + V_m)$ a contradiction. Now suppose $0 = v_1 + \cdots + v_m$ where $w_i \in V_i$ then we can write $v_j = \sum_{i=1}^{k_j} a_i^{(j)} w_i^{(j)}$ and $0 = \sum_{j=1}^m \sum_{i=1}^{k_j} a_i^{(j)} w_i^{(j)}$. Since $\{w_i^{(j)}\}_{1\le i\le k_j, 1\le j\le m}$ is linearly independent, we have $v_j = 0$ for all $1 \le j \le m$. □

Corollary 3.6. *$V_1 + V_2$ is a direct sum if and only if $V_1 \cap V_2 = \{0\}$.*

Example 3.7. $V_1 = \text{span}(v_1, v_2) = \{(a_1, a_2, 0)^t : a_1, a_2 \in \mathbb{R}\}$

$V_2 = \text{span}(v_3, v_4) = \{(a_3, a_3, a_4)^t : a_3, a_4 \in \mathbb{R}\}$.

$V_1 + V_2 = \mathbb{R}^3$ is not a direct sum: $0 = (v_1 + v_2) - v_3$ and $V_1 \cap V_2 = \text{span}(v_3)$ and $\dim(V_1 + V_2) = 3 < 4 = \dim V_1 + \dim V_2$.

$V_1 + W_1 = \mathbb{R}^3$ is a direct sum.

Lemma 3.8. *Let V be a finite dimensional space and $W \subset V$ be a subspace. Then there exists a subspace U of V such that $V = W \oplus U$.*

Proof. Let $w_1, \ldots, w_m$ be a basis of W. By Corollary 2.16, it can be extended to $w_1, \ldots, w_m, v_1, \ldots, v_{n-m}$ a basis of V. Let $U = \operatorname{span}(v_1, \ldots, v_{n-m})$. Then we have $V = W + U$. Since $\dim W + \dim U = \dim V$, we have that $V = W \oplus U$. $\square$

Definition 3.9. U is called the **complement** of W in V.

3.2 Quotient space

Let $W \subset V$ be a subspace. We are going to define a vector space V/W which is obtained from V by "collapsing" planes parallel to W to a point.

Definition 3.10. Let V be a vector space over F and $W \subset V$ be a subspace. We define the **coset** (**translate**) of W containing $x \in V$ to be $x + W = \{x + w : w \in W\}$. The **quotient space** V/W is the set of all cosets of W that is $V/W = \{x + W : x \in V\}$.

The **canonical projection** associated to a quotient space is $\pi : V \to V/W$, $\pi(x) = x + W$.

Lemma 3.11. *$x + W = y + W$ if and only if $x - y \in W$.*

Proof. Since $x + W = y + W$, we have $x \in y + W$. Then by definition of $y + W$, we have $x - y \in W$.

If $x - y \in W$, then there is $w \in W$ such that $x = y + w$. Hence for any $x_1 \in x + W$, $x_1 = x + w_1 = y + (w + w_1) \in y + W$. Similarly, for any $y_1 \in y + W$, $y_1 = y + w_2 = x - w + w_2 \in x + W$. $\square$

Definition 3.12. We define the addition and scalar multiplication on V/W as $(x + W) + (y + W) := (x + y) + W$ and $c(x + W) = (cx) + W$ for $x, y \in V, c \in F$.

It is not clear whether the definition above are well-defined, since it defines the addition of two cosets $(x + W)$ and $(y + W)$ as the coset $(x + y) + W$ which depends on the choice of x and y in $x + W$ and $y + W$. If we pick different representatives, does the addition give the same coset? This is answered affirmatively in the following lemma.

Lemma 3.13. *Addition and scalar multiplication on V/W are well-defined.*

Proof. For any $x_1 \in x + W$, $y_1 \in y + W$, we have $x_1 - x \in W$ and $y_1 - y \in W$. Thus $(x_1 + y_1) - (x + y) = (x_1 - x) + (y_1 - y) \in W$ and $cx_1 - cx = c(x_1 - x) \in W$. Thus we have $(x_1 + y_1) + W = x + y + W$, $(cx_1) + W = (cx) + W$. $\square$

Example 3.14. $W = \operatorname{span}((0, 1)^t) = \{(0, a)^t : a \in F\} \subset F^2$.

For any $x \in F^2$, $x + W$ the vertical line through x.

$V/W =$ the set of all vertical lines in F^2.

$\pi(x) = x + W$ the vertical line through x.

For example

$0 + W = W$

$x = (1, 0)^t \in F^2$ and $y = (2, 1)^t$

$x + W = \{(1, a)^t : a \in F\}$

$y+W = \{(2,b+1)^t : b \in F\} = \{(2,a)^t : a \in F\} = (2,0)^t + W$
$(x+y)+W = \{(3,a+1)^t : a \in F\} = \{(3,b)^t : b \in F\}$
$(-x)+W = \{(-1,a)^t : a \in F\}$
In general $(x_1,x_2)^t + W = \{(x_1,a)^t : a \in F\}$ and we have
$((x_1,x_2)^t + W) + ((y_1,y_2)^t + W) = \{(x_1+x_2,a)^t : a \in F\}$
$c((x_1,x_2)^t + W) = \{(cx_1,a)^t : a \in F\}$.

In other words, adding two cosets in this particular case is just adding the first component and scalar multiplying a coset is just scalar multiplying the first component.

Let $U = \operatorname{span}((1,0)^t)$. The addition and scalar multiplication are also purely on the first component. Thus the vector space structure of U is "the same as" that of V/W or in math notation $U \cong V/W$.

Proposition 3.15. *V/W is a vector space over F.*

Proof. We check the axioms of a vector space. For $x,y,z \in V$, $a,b \in F$ we have

I: $(x+W)+(z+W) = (x+z)+W = (z+x)+W = (z+W)+(x+W)$.

II: $((x+W)+(y+W))+(z+W) = ((x+y)+W)+(z+W) = ((x+y)+z)+W = (x+(y+z))+W = (x+W)+((y+W)+(z+W))$.

III: $(x+W)+(0+W) = (x+0)+W = x+W$.

IV: $(x+W)+((-x)+W) = (x+(-x))+W = 0+W$.

V: $(a+b)(x+W) = ((a+b)x)+W = (ax+bx)+W) = (ax)+W)+(bx)+W = a(x+W)+b(x+W)$.

VI: $(ab)(x+W) = (ab)x+W = a(bx)+W = a(bx+W) = a(b(x+W))$.

VII: $a((x+W)+(y+W)) = a((x+y)+W)) = a(x+y)+W = (ax+ay)+W = (ax+W)+(ay+W) = a((x+W))+a((y+W))$.

VIII: $1(x+W) = (1x)+W = x+W$. □

Theorem 3.16. *If V is a finite dimensional vector space and W is a subspace of V, then*
$$\dim V/W = \dim V - \dim W.$$

Proof. Let $w_1,\ldots,w_m$ be a basis of W. By Corollary 2.16, we can extend it to $w_1,\ldots,w_m,v_1,\ldots,v_{n-m}$ a basis of V. We show that $v_1+W,\ldots,v_{n-m}+W$ are linearly independent and span V/W.

Suppose $b_1(v_1+W)+\cdots+b_{n-m}(v_{n-m}+W) = 0+W$. Then $(b_1v_1+\cdots+b_{n-m}v_{n-m})+W = 0+W$. Therefore, $b_1v_1+\cdots+b_{n-m}v_{n-m} \in W$. Since $w_1,\ldots,w_m$ is a basis of W, there exists $a_1,\ldots,a_m \in F$ such that $b_1v_1+\cdots+b_{n-m}v_{n-m} = a_1w_1+\cdots+a_mw_m$. Since $w_1,\ldots,w_m,v_1,\ldots,v_{n-m}$ is a basis of V, we have $b_1 = \cdots = b_{n-m} = a_1 = \cdots = a_m = 0$. This shows that $v_1+W,\ldots,v_{n-m}+W$ is linearly independent.

Let $x \in V$. Since $w_1,\ldots,w_m,v_1,\ldots,v_{n-m}$ is a basis of V, there exists $a_1,\ldots,a_m,b_1,\ldots,b_{n-m} \in F$ such that $x = a_1w_1+\cdots+a_mw_m+b_1v_1+\cdots+b_{n-m}v_{n-m}$. Then $x+W = (b_1v_1+W+\cdots+b_{n-m}v_{n-m})+W$. This shows that $v_1+W,\ldots,v_{n-m}+W$ span V/W. □

Remark 3.17. In view of Corollary 3.8, we see that if U is the complement of W in V then $\dim U = \dim V/W$ and if $u_1,\ldots,u_k$ is a basis of U, then $u_1+W,\ldots,u_k+W$ is a basis

V/W. We can think of V/W "the same as" U. More precisely V/W is isomorphic to U. We will discuss isomorphic in the next section. The thing is V/W does not depend on a choice of a basis and the construction also works for infinite dimensional vector spaces.

3.3 Linear transformations

The key object we study in linear algebra is linear transformation which is a map between vector spaces preserving the vector space structure.

Definition 3.18. Let V, W be vector spaces over F. A map $T : V \to W$ is a **linear transformation** (**linear map/mapping/operator**) if for any $x, y \in V$ and $c \in F$ we have
$$T(x+y) = T(x) + T(y) \text{ and } T(cx) = cT(x).$$

Remark 3.19. If we take $c = 0$ in the second equation then we have $T(0) = 0$.

Example 3.20 (The identity map)**.** The map $\mathrm{id} : V \to V$ defined by $\mathrm{id}(x) = x$ for any $x \in V$ is a linear transformation called the **identity map**.

Example 3.21 (The zero map)**.** The map $0 : V \to W$, $0(x) = 0 \in W$ for any $x \in V$ is a linearly transformation called the **zero map**.

Example 3.22. The most important linear transformation in this course is $T : F^n \to F^m$, $T(x) = Ax$ where a matrix $A \in M_{m\times n}(F)$ is multiplying a vector $x \in F^n$ is defined by
$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{pmatrix}.$$
The matrix multiplication follows the rule "row by column". We will see in the next section that it is a natural consequence of linearity.

Example 3.23. $T : F^n \to F^m$, $T(x_1, \ldots, x_n)^t = (x_1, \ldots, x_r, 0, \ldots, 0)^t$ for some $0 \le r \le \min(m, n)$ is a linear transformation. What the map is doing is first forgetting the variables $x_{r+1}, \ldots, x_n$ and then adding $(m - r)$-0 components.

Example 3.24. For $0 \ne b \in F^n$, the map $T : F^n \to F^n$, $T(x) = x + b$ is not a linear transformation since $T(0) = b \ne 0$.

Example 3.25. $T : F \to F$ is linear if and only if $T(x) = ax$ for some $a \in F$. $T(x) = T(x1) = xT(1) = ax$ where we write $T(1) = a$.

$T : \mathbb{R} \to \mathbb{R}$, $T(x) = x^2$ is not linear since $T(1+1) = 4 \ne 2 = T(1) + T(1)$.

Example 3.26. Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be a differentiable map. The differential of f at $x \in \mathbb{R}^n$ $df_x : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation. Recall that the differential acting on a vector $v \in \mathbb{R}^n$ is defined by the directional derivative of f in the direction of v
$$df_x(v) = \lim_{h\to 0} \frac{f(x+hv) - f(x)}{h}.$$

By the chain rule, $df_x(v) = Df(x)v$ where the right hand side is the Jacobian matrix

$$Df(x) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix}$$

multiplying the vector $v \in \mathbb{R}^n$.

Example 3.27.

$$T : C(\mathbb{R}) \to \mathbb{R}, \quad T(f) = \int_a^b f(x)dx$$

is a linear transformation

$P : C^1(\mathbb{R}) \to C(\mathbb{R})$, $P(f)(x) = f'(x)$ is a linear transformation.

$Q : C(\mathbb{R}) \to C(\mathbb{R})$, $Q(f)(x) = xf(x)$ is a linear transformation.

The two linear transformations P, Q are the fundamental operators in quantum mechanics.

Also the operator T and P are the main operators studied in calculus/analysis. Of course, linearity is probably the less important property considered in that subject.

3.4 Operations on linear transformations

Definition 3.28 (Addition and scalar multiplication of linear transformations)**.** Let $T, S : V \to W$ be linear transformations and $c \in F$. The maps $T + S$ and cT are defined as: for any $x \in V$,

$$(T + S)(x) = T(x) + S(x), \quad (cT)(x) = cT(x).$$

It is easy to check that $T + S$ and cT are both linear transformations.

Let $\mathcal{L}(V, W)$ denote the set of all linear transformation. Then with the addition and scalar multiplication above, one can also check $\mathcal{L}(V, W)$ is a vector space. We also write $\mathcal{L}(V) = \mathcal{L}(V, V)$.

4 Lecture 4

4.1 Operations on linear transformations

Definition 4.1 (Composition of linear transformations)**.** Let $T : V \to W$ and $S : W \to U$ be linear transformations. The **composition** ST (or $S \circ T$) of T and S is defined as

$$(ST)(x) = S(T(x)).$$

One can also check that $ST : V \to U$ is a linear transformation.

Proposition 4.2. *Let R, S, T be linear transformations. Whenever the composition in each item is defined, we have*
(i) $R(ST) = (RS)T$
(ii) $R(S + T) = RS + RT$
(iii) $(R + S)T = RT + ST$
(iv) For $a \in F$, $a(ST) = (aS)T = S(aT)$
(iv) $T \circ \mathrm{id}_V = T$, $\mathrm{id}_W \circ T = T$

Proof. One can directly verify these properties following the definition. □

Remark 4.3. We note that in general $ST \neq TS$ since as maps $T : V \to W$ and $S : W \to U$ one can only be composed in the order ST but not in the order of TS. Even when $V = W = U$, we do have examples of $T, S : V \to V$ such that $ST \neq TS$. $T = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $S = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Then $ST = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ and $TS = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$.

4.2 Invertibility and rank-nullity theorem

Definition 4.4. Let $T : V \to W$ be a linear transformation.
T is **injective** (**one-to-one**) if $T(x) = T(y)$ implies $x = y$.
T is **surjective** (**onto**) if for any $y \in W$ there is $x \in V$ such that $T(x) = y$
T is **invertible** (**nonsingular**) if it is both injective and surjective. An invertible T is called an **isomorphism** between V and W.
V and W are **isomorphic** if there exists an isomorphism between them. We denote V and W are isomorphic by $V \cong W$.

Definition 4.5. The **kernel** (**null space**) of T is

$$\ker T = \{x \in V : T(x) = 0\}.$$

The **nullity** of T is $\operatorname{null} T = \dim \ker T$.
The **image** (**range**) of T is

$$\operatorname{im} T = \{T(x) : x \in V\}.$$

The **rank** of T is $\operatorname{rank} T = \dim \operatorname{im} T$.

Lemma 4.6. *Let $T : V \to W$ be a linear transformation. Then $\ker T$ is a subspace of V and $\operatorname{im} T$ is a subspace of W.*

Proof. Since $T(0_V) = 0_W$, $0_V \in \ker T$. Let $x, y \in \ker T$, $c \in F$. Then $T(x + y) = T(x) + T(y) = 0 + 0 = 0$ and $T(cx) = cT(x) = 0$. Thus $x + y \in \ker T$, $cx \in \ker T$.

Since $T(0_V) = 0_W$, $0_W \in \operatorname{im} T$. Let $u, v \in \operatorname{im} T$. Then there exist $x, y \in V$ such that $T(x) = u$ and $T(y) = v$. We have $u + v = T(x) + T(y) = T(x + y) \in \operatorname{im} T$ and $cu = cT(x) = T(cx) \in \operatorname{im} T$. □

Proposition 4.7. *We have the following characterization of injectivity and surjectivity of a linear transformation T.*

(i) T is injective if and only if $\ker T = \{0\}$.

(ii) T is surjective if and only if $\operatorname{im} T = W$.

Proof. (i) Since T is injective and $T(0) = 0$, we have $\ker T = \{0\}$.

Suppose $T(x) = T(y)$. Then by linearity of T, $T(x - y) = T(x) - T(y) = 0$. Thus $x - y \in \ker T = \{0\}$. Hence $x = y$.

(ii) This is just the definition of surjective. □

Example 4.8. $T : F^n \to F^m$, $T(x_1, \ldots, x_n)^t = (x_1, \ldots, x_r, 0, \ldots, 0)^t$ for some $0 \le r \le \min(m, n)$. We have $\ker T = \{(0, \cdots, 0, x_{r+1}, \ldots, x_n)^t : x_i \in F,\ r + 1 \le i \le n\}$ and $\operatorname{im} T = \{(x_1, \ldots, x_r, 0, \ldots, 0)^t : x_i \in F,\ 1 \le i \le r\}$. We have $\operatorname{null} T = n - r$ and $\operatorname{rank} T = r$. T is injective if $r = n$. T is surjective if $r = m$.

Proposition 4.9. *The following are equivalent:*

(i) $T : V \to W$ is invertible.

(ii) There is $S : W \to V$ linear such that $ST = \mathrm{id}_V$ and $TS = \mathrm{id}_W$.

(iii) T maps a basis into a basis.

If any of the above items is satisfied, then the linear transformation S in (ii) is unique.

Proof. (i)$\Longrightarrow$ (ii) Since T is both injective and surjective, we can define $S(y)$ to be the unique x such that $T(x) = y$. It follows from definition that $ST = \mathrm{id}_V$ and $TS = \mathrm{id}_W$.

We now show that S is linear. In $T(x + y) = T(x) + T(y)$, we take $x = S(u)$ and $y = S(v)$. Then we have $T(S(u) + S(v)) = TS(u) + T(S(v)) = u + v$. Composing S on both sides from the left, we get $S(u) + S(v) = S(u + v)$ Similarly we have $T(cS(u)) = cTS(u) = cu$ and composing S on both sides, we have $S(cu) = cS(u)$.

(ii)$\Longrightarrow$ (iii) Let $v_1, \ldots, v_n$ be a basis of V. We show that $T(v_1), \ldots, T(v_n)$ is a basis of W. To show that $T(v_1), \ldots, T(v_n)$ is linearly independent, let $a_1T(v_1) + \cdots + a_nT(v_n) = 0$. We compose S on both sides and use the linearity of S and $ST = \mathrm{id}_V$, we have $S(a_1T(v_1) + \cdots + a_nT(v_n)) = a_1v_1 + \cdots + a_nv_n = 0 = 0$. Since $v_1, \ldots, v_n$ is a basis, we have $a_1 = \cdots = a_n = 0$.

Let $y \in W$. Since $S(y) \in V$, there exist $a_1, \ldots, a_n \in F$ such that $S(y) = a_1v_1 + \cdots + a_nv_n$. Then $y = TS(y) = T(a_1v_1 + \cdots + a_nv_n) = a_1T(v_1) + \cdots + a_nT(v_n)$. Thus $W = \operatorname{span}(T(v_1), \ldots, T(v_n))$.

(iii) $\Longrightarrow$ (i) To show that T is injective suppose $T(x) = 0$. We write $x = a_1v_1 + \cdots + a_nv_n$. Then by linearity, $a_1T(v_1)+\cdots+a_nT(v_n) = T(x) = 0$. Since $T(v_1), \ldots, T(v_n)$ is a basis, $a_1 = \cdots = a_n = 0$. To show that T is surjective, let $y \in W$. Since $T(v_1), \ldots, T(v_n)$ is a basis of W, $y = a_1T(v_1) + \cdots + a_nT(v_n) = T(a_1v_1 + \cdots + a_nv_n) \in \operatorname{im} T$.

To show uniqueness of S, we assume S_1, S_2 are two maps satisfying (ii). We consider the composition S_1TS_2. On one hand, $(S_1T)S_2 = \mathrm{id}_V S_2 = S_2$. On the other hand $S_1(TS_2) = S_1\mathrm{id}_W = S_1$. Thus $S_1 = S_2$. $\square$

Definition 4.10. The linear transformation S in (ii) is called the **inverse** of T and is denoted by T^{-1}.

Corollary 4.11. *If $V \cong W$, then $\dim V = \dim W$.*

Proof. If $V \cong W$, then there is an isomorphism $T : V \to W$. By Proposition 4.9 (iii), T maps a basis into a basis and hence $\dim V = \dim W$. $\square$

Lemma 4.12. *If $T; V \to W$ and $S : W \to U$ are both invertible, then ST is invertible and $(ST)^{-1} = T^{-1}S^{-1}$.*

Proof. We have $(ST)(T^{-1}S^{-1}) = S(TT^{-1})S^{-1} = SS^{-1} = \mathrm{id}_U$ and $(T^{-1}S^{-1})(ST) = T^{-1}(S^{-1}S)T = T^{-1}T = \mathrm{id}_V$. The conclusion follows from Proposition 4.9. $\square$

Theorem 4.13 (First isomorphism theorem)**.** *Let $T : V \to W$ be a linear transformation. We define $\bar{T} : V/\ker T \to \operatorname{im} T$ to be $\bar{T}(x+\ker T) = T(x)$. Then $\bar{T}$ is a well-defined linear transformation and is an isomorphism.*

$$\begin{tikzcd}
V \arrow[r, "T"] \arrow[d, "\pi"'] & W \\
V/\ker T \arrow[r, dashed, "\bar{T}"'] & \operatorname{im} T \arrow[u, "i"']
\end{tikzcd}$$

Proof. We first show that $\bar{T}$ is well-defined. For any $x' \in x+\ker T$, we have $x-x' \in \ker T$. Then $T(x) = T(x') + T(x - x') = T(x')$.

$\bar{T}$ is linear follows from the linearity of T.

If $\bar{T}(x + \ker T) = T(x) = 0$ then $x \in \ker T$. Thus $x + \ker T = 0 + \ker T$. This shows $\bar{T}$ is injective.

For any $y \in \operatorname{im} T$, there exists $x \in V$ such that $T(x) = y$. Thus $\bar{T}(x + \ker T) = y$. This shows that $\bar{T}$ is surjective. Hence $\bar{T}$ is an isomorphism. $\square$

Corollary 4.14 (Rank-nullity theorem)**.** *Let $T : V \to W$ be a linear transformation. Then*

$$\dim V = \operatorname{null} T + \operatorname{rank} T.$$

Proof. By Theorem 4.13, $\bar{T} : V/\ker T \to \operatorname{im} T$ is an isomorphism. By we have $\dim(V/\ker T) = \dim \operatorname{im} T$. By Theorem 3.16, we have $\dim V - \dim \ker T = \dim(V/\ker T) = \dim \operatorname{im} T$. $\square$

Corollary 4.15 (Underdetermined system of linear equations)**.** *Let* $T : V \to W$ *be a linear transformation and* $\dim W < \dim V$*. Then there is* $0 \neq x \in V$ *such that* $T(x) = 0$.

Proof. $\operatorname{null} T = \dim V - \operatorname{rank} T \geq \dim V - \dim W > 0$. □

Corollary 4.16. *Let* V, W *be finite dimensional vector spaces such that* $\dim V = \dim W$ *and* $T : V \to W$ *be a linear transformation. Then the following are equivalent:*
(i) T *is an isomorphism.*
(ii) T *injective.*
(iii) T *surjective.*
(iv) There is $S : W \to V$ *such that* $ST = \mathrm{id}_V$.
(v) There is $S : W \to V$ *such that* $TS = \mathrm{id}_W$.

Proof. (i) $\Longrightarrow$ (ii): This follows directly from definition.
(ii) $\Longrightarrow$ (iii): By (ii) and Lemma 4.7, we have $\operatorname{null} T = 0$. By Corollary 4.14, we have $\operatorname{rank} T = \dim V = \dim W$. By Lemma 2.17, $\operatorname{im} T = V$.
(iii) $\Longrightarrow$ (i): By (iii) we have $\operatorname{rank} T = \dim W = \dim V$. By Corollary 4.14, we have $\operatorname{null} T = 0$ and hence $\ker T = \{0\}$. Thus T is an isomorphism.
(i) $\Longrightarrow$ (iv) and (i) $\Longrightarrow$ (v): This follows from Proposition 4.9.
(iv) $\Longrightarrow$ (ii): Let $x \in \ker T$. Then $x = ST(x) = S(0) = 0$. Thus $\ker T = \{0\}$.
(v) $\Longrightarrow$ (iii): Let $y \in W$. Then $y = TS(y) = T(S(y)) \in \operatorname{im} T$. Thus $\operatorname{im} T = W$. □

Remark 4.17. The infinite dimensional version of (ii) and (iii) in this Corollary is called Fredholm alternative. It is useful in solving PDEs cf. [1, Chapter 8].

4.3 Basis, coordinates and matrix representation

In Lecture 2, we saw that a finite dimensional vector space is built up by a basis. In this section, we will see that given a basis on a vector space, a linear transformation reduces to a matrix.

Lemma 4.18 (Basis and coordinates)**.** *Let* $v_1, \ldots, v_n$ *be a basis of* V*. Then for every* $v \in V$*, there exists unique* $x_1, \ldots, x_n \in F$ *such that* $v = x_1 v_1 + \cdots + x_n v_n$.

Proof. Since $v_1, \ldots, v_n$ spans V, there exist $x_1, \ldots, x_n \in F$ such that $x = x_1 v_1 + \cdots + x_n v_n$. To show uniqueness we assume that $x = y_1 v_1 + \cdots + y_n v_n$. Then we have $(x_1 - y_1)v_1 + \cdots + (x_n - y_n)v_n = 0$. Since $v_1, \ldots, v_n$ is linearly independent, $x_1 - y_1 = \cdots = x_n - y_n = 0$. Thus $x_i = y_i$ for all $1 \leq i \leq n$. □

Definition 4.19. Let $\beta = (v_1, \ldots, v_n)$ be a basis of V. We say that $(x_1, \ldots, x_n)^t$ is the **coordinates** of v under the basis and we write $[v]_\beta$ for $(x_1, \ldots, x_n)^t$.

Theorem 4.20. *Let* V *be a* n*-dimensional vector space over* F*. We define the map* $\phi_\beta : V \to F^n$ *by* $\phi_\beta(v) = [v]_\beta$*. Then* ϕ_β *is a linear isomorphism and* $\phi_\beta^{-1}(x_1, \ldots, x_n)^t = x_1 v_1 + \cdots + x_n v_n$.

Proof. We first show that ϕ_β is linear. Let $\phi_\beta(v) = (x_1, \dots, x_n)^t$ and $\phi_\beta(w) = (y_1, \dots, y_n)^t$. Then $v+w = (x_1+y_1)v_1+\cdots+(x_n+y_n)v_n$ and hence $\phi_\beta(v+w) = (x_1+y_1, \dots, x_n+y_n)^t = \phi_\beta(v) + \phi_\beta(w)$. Moreover, $cv = cx_1v_1 + \cdots + cx_nv_n$ then $\phi_\beta(cv) = c\phi_\beta(v)$.

Next we show that ϕ_β is injective. If $\phi_\beta(v) = \phi_\beta(w) = (x_1, \dots, x_n)^t$, then $v = x_1v_1 + \cdots + x_nv_n = w$. Hence ϕ_β is injective.

Finally we show that ϕ_β is surjective. For any $(x_1, \dots, x_n)^t \in F^n$ we take $v = x_1v_1 + \cdots + x_nv_n$. Then $\phi_\beta(x) = (x_1, \dots, x_n)^t$.

Thus ϕ_β is an isomorphism and $V \cong F^n$.

Since $\phi_\beta(v) = (x_1, \dots, x_n)^t$ implies that $v = x_1v_1+\cdots+x_nv_n$, we see that $\phi_\beta^{-1}(x_1, \dots, x_n)^t = x_1v_1 + \cdots + x_nv_n$. □

Lemma 4.21. *Let $T : V \to W$ be a linear transformations and $v_1, \dots, v_n$ be a basis of V. Then T is uniquely determined by its values at $T(v_1), \dots, T(v_n)$.*

Proof. Let $v \in V$. Since $v_1, \dots, v_n$ is a basis of V, we have $v = x_1v_1 + \cdots + x_nv_n$. Then $T(v) = T(x_1v_1 + \cdots + x_nv_n) = x_1T(v_1) + \cdots + x_nT(v_n)$. □

Theorem 4.22. *Let $T : V \to W$ be a linear transformation. Let $\beta = (v_1, \dots, v_n)$ be a basis of V and $\gamma = (w_1, \dots, w_m)$ be a basis of W. Then there exists a unique matrix $A \in M_{m\times n}(F)$ such that for any $x \in F^n$,*

$$\phi_\gamma T \phi_\beta^{-1}(x) = Ax,$$

where $\phi_\beta : V \to F^n$ and $\phi_\gamma : W \to F^m$ are the coordinates isomorphisms.

The j-th column of A is given by the coordinates of $T(v_j)$ under the basis γ. That is $A = ([T(v_1)]_\gamma, \dots, [T(v_n)]_\gamma)$.

Proof. Since $(w_1, \dots, w_m)$ is a basis, for each j there exist scalars $a_{1j}, \dots, a_{mj}$ such that

$$T(v_j) = \sum_{i=1}^{m} a_{ij} w_i. \tag{1}$$

Now for any $x = (x_1, \dots, x_n)^t \in F^n$,

$$\begin{aligned} T\phi_\beta^{-1}(x) &= T(x_1v_1 + \cdots + x_nv_n) = \sum_{j=1}^{n} x_j T(v_j) \\ &= \sum_{j=1}^{n} x_j \sum_{i=1}^{m} a_{ij} w_i = \sum_{i=1}^{m} \Big(\sum_{j=1}^{n} a_{ij} x_j \Big) w_i. \end{aligned}$$

Thus

$$\phi_\gamma T \phi_\beta^{-1}(x) = \begin{pmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{pmatrix} = Ax.$$

We note that by (1), the j-th column of A is given by the coordinates of $T(v_j)$ under the basis γ. This finishes the proof. □

Definition 4.23. The matrix A is called the **matrix representation** of T under the bases β of V and γ of W and is denoted by $[T]_\beta^\gamma$.

Corollary 4.24. *$\mathcal{L}(V,W) \cong M_{m\times n}(F)$. In particular,* $\dim \mathcal{L}(V,W) = (\dim V)(\dim W)$

Example 4.25 (The zero matrix)**.** For the zero map $0 : V \to W$, its matrix representation is the **zero matrix** under any basis β γ is $[0]_\beta^\gamma = 0_{m\times n}$.

Example 4.26 (The identity matrix)**.** For the identity map $\mathrm{id} : V \to V$, its matrix under a basis β is the **identity matrix** $[\mathrm{id}]_\beta^\beta = I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} = (\delta_{ij})$ where $n = \dim V$ and we used the **Kronecker delta** symbol $\delta_{ij} = \begin{cases} 1 \text{ if } i = j \\ 0 \text{ if } i \neq j \end{cases}$.

Example 4.27 (Diagonal matrix)**.** If $T(v_i) = \lambda_i v_i$ for all i, then

$$[T]_\beta^\beta = \mathrm{diag}(\lambda_1, \ldots, \lambda_n) = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}.$$

Example 4.28. Let $T : \mathcal{P}_n(\mathbb{R}) \to \mathcal{P}_n(\mathbb{R})$ $Tf = f'$ and $\beta = (1, x, \ldots, x^n)$ be the standard basis. To compute $[T]_\beta^\beta$, we find $T(x^k) = kx^{k-1}$. Thus $[T(x^k)]_\beta = (0, \ldots, 0, k, 0 \ldots, 0)^t$ where the k is at the $(k-1)$-th position. Then

$$[T]_\beta^\beta = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 2 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & n \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix}.$$

References

[1] Gilbarg D, Trudinger NS. Elliptic partial differential equations of second order. Berlin: springer; 1977.

5 Lecture 5

5.1 Matrix multiplication

Lemma 5.1. *$T, S : V \to W$ be linear transformations. Let $\beta = (v_1, \ldots, v_n)$ be a basis of V and $\gamma = (w_1, \ldots, w_m)$ be a basis of W. Then $[T+S]_\beta^\gamma = [T]_\beta^\gamma + [S]_\beta^\gamma$ and $[cT]_\beta^\gamma = c[T]_\beta^\gamma$.*

Proof. Use linearity of $T, S, \phi_\beta, \phi_\gamma$ to verify. $\square$

Corollary 5.2. *$\mathcal{L}(V, W) \cong M_{m\times n}(F)$ under the map $T \mapsto [T]_\beta^\gamma$. In particular,* $\dim \mathcal{L}(V, W) = (\dim V)(\dim W)$.

Proof. Show injectivity and surjectivity of the map $T \mapsto [T]_\beta^\gamma$ using β and γ are bases. $\square$

Definition 5.3. Let $A \in M_{m\times n}(F)$ and $B \in M_{n\times p}(F)$. Then the **matrix multiplication** $C = AB \in M_{m\times p}(F)$ is given by

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}.$$

The i-th row, j-th column of C is given by multiplying the i-th row of A and j-th column of B.

Proposition 5.4 (Matrix multiplication is composition of linear transformation)**.** *Let $T : V \to W$ and $S : W \to U$ be linear transformations, and β, γ and δ are bases of V, W, U respectively. Then*

$$[ST]_\beta^\delta = [S]_\gamma^\delta \, [T]_\beta^\gamma.$$

Proof. We write $[S]_\gamma^\delta = (a_{ij}) \in M_{m\times n}(F)$ and $[T]_\beta^\gamma = (b_{ij}) \in M_{n\times p}(F)$ and $\beta = (v_1, \ldots, v_p)$, $\gamma = (w_1, \ldots, w_n)$ and $\delta = (u_1, \ldots, u_m)$. Then

$$\begin{aligned} ST(v_j) =& S\left(\sum_{k=1}^{n} b_{kj} w_k\right) \\ =& \sum_{k=1}^{n} b_{kj} S(w_k) \\ =& \sum_{k=1}^{n} b_{kj} \sum_{i=1}^{m} a_{ik} u_i \\ =& \sum_{i=1}^{m} \left(\sum_{k=1}^{n} a_{ik} b_{kj}\right) u_i. \end{aligned}$$

Thus $[ST]_\beta^\gamma = (c_{ij})$ where $c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$. $\square$

We can translate definitions and results about linear transformations into matrices.

Definition 5.5. For $A \in M_{m\times n}(F)$, we define $L_A : F^n \to F^m$ to be the linear transformation $L_A(x) = Ax$. (We sometimes do not distinguish A and L_A.) The following notation are inherited from those of linear transformations

$\ker A = \ker L_A$,
$\operatorname{null} A = \operatorname{null} L_A$,
$\operatorname{im} A = \operatorname{im} L_A$,
$\operatorname{rank} A = \operatorname{rank} L_A$.

The following properties of matrix multiplication follows directly from Proposition 5.4 and Proposition 4.2 applied to L_A, L_B, L_C.

Proposition 5.6 (Matrix algebra rules)**.** *For any matrices of compatible sizes and any scalar $a \in F$:*

(i) $A(BC) = (AB)C$
(ii) $A(B + C) = AB + AC$
(iii) $(A + B)C = AC + BC$
(iv) $a(BC) = (aB)C = B(aC)$
(iv) $AI_n = A$, $I_m A = A$

The following rank nullity theorem for matrix follows from applying Corollary 4.14 to L_A.

Theorem 5.7. *Let $A \in M_{m\times n}(F)$. Then*

$$n = \operatorname{rank} A + \operatorname{null} A.$$

Definition 5.8. A matrix A is called **invertible** if L_A is invertible.

By Corollary 4.11, if A is invertible, then A is a square matrix. By Corollary 4.16, we have the following invertiblity criterion.

Corollary 5.9. *Let $A \in M_{n\times n}(F)$. The following are equivalent.*

(i) A is invertible.
(ii) $\operatorname{null} A = 0$.
(iii) $\operatorname{rank} A = n$.
(iv) There is $B \in M_{n\times n}(F)$ such that $AB = I$.
(v) There is $B \in M_{n\times n}(F)$ such that $BA = I$.
If any of the above holds, then the matrix B in (iv) (v) are the same and unique.

Definition 5.10. If A is invertible, then the B in (iv), (v) above are called the **inverse matrix** of A denoted by A^{-1}.

Corollary 5.11. *Let $T : V \to W$ be a linear transformation, β be a basis of V and γ be a basis of W. Then T is invertible if and only if $[T]^\gamma_\beta$ is invertible. Furthermore,*

$$[T^{-1}]^\beta_\gamma = ([T]^\gamma_\beta)^{-1}.$$

Proof. If T is invertible, then $I = [\mathrm{id}_V]^\beta_\beta = [T^{-1}T]^\beta_\beta = [T^{-1}]^\beta_\gamma [T]^\gamma_\beta$. Then $[T]^\gamma_\beta$ is invertible and we have $([T]^\gamma_\beta)^{-1} = [T^{-1}]^\beta_\gamma$.

If $A = [T]^\gamma_\beta$ is invertible, then since $\phi_\gamma T \phi_\beta^{-1} = L_A$, we have $T = \phi_\gamma^{-1} L_A \phi_\beta$. Since the right hand side are all invertible, we have T is invertible by Lemma 4.12. □

5.2 Change of basis formula

Lemma 5.12. *Let $\beta = (v_1, \dots, v_n)$ and $\gamma = (w_1, \dots, w_n)$ be two bases of V. Let $P = (p_{ij}) = [\mathrm{id}]_\beta^\gamma \in M_{n\times n}(F)$. Then $v_j = \sum_{i=1}^m p_{ij} w_i$ for $1 \le j \le n$ and for any $v \in V$, $[v]_\gamma = P[v]_\beta$.*

Proof. This is a direct consequence of Theorem 4.22 with $T = \mathrm{id}$. □

Definition 5.13. The **change of coordinates matrix** from β-coordinates to γ-coordinates is the matrix $P = (p_{ij}) \in M_{n\times n}(F)$ such that $[v]_\gamma = P[v]_\beta$ for any $v \in V$. In other words, $P = [\mathrm{id}]_\beta^\gamma = ([v_1]_\gamma, \dots, [v_n]_\gamma)$.

The matrix P is also called the **change of basis matrix** (**transition matrix**) from γ to β in view of the property $v_j = \sum_{i=1}^n p_{ij} w_i$ for any $1 \le j \le n$. .

Remark 5.14. We note here that if P is the change of basis matrix from γ to β, then it is the change of coordinates matrix from β-coordinates to γ-coordinates. The role of β and γ are switched when we switch our view point from change of basis to change of coordinates.

Remark 5.15. We also note difference between the definition of change of basis matrix, $v_j = \sum_{i=1}^m p_{ij} w_i$ and the usual definition of a matrix A multiplying a vector x, $\sum_{j=1}^n a_{ij} x_j$. In the first case, the first index of the matrix P is summed over with w_j which is a vector, while in the second case, the second index of the matrix A is summed over with x_j which is a component of a vector. This is why some people will write $(v_1, \dots, v_n) = (w_1, \dots, w_n)P$ for the change of basis matrix thought of as the "row vector" $(w_1, \dots, w_n)$ multiplying the matrix P in the usual way.

Lemma 5.16. *Let $P = [\mathrm{id}]_\beta^\gamma$ be the change of basis matrix from γ to β. Then P is invertible and $P^{-1} = [\mathrm{id}]_\gamma^\beta$ is the change of basis matrix from β to γ.*

Proof. Since id is invertible and $\mathrm{id}^{-1} = \mathrm{id}$, by Corollary 5.11 we have $P = [\mathrm{id}]_\beta^\gamma$ is invertible and $P^{-1} = [\mathrm{id}]_\gamma^\beta$. □

Example 5.17. Let β be the standard basis of F^3 and $\gamma = ((1,0,0)^t, (1,1,0)^t, (1,1,1)^t)$, then the change of basis matrix from β to γ is $[\mathrm{id}]_\gamma^\beta = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$. In other words, $w_1 = v_1$, $w_2 = v_1 + v_2$, $w_3 = v_1 + v_2 + v_3$. To find the inverse matrix, we have $v_1 = w_1$, $v_2 = w_2 - v_1 = w_2 - w_1$, $v_3 = w_3 - v_1 - v_2 = w_3 - w_2$. Then $[\mathrm{id}]_\beta^\gamma = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$.

Theorem 5.18. *Let $T : V \to W$ be a linear transformation, β, β' be bases of V and γ, γ' be bases of W. Suppose $A = [T]_\beta^\gamma$, $B = [T]_{\beta'}^{\gamma'}$, $P = [\mathrm{id}_V]_{\beta'}^\beta$ and $Q = [\mathrm{id}_W]_{\gamma'}^\gamma$. Then we have*

$$B = Q^{-1}AP.$$

Proof. By Proposition 5.4, $B = [T]_{\beta'}^{\gamma'} = [\mathrm{id}]_{\gamma}^{\gamma'}[T]_{\beta}^{\gamma}[\mathrm{id}]_{\beta'}^{\beta} = Q^{-1}AP$. □

Corollary 5.19. *Let $T : V \to V$ be a linear transformation and β ,β' be two bases of V. Suppose $A = [T]_{\beta}^{\beta}$, $B = [T]_{\beta'}^{\beta'}$ and $P = [\mathrm{id}]_{\beta'}^{\beta}$ is the change of basis matrix from β to β'. Then*

$$B = P^{-1}AP.$$

Definition 5.20. Given $A, B \in M_{n\times n}(F)$, we say that A and B are **similar** if there is $P \in M_{n\times n}(F)$ invertible such that $B = P^{-1}AP$.

Finding the simplest matrix which is similar to a given $A \in M_{n\times n}(F)$ is the main reason to introduce eigenvalues and eigenvectors which will be discussed later in the notes. However, if one allows different bases of V and W as in Theorem 5.18, then we have the following theorem.

Theorem 5.21. *Let $T : V \to W$ be a linear transformation. Then there exists a basis β of V and a basis γ of W such that*

$$[T]_{\beta}^{\gamma} = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix},$$

where $r = \operatorname{rank} T$. In particular, the rank is the only invariant of a linear transformation under change of basis on both the domain and the target.

In matrix language, for any $A \in M_{m\times n}(F)$, then there exist invertible matrices $P \in M_{n\times n}(F)$ and $Q \in M_{m\times m}(F)$ such that $Q^{-1}AP = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$.

Proof. Let $v_1, \dots, v_n$ be a basis of V. Then $\operatorname{im} T = \operatorname{span}(T(v_1), \dots, T(v_n))$. By Lemma 2.11, $(T(v_1), \dots, T(v_n))$ contains a basis of $\operatorname{im} T$ which after relabeling the indices, we assume to be $T(v_1), \dots, T(v_r)$ for $r = \operatorname{rank} T$. In particular, $T(v_1), \dots, T(v_r)$ is linearly independent. By Corollary 2.16, we can extend it to be $\gamma = (T(v_1), \dots, T(v_r), w_{r+1}, \dots, w_m)$ which is a basis of W. Since $\operatorname{span}(T(v_1), \dots, T(v_r)) = \operatorname{im} T$, there exist $a_{ij} \in F$ for $1 \le i \le r$ and $r + 1 \le j \le n$ such that $T(v_j) = \sum_{i=1}^{r} a_{ij}T(v_i)$. We define $e_j = v_j - \sum_{i=1}^{r} a_{ij}v_i$ for $r+1 \le j \le n$. Apply Lemma 2.12 to $e_{r+1}, \dots, e_n$ one at a time, we see that $\beta = (v_1, \dots, v_r, e_{r+1}, \dots, e_n)$ is a basis of V. By definition of e_j, we have $T(e_j) = 0$ for $r+1 \le j \le n$. Then under the basis β and γ, we have $[T]_{\beta}^{\gamma} = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$. □

5.3 Dual space

Definition 5.22. A **linear function** (**linear functional/ 1-form/covector**) on V is a linear transformation $\varphi : V \to F$. The **dual space** V^* of V is the set of all linear functions on V. In other words $V^* = \mathcal{L}(V, F)$. In particular, V^* is a vector space. The **natural pairing** $\langle \cdot, \cdot \rangle : V^* \times V \to F$ is defined as $\langle \varphi, x\rangle := \varphi(x)$ for $x \in V, \varphi \in V^*$.

Remark 5.23. In quantum mechanics, a linear function is also called a bra-vector denoted by $\langle \varphi|$ while a usual vector is a ket-vector denoted by $|\psi\rangle$. To pair them, just write them together $\langle \varphi|\psi\rangle$ and it becomes a "bracket" which is the combined word of bra and ket.

Remark 5.24. When we don't have an inner product on a vector space, e.g. for vector spaces over finite fields, the natural paring serves as an "inner product". This is why we write it the same way as an inner product.

Example 5.25. For $V = F^n$, every linear functional $\varphi : F^n \to F$ is given by

$$\varphi \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = (a_1, \dots, a_n) \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = a_1x_1 + \cdots + a_nx_n.$$

for some row vector $(a_1, \dots, a_n) \in M_{1\times n}(F)$. Thus $(F^n)^*$ the space of row vectors while F^n is the space of column vectors.

Definition 5.26. Given a basis $\beta = (v_1, \dots, v_n)$ of V, the **dual basis** $\beta^* = (\varphi_1, \dots, \varphi_n)$ of V^* is defined by $\varphi_i(v_j) = \delta_{ij}$ for $1 \le i, j \le n$. By Lemma 4.21, φ_i is uniquely determined by its value at a basis.

Proposition 5.27. *The dual basis is a basis of V^*. In particular, $\dim V^* = \dim V$.*

Proof. Let $\varphi_1, \dots, \varphi_n$ be the dual basis of $\beta = (v_1, \dots, v_n)$. We first show that $\varphi_1, \dots, \varphi_n$ is linearly independent. Suppose $a_1\varphi_1 + \cdots + a_n\varphi_n = 0$. We evaluate both sides on v_i for $1 \le i \le n$ and get $a_1\varphi_1(v_i) + \cdots + a_n\varphi_n(v_i) = 0$. Since $\varphi_i(v_j) = \delta_{ij}$, we have $a_i = 0$ for all $1 \le i \le n$. Hence $\varphi_1, \dots, \varphi_n$ is linearly independent.

Next we show that $\operatorname{span}(\varphi_1, \dots, \varphi_n) = V^*$. Let $\varphi \in V^*$. For any $v \in V$ with $[v]_\beta = (x_1, \dots, x_n)^t$, we have $\varphi_i(v) = x_i$ for $1 \le i \le n$. We write $a_1 = \varphi(v_1), \dots, a_n = \varphi(v_n)$. Then by Lemma 4.21, $\varphi(v) = a_1x_1 + \cdots + a_nx_n = a_1\varphi_1(v) + \cdots + a_n\varphi_n(v) = (a_1\varphi_1 + \cdots + a_n\varphi_n)(v)$. Since this is true for any $v \in V$ we have $\varphi = a_1\varphi_1 + \cdots + a_n\varphi_n$. $\square$

Remark 5.28. From the proof above, we see that the coordinates of v under the basis β is given by $\phi_\beta(v) = (\varphi_1(v), \dots, \varphi_n(v))^t$.

5.4 Transpose of a linear transformation

Definition 5.29. Let $T : V \to W$ be a linear transformation. We define the **transpose map** (**dual map/pull-back**) $T^t : W^* \to V^*$ as $T^t(\varphi) = \varphi \circ T$. That is, for any $\varphi \in V^*$ and $v \in V$, $(T^t(\varphi))(v) = \varphi(T(v))$ or $\langle T^t\varphi, v\rangle = \langle \varphi, Tv\rangle$. Since $T^t(\varphi) = \varphi \circ T$, $T^t(\varphi)$ is a linear function on V since it is a composition of linear transformations. The map T^t is linear due to the algebraic rules of composition by linear transformations in Proposition 4.2.

Proposition 5.30. *Let $T : V \to W$ be a linear transformation. Let $\beta = (v_1, \dots, v_m)$ be a basis of V and $\gamma = (w_1, \dots, w_m)$ be a basis of W. Let β^* be the dual basis of β and γ^* be the dual basis of γ. Then $[T^t]_{\gamma^*}^{\beta^*} = ([T]_\beta^\gamma)^t$.*

Proof. Let $\gamma^* = (\psi_1, \ldots, \psi_m)$ be the dual basis of γ and $[T]_\beta^\gamma = (a_{ij})$. We want to find the coordinates of $T^t(\psi_i)$ under $\beta^* = (\varphi_1, \ldots, \varphi_n)$, the dual basis of β. Let $v \in V$ and $v = x_1 v_1 + \cdots + x_n v_n$. Note that $x_i = \varphi_i(v)$. Then we have $T^t(\psi_i)(v) = \psi_i(T(v)) = \psi_i(\sum_{k=1}^m (\sum_{j=1}^n a_{kj} x_j) w_k) = \sum_{j=1}^n a_{ij} x_j = \sum_{j=1}^n a_{ij} \varphi_j(v) = (\sum_{j=1}^n a_{ij} \varphi_j)(v)$. Thus $T^t(\psi_i) = \sum_{j=1}^n a_{ij}\varphi_j$. That is $[T^t(\psi_i)]_{\gamma^*} = \begin{pmatrix} a_{i1} \\ \vdots \\ a_{in} \end{pmatrix}$. Hence $[T^t]_{\gamma^*}^{\beta^*} = ([T]_\beta^\gamma)^t$. $\square$

6 Lecture 6

6.1 Transpose of a linear transformation continued

Proposition 6.1. *Let $T, S : V \to W$ be linear transformations. Then $(T+S)^t = T^t + S^t$ $(cT)^t = cT^t$.*

Proposition 6.2. *Let $T : V \to W$ and $S : W \to U$ be linear transformations. Then*

$$(ST)^t = T^t S^t.$$

Proof. Let $\varphi \in U^*$ and $v \in V$. Then $(ST)^t(\varphi)(v) = \varphi(ST(v)) = \varphi(S(T(v))) = S^t(\varphi)(T(v)) = T^t(S^t(\varphi))(v) = (T^tS^t)(\varphi)(v)$. Hence $(ST)^t(\varphi) = (T^tS^t)(\varphi)$ for any $\varphi \in U^*$. We get $(ST)^t = T^tS^t$. $\square$

Proposition 6.3. *Let $P = [\mathrm{id}]_\beta^\gamma$ be the change of basis matrix from γ to β. Then the change of basis matrix from β^* to γ^* is P^t. And we have for any $\varphi \in V^*$, $[\varphi]_{\beta^*} = P^t[\varphi]_{\gamma^*}$.*

Proof. By Proposition 5.30, we have $[\mathrm{id}]_{\gamma^*}^{\beta^*} = [\mathrm{id}^t]_{\gamma^*}^{\beta^*} = ([\mathrm{id}]_\beta^\gamma)^t = P^t$. The rest of the proposition follows from Lemma 5.12. $\square$

Remark 6.4. By Lemma 5.12, for $v \in V$ we have $[v]_\gamma = P[v]_\beta$ which is the opposite of the change of basis. Thus vectors are also called *contravariant vectors*. However, for a linear function $\varphi \in V^*$, the coordinate change satisfies $[\varphi]_{\beta^*} = P^t[\varphi]_{\gamma^*}$ which aligns with the change of basis of V. So linear functions are also called *covariant vectors*. Physicists use how components transform under a change of basis to define vectors and covectors (linear functions). This distinction is also important in differential geometry; see, for example, [1].

6.2 Double dual space

The dual space V^* consists of linear functions on V. If $\varphi \in V^*$, then φ maps a vector $v \in V$ to a scalar $\varphi(v) \in F$. The double dual space V^{**} is the dual space of V^*. If $\lambda \in V^{**}$, then λ maps a linear function $\varphi \in V^*$ to a scalar $\lambda(\varphi) \in F$.

Example 6.5. Let $V = F^3$ and $\psi_1, \psi_2, \psi_3 \in V^*$ be $\psi_1(x_1, x_2, x_3)^t = x_1$, $\psi_2(x_1, x_2, x_3)^t = x_1 + x_2$ and $\psi_3(x_1, x_2, x_3)^t = x_1 + x_2 + x_3$. We define $\lambda \in V^{**}$ as $\lambda(\varphi) = \varphi(1, 1, 1)^t$ for any $\varphi \in V^*$ which is the evaluation of a linear function at the vector $(1, 1, 1)^t$. Then $\lambda(\psi_1) = \psi_1(1, 1, 1)^t = 1$, $\lambda(\psi_2) = \psi_2(1, 1, 1)^t = 2$ and $\lambda(\psi_3) = \psi_3(1, 1, 1)^t = 3$. This λ is one element in V^{**}. In fact for any $v \in V$, we can define $J(v) \in V^{**}$ by $J(v)(\varphi) = \varphi(v)$ which is the evaluation of φ at $v \in V$. For example $J(0)$ is the zero vector in V^{**} since $J(0)(\varphi) = \varphi(0) = 0$ by linearity of φ. This is true for any $\varphi \in V^*$ and hence $J(0)$ is the zero function on V^* i.e. the zero vector in V^{**}. Theorem 6.7 shows that $J(v)$ for some $v \in V$ are all possible form of vectors that can appear in V^{**}.

Lemma 6.6. *Let V be a finite-dimensional vector space over a field F. Then* $\dim V = \dim V^* = \dim V^{**}$.

Proof. We apply the fact $\dim W = \dim W^*$ to $W = V^*$ and get $\dim V^* = \dim V^{**}$. Since $\dim V = \dim V^*$, we have the lemma. $\square$

Theorem 6.7 (Double dual isomorphism)**.** *Let V be a finite-dimensional vector space over a field F. Define the evaluation map $J : V \to V^{**}$, $J(v)(\varphi) = \varphi(v)$ for $\varphi \in V^*$. Then J is a linear isomorphism.*

Proof. First we show that J is well-defined, that is we show that $J(v) \in V^{**}$ for any $v \in V$. We need to show that $J(v)$ is a linear function on V^*. Let $\phi, \psi \in V^*$, $c \in F$. Then $J(v)(\varphi + \psi) = (\varphi + \psi)(v) = \varphi(v) + \psi(v) = J(v)(\varphi) + J(v)(\psi)$. And $J(v)(c\varphi) = c\varphi(v) = cJ(v)(\varphi)$.

Next we check linearity of J. For $v, w \in V$, $c \in F$, and $\varphi \in V^*$ we have $J(v + w)(\varphi) = \varphi(v + w) = \varphi(v) + \varphi(w) = J(v)(\varphi) + J(w)(\varphi) = (J(v) + J(w))(\varphi)$. In the last equality, we used the definition of the sum of two linear functions on V^*, $J(v)$ and $J(w)$. Thus $J(v+w) = J(v)+J(w)$. And $J(cv)(\varphi) = \varphi(cv) = c\varphi(v) = cJ(v)(\varphi)$. Hence $J(cv) = cJ(v)$. Therefore, J is linear.

Next we show injectivity of J. Suppose $J(v) = 0$. Then for every $\varphi \in V^*$, $J(v)(\varphi) = \varphi(v) = 0$. If $v \neq 0$, we can extend v to be a basis $v, v_2, \ldots, v_n$ and let $\varphi = \varphi_1$ be the first linear function in the dual basis $(\varphi_1, \ldots, \varphi_n)$. Then $\varphi(v) = 1$ while $\varphi(v_i) = 0$ for $i \geq 2$. Thus we have $\varphi \in V^*$ with $\varphi(v) \neq 0$. This is a contradiction. Thus $v = 0$, and J is injective.

Finally by Lemma 6.6 $\dim V = \dim V^{**}$, by Corollary 4.16 an injective linear map $V \to V^{**}$ is automatically surjective. Therefore J is an isomorphism. $\square$

Remark 6.8. If V is infinite dimensional, then J is in general an injective map and not necessarily an isomorphism. When it is an isomorphism, V is called *reflexive* and such spaces are important in Functional Analysis cf. [2].

Remark 6.9. Unlike the coordinates isomorphism ϕ_β which requires a choice of a basis of V, the isomorphism J is naturally defined out of the structure of dual spaces and is independent of a choice of a basis V. Thus we call J the *natural isomorphism* between V and V^{**} and we regard $V = V^{**}$. In category theory, the idea of being natural can be mathematically defined and in that language, "J is a component of a natural transformation from the identity functor to the double dual functor." This is not required in this course.

Proposition 6.10. *Let $T : V \to W$ be a linear transformation. Then $T^{tt} \circ J_V = J_W \circ T$. That is*

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "J_V"'] & W \arrow[d, "J_W"] \\ V^{**} \arrow[r, "T^{tt}"'] & W^{**} \end{tikzcd}$$

Thus if we think of J as the natural identification, then $T^{tt} = T$.

Proof. Let $v \in V$ and let $\varphi \in W^*$. We compute both sides applied to $\varphi \in W^*$. For the left-hand side we have $T^{tt}(J_V(v))(\varphi) = J_V(v)(T^t(\varphi)) = T^t(\varphi)(v) = \varphi(T(v))$. For the right-hand side, $J_W(T(v))(\varphi) = \varphi(T(v))$. Thus $J_W(Tv)(\varphi) = T^{tt}(J_V(v))(\varphi)$, for all $\varphi \in W^*$. Hence $J_W(T(v)) = T^{tt}(J_V(v))$ for any $v \in V$. Therefore, $T^{tt} \circ J_V = J_W \circ T$. □

6.3 Annihilators

Definition 6.11. For a subspace $W \subseteq V$, the **annihilator** of W is defined by $W^\perp = \{\varphi \in V^* : \varphi(w) = 0 \text{ for any } w \in W\}$ which is the set of all linear functions on V which vanish on W.

Proposition 6.12. *Let $\pi : V \to V/W$ be the canonical projection. Then $\pi^t : (V/W)^* \to V^*$ is injective and* $\operatorname{im} \pi^t = W^\perp$. *In particular, $W^\perp \cong (V/W)^*$ and* $\dim W^\perp = \dim V - \dim W$.

Proof. Suppose $\pi^t(\varphi) = 0$ for some $\varphi \in (V/W)^*$. Then for all $x \in V$, $0 = \pi^t(\bar{\varphi})(x) = \varphi(\pi(x)) = \varphi(x + W)$. Hence, $\bar{\varphi} = 0$. Thus $\ker(\pi^t) = \{0\}$ and π^t is injective.

Let $\varphi \in (V/W)^*$. Then for any $w \in W$, $\pi^t(\varphi)(w) = \varphi(w + W) = \varphi(0 + W) = 0$. Therefore, $\pi^t(\varphi) \in W^\perp$, and so $\operatorname{im}(\pi^t) \subset W^\perp$.

Let $\varphi \in V^*$ such that $\varphi \in W^\perp$, i.e., $\varphi(w) = 0$ for all $w \in W$. Define a functional $\tilde{\varphi} : V/W \to F$ by $\tilde{\varphi}(x + W) = \varphi(x)$. We must check that $\tilde{\varphi}$ is well-defined: If $x + W = x' + W$, then $x - x' \in W$, so $x = x' + w$ for some $w \in W$. Then $\tilde{\varphi}(x + W) = \varphi(x) = \varphi(x') + \varphi(w) = \varphi(x') + 0 = \varphi(x') = \tilde{\varphi}(x' + W)$. Thus, $\tilde{\varphi}$ is well-defined.

Linearity of $\tilde{\varphi}$ follows from the linearity of φ.

Now, for all $x \in V$, $\pi^t(\tilde{\varphi})(x) = \tilde{\varphi}(x + W) = \varphi(x)$. Hence, $\varphi = \pi^t(\tilde{\varphi})$. Thus so every element of $W^\perp$ lies in the image of π^t, and therefore $\operatorname{im}(\pi^t) = W^\perp$.

By the first isomorphism theorem Theorem 4.13, $\pi^t : (V/W)^* \to W^\perp$ is an isomorphism. So $(V/W)^* \cong W^\perp$ and $\dim W^\perp = \dim(V/W)^* = \dim V/W = \dim V - \dim W$. □

Corollary 6.13. $W^{\perp\perp} = J(W) = W$

Proof. For any $\varphi \in W^\perp$ and $v \in W$, we have $J(v)(\varphi) = \varphi(v) = 0$. Thus $J(W) \subset W^{\perp\perp}$. By Proposition 6.12, $\dim W^{\perp\perp} = \dim V^* - \dim W^\perp = \dim V - (\dim V - \dim W) = \dim W$. Since J is an isomorphism, $\dim J(W) = \dim W = \dim W^{\perp\perp}$. Thus by Lemma 2.17, $J(W) = W^{\perp\perp}$. □

Remark 6.14. There is a intuitive way to think of the dual space as the mirror of V. The fact that $V^{**} = V$ means if you take the mirror twice, you get back to yourself. If you have a subspace $W \subset V$, then its mirror is W^* (strictly speaking it consists of $\varphi \in W^*$ extended by 0 to all of V, see Homework 6). And $W^\perp$ is the complement of W^* in V^*. So taking $W^\perp$ is taking the complement of the mirror of W. The fact that $W^{\perp\perp} = W$ is saying that if you take complement and mirror twice then you get back to W.

Theorem 6.15. *Let $T : V \to W$ be a linear transformation. Then*

(i) $\ker T^t = (\operatorname{im} T)^\perp$

(ii) $\ker T = (\operatorname{im} T^t)^\perp$

(iii) $\operatorname{im} T^t = (\ker T)^\perp$

(iv) $\operatorname{im} T = (\ker T^t)^\perp$

Remark 6.16. In particular, if $T: V \to V$ is a linear transformation, then T is injective if and only if T^t is surjective. T is surjective if and only if T^t is injective.

Proof. Since for a subspace $W \subset V$ we have $W^{\perp\perp} = W$, (i) and (iv) are equivalent by taking $\perp$. Similarly, (ii) and (iii) are equivalent. Moreover (ii) is exactly (i) applied to the operator T^t using $T^{tt} = T$ from Proposition 6.10. So we only need to prove (i).

Let $\varphi \in (\operatorname{im} T)^\perp$. Then $\varphi(T(v)) = 0$ for all $v \in V$. Since $T^t(\varphi)(v) = \varphi(T(v)) = 0$, we have $T^t(\varphi) = 0$ i.e. $\varphi \in \ker(T^t)$. Thus $(\operatorname{im} T)^\perp \subset \ker T^t$.

Let $\varphi \in \ker(T^t)$. Then $T^t(\varphi) = 0$. That is for any $v \in V$, $0 = T^t(\varphi)(v) = \varphi(T(v))$. Hence $\varphi \in (\operatorname{im} T)^\perp$. Thus $\ker T^t \subset (\operatorname{im} T)^\perp$. □

Corollary 6.17. *Let $T: V \to W$ be a linear transformation and $T^t: W^* \to V^*$ be its transpose. Then we have $\operatorname{rank} T^t = \operatorname{rank} T$. In matrix language, we have $\operatorname{rank} A^t = \operatorname{rank} A$ for any $A \in M_{m\times n}(F)$.*

Proof. $\operatorname{rank}(T^t) = \dim \operatorname{im} T^t = \dim(\ker T)^\perp = \dim V^* - \dim \ker T = \dim V - (\dim V - \dim \operatorname{im} T) = \dim \operatorname{im} T = \operatorname{rank} T$. □

Remark 6.18. We define $\operatorname{rank} A$ as $\dim \operatorname{im} L_A$ which is the dimension of the span of columns of A and thus this is also called the column rank of A. $\operatorname{rank} A^t$ is the dimension of the span of rows of A and hence named row rank. The last conclusion the the corollary is also referred to as "row rank = column rank".

6.4 Linear systems and Gaussian elimination

Definition 6.19. The **preimage** of a set $Y \subset W$ under T to be $T^{-1}(Y) = \{x \in V : T(x) \in Y\}$. For example $\ker T = T^{-1}(0)$.

A system of linear equation is of the form

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \quad\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

In matrix notation, the equation above is written as $Ax = b$ where $A \in M_{m\times n}(F)$, $x \in F^n$ and $b \in F^m$. In the language of preimage, finding solutions to $Ax = b$ is finding the preimage $L_A^{-1}(b)$.

Lemma 6.20. *For any $y \in \operatorname{im} T$, either $T^{-1}(y) = \emptyset$ or there is $x \in V$ such that $T(x) = y$ and $T^{-1}(y) = x + \ker T$.*

Proof. If $y \notin \operatorname{im} T$, then $T^{-1}(y) = \emptyset$.

If $y \in \operatorname{im} T$, then there is $x \in V$ such that $T(x) = y$. For any $x' \in V$ such that $T(x') = y$, we have $T(x' - x) = T(x') - T(x) = y - y = 0$. Thus $x' \in x + \ker T$. On the other hand, if $x' \in x + \ker T$, then $x' - x \in \ker T$ and $T(x') = T(x) + T(x' - x) = y + 0 = y$. Thus $T^{-1}(y) = x + \ker T$. □

This gives the structure for the preimage $L_A^{-1}(b)$ being either $\emptyset$ or $x + \ker A$ for some $x \in F^n$ such that $Ax = b$. We would like to to develop an algorithm to explicitly find this preimage. So we would need to consider the following problems.

1. Given $A \in M_{m\times n}(F)$, find a basis for $\ker A$, i.e., all solutions to $Ax = 0$.
2. Determine if $b \in \operatorname{im} A$ or $Ax = b$ has a solution.
3. If a solution exists, find a specific x such that $Ax = b$.

The starting point of the algorithm is the following simple lemma.

Lemma 6.21. *Let $P \in M_{m\times m}(F)$ be invertible. Then $Ax = 0$ if and only if $PAx = 0$ and $Ax = b$ if and only if $PAx = Pb$.*

Definition 6.22. An **elementary row operation** on a matrix A is one of the following operations, each corresponding to multiplying A on the left by an appropriate **elementary matrix**.

(i) Swap rows i and j of A: $A \mapsto P_{ij}A$

$$P_{ij} = \begin{array}{cc} \begin{pmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 0 & \cdots & 1 & & \\ & & \vdots & \ddots & \vdots & & \\ & & 1 & \cdots & 0 & & \\ & & & & & \ddots & \\ & & & & & & 1 \end{pmatrix} & \begin{matrix} \\ \vdots \\ \text{row } i \\ \vdots \\ \text{row } j \\ \vdots \\ \\ \end{matrix} \\ \begin{matrix} \cdots & \text{col } i & \cdots & \text{col } j & \cdots \end{matrix} & \end{array}$$

(ii) Multiply row i by $\lambda \neq 0$: $A \mapsto D_i(\lambda)A$

$$D_i(\lambda) = \begin{array}{cc} \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & \lambda & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix} & \begin{matrix} \\ \vdots \\ \text{row } i \\ \vdots \\ \\ \end{matrix} \\ \begin{matrix} \cdots & \text{col } i & \cdots \end{matrix} & \end{array}$$

(iii) Replace row i with row $i + \lambda \cdot$ row j: $A \mapsto E_{ij}(\lambda)A$

$$E_{ij}(\lambda) = \begin{array}{cc} \begin{pmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & \cdots & \lambda & & \\ & & & \ddots & \vdots & & \\ & & & & 1 & & \\ & & & & & \ddots & \\ & & & & & & 1 \end{pmatrix} & \begin{matrix} \\ \vdots \\ \text{row } i \\ \vdots \\ \text{row } j \\ \vdots \\ \\ \end{matrix} \\ \begin{matrix} \cdots & \text{col } i & \cdots & \text{col } j & \cdots \end{matrix} & \end{array}$$

Remark 6.23. If we multiply the elementary matrix to the right of A then we are performing an elementary column operation:
$A \mapsto AP_{ij}$: swaps column i and column j of A
$A \mapsto AD_i(\lambda)$: multiply a column i by λ
$A \mapsto AE_{ij}(\lambda)$: replace *column* j with column j + $\lambda \cdot$ column i

Definition 6.24. Let A be a matrix. We now describe a systematic procedure for solving linear systems by row reduction, known as **Gaussian elimination**.

1. **Pivot selection:** Start with the leftmost column that contains a nonzero entry. Choose a nonzero entry in this column as the *pivot*. If necessary, interchange rows so that the pivot is at the top of the remaining submatrix.

2. **Normalize pivot:** Multiply the pivot row by a nonzero scalar so that the pivot entry becomes 1.

3. **Eliminate below pivot:** For each row below, subtract a suitable multiple of the pivot row so that all entries below the pivot are 0.

4. **Move right and down:** Restrict to the submatrix obtained by ignoring the pivot row and all columns to its left. Repeat steps (1)–(3) until no further pivots can be chosen.

5. **Eliminate above pivot:** Use row operations to eliminate all nonzero entries above the pivot, we obtain the **row reduced echelon form (RREF)**.

Definition 6.25. A matrix is in **row reduced echelon form (RREF)** if
(i) All zero rows, if any, are at the bottom.
(ii) In each nonzero row, the first nonzero entry (called a **pivot**) appears to the right of the pivot in the row above.
(iii) Each pivot equals 1.
(iv) Each pivot is the only nonzero entry in its column.
The non-pivot columns are called **free variables**.

References

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[2] Brezis H. Functional analysis, Sobolev spaces and partial differential equations. New York: Springer; 2011.

7 Lecture 7

7.1 Gauss elimination continued

Definition 7.1. A **system of linear equations** is of the form

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases} \iff x_1 \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix} + \cdots + x_n \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}.$$

In matrix notation, the equation above is written as $Ax = b$ where $A \in M_{m\times n}(F)$, $x \in F^n$ and $b \in F^m$. If $b = 0$, then the system is called **homogeneous**. If $b \neq 0$, the system is called **inhomogeneous**. We identify the system with the matrix A so we will refer to some columns as variables and variables as columns.

Definition 7.2. A matrix is in **row reduced echelon form (RREF)** if

(i) All zero rows, if any, are at the bottom.

(ii) In each nonzero row, the first nonzero entry (called a **pivot**) appears to the right of the pivot in the row above.

(iii) Each pivot equals 1.

(iv) Each pivot is the only nonzero entry in its column.

The non-pivot columns are called **free variables**.

Theorem 7.3. *For every matrix $A \in M_{m\times n}(F)$, one can apply elementary row operations to reduce A to a matrix R in RREF.*

(i) The number of pivot columns in R is $\operatorname{rank} A$. *The columns of the original matrix A corresponding to the pivot columns form a basis for* $\operatorname{im} A$.

(ii) The number of free variables is $n - \operatorname{rank} A = \operatorname{null} A$. To find a basis for $\ker A$, *assign 1 to one free variable and 0 to the others, then solve the resulting system for the pivot variables, repeat this for each free variable. The resulting vectors form a basis for* $\ker A$.

(iii) The nonzero rows of R form a basis for $\operatorname{im} A^t$.

(iv) The RREF of a matrix A is unique.

Remark 7.4. The pivot column must be a basis of $\operatorname{im} A$ but a basis of $\operatorname{im} A$ does not have to be the pivot columns.

Definition 7.5. The **augmented matrix** $[A \mid b]$ is the matrix obtained by appending the column vector $b \in F^m$ on the right to a matrix $A \in M_{m\times n}(F)$.

Theorem 7.6. *The system $Ax = b$ has a solution if and only if the last column in the RREF of $[A \mid b]$ is not a pivot column.*

If $Ax = b$ has a solution, then solutions of $Ax = b$ is given by solving the pivot variables in terms of the free variables.

We recall that Theorem 6.15 (ii) says $\operatorname{im} A = (\ker A^t)^\perp$. This is a dual space criterion which we can test solvability of $Ax = b$.

Theorem 7.7. *The equation $Ax = b$ has a solution if and only if for any $y \in F^m$ such that $A^t y = 0$ we have $y^t b = 0$.*

Proof. This is the content of Theorem 6.15 (ii). $\square$

Example 7.8. Let

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 2 \end{pmatrix}, \quad b = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$$

We will compute:
(i) The kernel $\ker(A)$.
(ii) The image $\operatorname{im}(A^t) = \operatorname{im}(A)$.
(iii) The solution to the inhomogeneous system $Ax = b$.
(iv) Check $b \in \operatorname{im} A$ using the dual space criterion.
(v) A basis for $\operatorname{span} \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \bigcap \operatorname{span} \left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} \right\}$.
We start with the augmented matrix and perform Gauss Elimination:

$$\left(\begin{array}{ccc|c} 2 & 1 & -1 & 2 \\ 1 & 2 & 1 & 3 \\ -1 & 1 & 2 & 1 \end{array}\right)$$

Step 1: Swap R_1 and R_2 to get a leading 1 in the top-left corner.

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 2 & 1 & -1 & 2 \\ -1 & 1 & 2 & 1 \end{array}\right)$$

Step 2: Eliminate the entries below the pivot in column 1:

$$R_2 \leftarrow R_2 - 2R_1, \quad R_3 \leftarrow R_3 + R_1$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 0 & -3 & -3 & -4 \\ 0 & 3 & 3 & 4 \end{array}\right)$$

Step 3: Make the pivot in row 2 a 1:

$$R_2 \leftarrow -\frac{1}{3} R_2$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 0 & 1 & 1 & \frac{4}{3} \\ 0 & 3 & 3 & 4 \end{array}\right)$$

Step 4: Eliminate above and below the pivot in column 2:

$$R_1 \leftarrow R_1 - 2R_2, \quad R_3 \leftarrow R_3 - 3R_2$$

$$\left(\begin{array}{ccc|c} 1 & 0 & -1 & \frac{1}{3} \\ 0 & 1 & 1 & \frac{4}{3} \\ 0 & 0 & 0 & 0 \end{array}\right)$$

(i) $Ax = 0$ is equivalent to

$$x_1 - x_3 = 0, \quad x_2 + x_3 = 0$$

So

$$\ker(A) = \left\{ \begin{pmatrix} x_3 \\ -x_3 \\ x_3 \end{pmatrix} : x_3 \in F \right\} = \operatorname{span} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}.$$

(ii) The first two columns of A are pivot columns, so

$$\operatorname{im} A = \operatorname{span} \left\{ \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right\}.$$

Since A is symmetric, $\operatorname{im} A = \operatorname{im} A^t$. So we also know that

$$\operatorname{im} A = \operatorname{im} A^t = \operatorname{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

The two bases are equivalent.

(iii) $Ax = b$ is equivalent to

$$x_1 - x_3 = \frac{1}{3}, \quad x_2 + x_3 = \frac{4}{3}$$

Thus the set of all solutions to $Ax = b$ are

$$\left\{ \begin{pmatrix} \frac{1}{3} + x_3 \\ \frac{4}{3} - x_3 \\ x_3 \end{pmatrix} : x_3 \in \mathbb{R} \right\} = \begin{pmatrix} \frac{1}{3} \\ \frac{4}{3} \\ 0 \end{pmatrix} + \operatorname{span} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}.$$

(iv) We test whether $Ax = b$ is consistent using the criterion:

$$Ax = b \iff y^t b = 0 \text{ for any } A^t y = 0$$

Since $A = A^t$, we already found:

$$\ker(A^t) = \ker(A) = \operatorname{span} \left\{ \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right\}$$

Compute

$$\begin{pmatrix}1 & -1 & 1\end{pmatrix}\begin{pmatrix}2\\3\\1\end{pmatrix} = 1\cdot 2 + (-1)\cdot 3 + 1\cdot 1 = 2-3+1 = 0$$

Therefore $b \in \operatorname{im} A$.

(v) Let the columns of A be v_1, v_2, v_3. Vectors in $\operatorname{span} v_1 \cap \operatorname{span}\{v_2, v_3\}$ are of the form x_1v_1 such that there are $x_2, x_3 \in F$ with $x_1v_1 = x_2v_2 + x_3v_3$. That is $(x_1, -x_2, -x_3)^t$ is a solution to $Ax = 0$. We know that $\ker A = \{c(1,-1,1)^t : c \in F\}$ and hence $(x_1, -x_2, -x_3)^t = c(1,-1,1)^t$. So we have $\operatorname{span} v_1 \cap \operatorname{span}\{v_2, v_3\} = \{cv_1 : c \in F\}$.

Proof of Theorem 7.3. Let $A = (v_1 \dots v_n)$ and R be an RREF obtained from A. Reordering the columns if necessary, we assume the pivot columns of R occur in positions 1 to r then $R = \begin{pmatrix} I_r & C \\ 0 & 0\end{pmatrix}$ where $C = (c_{ij}) \in M_{r\times(n-r)}(F)$.

(i) We are going to show that $v_1, \dots, v_r$ is a basis of $\operatorname{im} A$. Let $x = (x_1, \dots, x_r, 0, \dots, 0)^t$. Suppose $x_1v_1 + \cdots + x_rv_r = Ax = 0$. Then $0 = PAx = Rx = (x_1, \dots, x_r, 0, \dots, 0)^t$. Thus $x_1 = \cdots = x_r = 0$ and $v_1, \dots, v_r$ are linearly independent. We write $R = (w_1 \dots w_n)$. By the structure of RREF, for any $r+1 \le j \le n$ $w_j = \sum_{i=1}^r c_{ij}w_i$. Thus $x = (-c_{1j}, \dots, -c_{rj}, 0, \dots, 1, \dots, 0)^t$ where 1 appears at the j-th component satisfies $Rx = 0$. Hence $Ax = 0$ and $v_j = \sum_{i=1}^r c_{ij}v_i$. This shows $\operatorname{span}(v_1, \dots, v_r) = \operatorname{im} A$.

(ii) For a general solution of $Ax = 0$, solving $Rx = 0$ gives $x_i = -\sum_{j=r+1}^n c_{ij}x_j$ for $1 \le i \le r$. Hence

$$x = \begin{pmatrix} -\sum_{j=r+1}^n c_{1j}x_j \\ \vdots \\ -\sum_{j=r+1}^n c_{rj}x_j \\ x_{r+1} \\ \vdots \\ x_n \end{pmatrix} = x_{r+1}\begin{pmatrix} -c_{1(r+1)} \\ \vdots \\ -c_{r(r+1)} \\ 1 \\ \vdots \\ 0\end{pmatrix} + \cdots + x_n \begin{pmatrix} -c_{1n} \\ \vdots \\ -c_{rn} \\ 0 \\ \vdots \\ 1\end{pmatrix}.$$

For $r+1 \le j \le n$, let $u_j = (-c_{1j}, \dots, -c_{rj}, 0, \dots, 1, \dots, 0)^t$. Then $u_j \in \ker A$ by construction. We know that $\ker A = \operatorname{span}(u_{j+1}, \dots, u_n)$. Suppose $x_{r+1}u_{r+1} + \cdots + x_nu_n = 0$. By construction, last $n-r$ coordinates of x are $(x_{r+1}, \dots, x_n)^t$. So $x_{r+1}u_{r+1} + \cdots + x_nu_n = 0$ implies $x_{r+1} = \cdots = x_n = 0$. Thus, the vectors are linearly independent. Therefore, $(u_{r+1}, \dots, u_n)$ is a basis for $\ker A$.

(iii) Since $(PA)^t = A^tP^t$, we have $\operatorname{im}(R^t) = \operatorname{im}(A^tP^t) \subset \operatorname{im} A^t$. On the other hand $\operatorname{im}(A^t) = \operatorname{im}(R^t(P^{-1})^t) \subset \operatorname{im}(R^t)$. Hence $\operatorname{im} A^t = \operatorname{im} R^t$. For $1 \le i \le r$, we define $\varphi_i = (0, \dots, 1, \dots, 0, c_{i(r+1)}, \dots, c_{in})^t$. Then $R^t = (\varphi_1 \dots \varphi_r\, 0 \dots 0)$. If $y_1, \dots, y_r \in F$ are such that $y = y_1\varphi_1 + \cdots + y_r\varphi_r = 0$. Since the first r coordinates of y is $(y_1, \dots, y_r)^t$, the equation implies $y_1 = \cdots = y_r = 0$. Since other rows of R are 0, the first r rows of R is a basis of $\operatorname{im} R^t = \operatorname{im} A^t$.

(iv) Suppose R and R' are two RREF of A. Then $R' = P'A$ and $R = PA$ where P and P' are products of elementary matrices. Then $R' = QR$ and where $Q = P'P^{-1}$

is invertible. As above, we assume $R = \begin{pmatrix} I_r & C \\ 0 & 0 \end{pmatrix}$ where $C = (c_{ij}) \in M_{r\times(n-r)}(F)$. We write $Q = \begin{pmatrix} Q_1 & Q_2 \\ Q_3 & Q_4 \end{pmatrix}$ where $Q_1 \in M_{r\times r}(F)$, $Q_2 \in M_{r\times(n-r)}(F)$, $Q_3 \in M_{(n-r)\times r}$ and $Q_4 \in M_{(n-r)\times(n-r)}(F)$. Then $R' = QR = \begin{pmatrix} Q_1 & Q_1C \\ Q_3 & Q_3C \end{pmatrix}$. By (i), the number of pivots is $\operatorname{rank} A = r$. Thus by definition of RREF, the last $m-r$ rows of R' are 0. Hence $Q_3 = 0$. We claim that Q_1 is invertible. Indeed, if this is not the case, then the first r columns of $Q = \begin{pmatrix} Q_1 & Q_2 \\ 0 & Q_4 \end{pmatrix}$ are linearly dependent. This contradicts the fact that Q is invertible. Thus Q_1 is invertible. Since $R' = \begin{pmatrix} Q_1 & Q_1C \\ 0 & 0 \end{pmatrix}$ is in RREF and the first r columns are linearly independent, they must all be the pivot columns since any non-pivot column is a linear combination of the pivot columns to the left of it. Thus using the definition of RREF, $Q_1 = I_r$. Hence $R' = R$. □

Corollary 7.9. *Let $A \in M_{n\times n}(F)$ be invertible. Then there exists elementary matrices $E_1, \dots, E_m$ such that $A = E_1 \cdots E_m$.*

Proof. Since A is invertible, $\operatorname{rank} A = n$. Then by Theorem 7.3, every column of A is a pivot column. Since pivot column in an RREF consist of 1 at the pivot and 0 elsewhere, the RREF for A is I. Since the Gauss Elmination is multiplying elementary matrices $M_1, \dots, M_m$ to the left of A, we have $M_m \cdots M_1 A = I$ and hence $A = E_1 \cdots E_m$ where $E_i = M_i^{-1}$ is also an elementary matrix. □

Corollary 7.10. *Compute A^{-1} using Gauss elimination:*

1. *Start with the augmented matrix $[A \mid I]$*
2. *Perform row operations to reduce the left side A to the identity matrix.*
3. *The result will be $[I \mid A^{-1}]$ where the right block becomes the inverse of A.*

Computationally, row operations correspond to multiplying by elementary matrices, so the whole process is: $E_k \cdots E_2E_1A = I \quad\Longrightarrow\quad A^{-1} = E_k \cdots E_2E_1$.

If at any step we cannot reduce A to the identity (e.g., a pivot is zero and cannot be fixed by swapping rows), then A is not invertible.

Proof of Theorem 7.6. If the last column of $[A \mid b]$ is a pivot, then the last nonzero rows of the RREF of $[A \mid b]$ contains a row of the form $(0 \quad 0 \quad \cdots \quad 0 \mid 1)$. This corresponds to the equation $0x_1 + 0x_2 + \cdots + 0x_n = 1$ which has no solution.

If the last column in the RREF of $[A \mid b]$ is not a pivot column, then the pivot appears in columns of A. Hence a basis of $\operatorname{im}[A \mid b]$ is a basis of $\operatorname{im} A$ i.e. $\operatorname{im}[A \mid b] = \operatorname{im} A$. Hence $b = Ax$ for some $x \in F^n$.

To find all solutions to $Ax = b$ we write the RREF of $[A \mid b]$ as $\left(\begin{array}{cc|c} I_r & (c_{ij}) & \tilde{b} \\ 0 & 0 & 0 \end{array}\right)$ after permuting columns. We solve the pivots as $x_i = \tilde{b}_i + \sum_{i=r+1}^{n} c_{ij}x_j$ for $1 \le i \le r$.

Then

$$x = \begin{pmatrix} \tilde{b}_1 - \sum_{j=r+1}^n c_{1j}x_j \\ \vdots \\ \tilde{b}_r - \sum_{j=r+1}^n c_{rj}x_j \\ x_{r+1} \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \tilde{b}_1 \\ \vdots \\ \tilde{b}_r \\ 0 \\ \vdots \\ 0 \end{pmatrix} + x_{r+1} \begin{pmatrix} -c_{1(r+1)} \\ \vdots \\ -c_{r(r+1)} \\ 1 \\ \vdots \\ 0 \end{pmatrix} + \cdots + x_n \begin{pmatrix} -c_{1n} \\ \vdots \\ -c_{rn} \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

□

Remark 7.11. What if $b \notin \operatorname{im} A$ but we want to find the x such that Ax is closest to b? You would need the least square method which is covered in Linear Algebra II.

7.2 Determinant

Let $v_1, \dots, v_3 \in \mathbb{R}^3$. We write $vol(v_1, v_2, v_3)$ to be the volume of the parallelepiped spanned by v_1, v_2, v_3. The *signed volume* of v_1, v_2, v_3, denoted by $\det(v_1, v_2, v_3)$ is a function that gives $vol(v_1, v_2, v_3)$ if v_1, v_2, v_3 are right handed and $-vol(v_1, v_2, v_3)$ if v_1, v_2, v_3 are left handed. The determinant is the n-dimensional generalization of this which satisfies the properties in the following definition.

Definition 7.12. The **determinant** is a function $\det : M_{n\times n}(F) \to F$ satisfying the following properties. We write $A = (v_1 \dots v_n)$ where $v_i \in F^n$ are the columns of A.

(i) det is multilinear in columns: If for some $1 \le j \le n$, $v_j = ax + by$ where $a, b \in F$ $x, y \in F^n$, then

$$\det(v_1, \dots, ax + by, \dots, v_n) = a \det(v_1, \dots, x, \dots, v_n) + b \det(v_1, \dots, y, \dots, v_n).$$

(ii) det alternative in column: If $v_i = v_j$ for $i \neq j$, then $\det(v_1 \dots v_n) = 0$.

(iii) $\det I = 1$.

Lemma 7.13. *Suppose* $\det : M_{n\times n}(F) \to F$ *is a function satisfying (i), (ii), (iii) of Definition 7.12. Then we have the following*

(i) $\det(v_1 \dots 0 \dots v_n) = 0$.

(ii) $\det(\dots v_i \dots v_j \dots) = -\det(\dots v_j \dots v_i \dots)$.

(iii)

$$\det(v_1 \dots v_j + \sum_{i\neq j} \lambda_i v_i \dots v_n) = \det(v_1 \dots v_j \dots v_n)$$

Proof. (i) By multilinearity, $\det(v_1 \dots 0 \dots v_n) = 0 \det(v_1 \dots 0 \dots v_n) = 0$.

(ii)

$$\begin{aligned}
&\det(\dots, v_i \dots v_j, \dots) + \det(\dots, v_j \dots v_i, \dots)\\
=&\det(\dots, v_i \dots v_j, \dots) + \det(\dots, v_i \dots v_i, \dots)\\
&+ \det(\dots, v_j \dots v_j, \dots) + \det(\dots, v_j \dots v_i, \dots)\\
=&\det(\dots, v_i \dots v_i + v_j, \dots) + \det(\dots, v_j \dots v_i + v_j, \dots)\\
=&\det(\dots, v_i + v_j \dots v_i + v_j, \dots)\\
=&0.
\end{aligned}$$

(iii) By multilinearity in the j-th column, we can expand:

$$\det(v_1 \dots v_j + \sum_{i \neq j} \lambda_i v_i \dots v_n) = \det(v_1 \dots v_j \dots v_n) + \sum_{i \neq j} \lambda_i \det(v_1 \dots v_i \dots v_n),$$

where in each term of the sum, the j-th column has been replaced by v_i. By the alternating property, each determinant in the sum vanishes, since two columns are equal in each: $\det(\dots, v_i \dots v_i, \dots) = 0$. Therefore,

$$\det(v_1 \dots v_j + \sum_{i \neq j} \lambda_i v_i \dots v_n) = \det(v_1 \dots v_j \dots v_n)$$

$\square$

8 Lecture 8

8.1 Existence, uniqueness and basic properties of the determinant

Theorem 8.1. *There is a unique function* $\det : M_{n\times n}(F) \to F$ *satisfying the properties in Definition 7.12. In fact, the determinant is given by*

$$\det(A) = \sum_{\sigma\in S_n} \operatorname{sign}(\sigma) a_{\sigma(1)1}a_{\sigma(2)2}\cdots a_{\sigma(n)n}.$$

This is called the ***Leibniz formula for determinant****.*

Proof. Let $e_1, \dots, e_n$ be the standard basis of F^n. By multilinearity,

$$\begin{aligned}\det(A) &= \det\Big(\sum_{i_1=1}^{n} a_{i_1 1}e_i \ \dots \ \sum_{i_n=1}^{n} a_{i_n n}e_i\Big)\\ &= \sum_{i_1=1}^{n}\cdots\sum_{i_n=1}^{n} a_{i_1 1}a_{i_2 2}\dots a_{i_n n}\det(e_{i_1} \ \dots \ e_{i_n})\end{aligned}$$

By the alternating property, $\det(e_{i_1} \ \dots \ e_{i_n}) = 0$ if $i_k = i_l$ for $k \neq l$. Thus the summation is over all $i_1, \dots, i_n$ which are distinct. We would like to permute $e_{i_1} \ \dots \ e_{i_n}$ to $e_1, \dots, e_n$ and use the fact that $\det I = 1$. By Lemma 7.13, each swap will alter the sign of det. The question is how many swaps do we need to permute $e_{i_1} \ \dots \ e_{i_n}$ to $e_1, \dots, e_n$?

Definition 8.2. A **permutation of n elements** is a bijective map $\sigma : \{1, \dots, n\} \to \{1, \dots, n\}$. To write a permutation, we just list all its values $\sigma(1)$ to $\sigma(n)$. We use the one-line notation to write $\sigma = \sigma(1)\cdots\sigma(n)$. The set of all permutation is denoted by S_n which is called the **symmetric group**. There are $n! = n(n-1)\cdots 1$ elements in S_n.

An **inversion** in σ is a pair of indices (i, j) such that $i < j$ and $\sigma(i) > \sigma(j)$.

The **inversion number** of σ, denoted $\operatorname{inv}(\sigma)$, is the total number of inversions in σ.

The **sign** of a permutation σ is $\operatorname{sign}(\sigma) = (-1)^{\operatorname{inv}(\sigma)}$.

Example 8.3. $\operatorname{inv}(123) = 0$, $\operatorname{inv}(231) = 2$, $\operatorname{inv}(312) = 2$, $\operatorname{inv}(213) = 1$, $\operatorname{inv}(132) = 1$, $\operatorname{inv}(321) = 3$.

Lemma 8.4. *The number of adjacent swaps needed to sort σ into the identity permutation in the bubble sort algorithm is* $\operatorname{inv}(\sigma)$*. In particular,* $\det(e_{\sigma(1)} \ \dots \ e_{\sigma(n)}) = \operatorname{sign}(\sigma)$.

Proof. Given a list $a_1, a_2, \dots, a_n$, the bubble sort proceeds as follows:

(i) For each $1 \le i \le n-1$, compare a_i and a_{i+1}.

(ii) If $a_i > a_{i+1}$, swap them. If $a_i < a_{i+1}$, keep them unchanged.

(iii) After one full pass through the list, the largest element "bubbles up" to the end of the list.

(iv) Repeat the above passes on the first $n-1$, then $n-2$, ..., elements, until no swaps occur during a pass.

Consider bubble sort applied to the sequence $\sigma(1), \dots, \sigma(n)$. We claim that each swap removes exactly one inversion. Bubble sort only swaps adjacent elements $\sigma(k)$ and

$\sigma(k+1)$ when $\sigma(k) > \sigma(k+1)$. This removes the inversion $(k, k+1)$. We now verify that the number of other inversions remains the same. Let $(i, j) \neq (k, k+1)$ be an inversion before the swap. Consider the following cases:

If $i < k$ and $j > k+1$, then $\sigma(i)$ and $\sigma(j)$ are unchanged — the inversion remains.

If $i < k$ and $j = k$, the inversion becomes $(i, k+1)$ after the swap.

If $i < k$ and $j = k+1$, it becomes (i, k).

If $i = k$ and $j > k+1$, it becomes $(k+1, j)$.

If $i = k+1$ and $j > k+1$, it becomes (k, j).

Therefore, each swap in bubble sort decreases the total number of inversions by exactly one. Since bubble sort terminates when the sequence is sorted (i.e., has zero inversions), the total number of swaps performed is equal to the number of inversions in the original sequence. □

Coming back to the proof since $i_1, \ldots, i_n$ are distinct, there is $\sigma \in S_n$ such that $i_k = \sigma(k)$ for all k. Therefore

$$\begin{aligned}\det(A) &= \sum_{\sigma \in S_n} a_{\sigma(1)1} a_{\sigma(2)2} \cdots a_{\sigma(n)n} \det(e_{\sigma(1)} \ \ldots \ e_{\sigma(n)}) \\ &= \sum_{\sigma \in S_n} \operatorname{sign}(\sigma) a_{\sigma(1)1} a_{\sigma(2)2} \cdots a_{\sigma(n)n}.\end{aligned}$$

The above argument show that any function satisfying Definition 7.12 has to have the above expression. We now check that the function given by the expression above does satisfies Definition 7.12.

(i) Fix all columns except column j. Suppose $v_j = ax + by$. Then each entry in column j satisfies $a_{ij} = ax_i + by_i$ for all $i = 1, \ldots, n$.

$$\begin{aligned}&\sum_{\sigma \in S_n} \operatorname{sign}(\sigma) a_{\sigma(1)1} \cdots a_{\sigma(j)j} \cdots a_{\sigma(n)n} \\ =&\sum_{\sigma \in S_n} \operatorname{sign}(\sigma) a_{\sigma(1)1} \cdots (ax_{\sigma(j)} + by_{\sigma(j)}) \cdots a_{\sigma(n)n} \\ =&a \sum_{\sigma \in S_n} \operatorname{sign}(\sigma) a_{\sigma(1)1} \cdots x_{\sigma(j)} \cdots a_{\sigma(n)n} + b \sum_{\sigma \in S_n} \operatorname{sign}(\sigma) a_{\sigma(1)1} \cdots y_{\sigma(j)} \cdots a_{\sigma(n)n}.\end{aligned}$$

(ii) Suppose two columns are equal: $v_i = v_j$ for $i \neq j$. For any $\sigma \in S_n$ let $\sigma' \in S_n$ such that $\sigma'(i) = \sigma(j)$ and $\sigma'(j) = \sigma(i)$ and $\sigma'(k) = \sigma(k)$ for $k \neq i, j$. We have seen in the proof of Lemma 8.4 that swapping two adjacent elements in a permutation change the inversion by ± 1. Since we need $2|j-i|+1$ adjacent swaps to permute σ' to σ we have $\operatorname{sign}(\sigma') = (-1)^{2|j-i|+1} \operatorname{sign}(\sigma) = -\operatorname{sign}(\sigma)$. Since columns i and j are equal, $a_{\sigma(1)1} \cdots a_{\sigma(n)n} = a_{\sigma'(1)1} \cdots a_{\sigma'(n)n}$. Thus we have

$$\begin{aligned}&\operatorname{sign}(\sigma) a_{\sigma(1)1} \cdots a_{\sigma(n)n} + \operatorname{sign}(\sigma') a_{\sigma'(1)1} \cdots a_{\sigma'(n)n} \\ =&(\operatorname{sign}(\sigma) + \operatorname{sign}(\sigma')) a_{\sigma(1)1} \cdots a_{\sigma(n)n} = 0.\end{aligned}$$

Thus the terms corresponding to σ and σ' cancel in the sum. As every permutation σ corresponds to exactly one such pair, we have $\sum_{\sigma\in S_n} \operatorname{sign}(\sigma) a_{\sigma(1)1}a_{\sigma(2)2}\cdots a_{\sigma(n)n} = 0$.

(iii) Recall the identity matrix $I = (\delta_{ij})$, so we have

$$\delta_{\sigma(1)1}\delta_{\sigma(2)2}\cdots\delta_{\sigma(n)n} = \begin{cases} 1 & \text{if } \sigma = \mathrm{id} \\ 0 & \text{otherwise} \end{cases}$$

Only the identity permutation contributes, and $\operatorname{sign}(\mathrm{id}) = 1$, so:

$$\sum_{\sigma\in S_n} \operatorname{sign}(\sigma)\delta_{\sigma(1)1}\delta_{\sigma(2)2}\cdots\delta_{\sigma(n)n} = 1. \qquad \square$$

Example 8.5. $\det\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = \operatorname{sign}(12)a_{11}a_{22} + \operatorname{sign}(21)a_{21}a_{12} = a_{11}a_{22} - a_{21}a_{12}$.

$$\begin{aligned}\det\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} =& \operatorname{sign}(123)a_{11}a_{22}a_{33} + \operatorname{sign}(231)a_{21}a_{31}a_{13} + \operatorname{sign}(312)a_{31}a_{12}a_{23} \\ &+ \operatorname{sign}(213)a_{21}a_{12}a_{33} + \operatorname{sign}(132)a_{11}a_{32}a_{23} + \operatorname{sign}(321)a_{31}a_{22}a_{13} \\ =& a_{11}a_{22}a_{33} + a_{21}a_{31}a_{13} + a_{31}a_{12}a_{23} - a_{21}a_{12}a_{33} - a_{11}a_{32}a_{23} - a_{31}a_{22}a_{13}\end{aligned}$$

The Leibniz formula is of theoretical importance but it's not a good idea to compute the determinant using this formula for large matrices.

Corollary 8.6.

$$\det(P_{ij}) = -1$$
$$\det(D_i(\lambda)) = \lambda$$
$$\det(E_{ij}(\lambda)) = 1$$

For any $A \in M_{n\times n}(F)$ and elementary matrix E

$$\det(AE) = \det(A)\det(E)$$

Proof. Since P_{ij} is obtained by swaping column i and j of I, we have $\det(P_{ij}) = -1$.

Since $D_i(\lambda)$ is obtained by multiplying column j of I by λ, we have $\det(D_i(\lambda)) = \lambda$.

Since $E_{ij}(\lambda)$ is obtained by adding $\lambda\cdot$ column i to column j of I, we have $\det(E_{ij}(\lambda)) = 1$.

By Remark 6.23, multiplying elementary matrices to the right of a matrix corresponds to performing elementary column operations. Since each elementary column operation corresponds to multiplying the determinant of A by $\det(E)$, we have $\det(AE) = \det(A)\det(E)$. $\square$

Corollary 8.7. *Let $A \in M_{n\times n}(F)$ be singular. Then* $\det(A) = 0$.

Proof. Since $\operatorname{rank} A < n$, the columns of A are linearly dependent. We write $A = (v_1 \ldots v_n)$. By Lemma 2.8, there is a column v_k which is a linear combination of $v_1, \ldots, v_{k-1}$ i.e. $v_k = a_1v_1 + \cdots + a_{k-1}v_{k-1}$. Thus by Lemma 7.13, adding $-a_1v_1 - \cdots - a_{k-1}v_{k-1}$ to v_k we get $\det(A) = \det(v_1 \ldots v_{k-1}, 0, v_{k+1} \ldots v_n) = 0$. □

Proposition 8.8. *Let $A, B \in M_{n\times n}(F)$. Then $\det(AB) = \det(A)\det(B)$.*

Proof. In case B is not invertible, we have first that $\det(B) = 0$. Next, we know that AB is not invertible by Homework 4 (ST invertible $\Longrightarrow S, T$ are both invertible). Hence $\det(AB) = 0 = \det(A)\det(B)$.

We have seen in Corollary 8.6 that $\det(AE) = \det(A)\det(E)$ for any $A \in M_{n\times n}(F)$ and elementary matrix E. By Corollary 7.9 for any invertible matrix B, we have $B = E_1 \cdots E_m$ for some elementary matrices $E_1, \ldots, E_m$. Hence

$$\begin{aligned}\det(AB) &= \det(AE_1 \cdots E_{m-1}E_m) = \det(AE_1 \cdots E_{m-1})\det(E_m)\\ &= \cdots\\ &= \det(A)\det(E_1)\cdots\det(E_m)\\ &= \det(A)\det(E_1 \cdots E_m) = \det(A)\det(B).\end{aligned}$$

□

Corollary 8.9. *$A \in M_{n\times n}(F)$ is invertible if and only if $\det(A) \neq 0$. If A is invertible, then $\det(A)^{-1} = \frac{1}{\det(A)}$.*

Proof. If A is not invertible then by Corollary 8.7, $\det(A) = 0$.

If A is invertible Then $A^{-1}A = I$. By Proposition 8.8, $\det(A)\det(A^{-1}) = \det(AA^{-1}) = \det I = 1$. Thus $\det(A) \neq 0$ and $\det(A^{-1}) = \frac{1}{\det(A)}$. □

Remark 8.10. This Corollary shows that computing the determinant is a good way to tell whether a matrix is invertible or not.

Corollary 8.11. *If $B = P^{-1}AP$, then $\det(B) = \det(A)$. The determinant is independent of the choice of a basis. That is if $T : V \to V$ is linear then for any bases β and γ of V, we have $\det([T]_\beta^\beta) = \det([T]_\gamma^\gamma)$. Thus we define $\det(T) = \det([T]_\beta^\beta)$.*

Proposition 8.12. *Let $A \in M_{n\times n}(F)$. Then $\det(A^t) = \det(A)$.*

Proof. If A is not invertible, then by Corollary 6.17 A^t is not invertible. Thus $\det(A^t) = 0 = \det(A)$.

If A is invertible then by Corollary 7.9, $A = E_1 \cdots E_m$ for some elementary matrices $E_1, \ldots, E_m$. Note that for any elementary matrix E, E^t is also an elementary matrix and $\det(E) = \det(E^t)$. Then

$$\begin{aligned}\det(A) &= \det(E_1)\cdots\det(E_m) = \det(E_m^t)\cdots\det(E_1^t)\\ &= \det(E_m^t \cdots E_1^t) = \det((E_1 \cdots E_m)^t) = \det(A^t).\end{aligned}$$

□

Remark 8.13. Thus the properties in Definition 7.12 and Lemma 7.13 are also true for rows of A.

8.2 Laplace expansion

Proposition 8.14. *Let $B \in M_{(n-1)\times(n-1)}(F)$ and $u \in F^{n-1}$. Then*

$$\det\begin{pmatrix} 1 & u^t \\ 0 & B \end{pmatrix} = \det(B).$$

In particular

$$\det\begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \lambda_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} = \lambda_1 \dots \lambda_n.$$

Proof. By Lemma 7.13, we can perform elementary column operations to eliminate the u^t using the first column. For instance one first add $-u_1$ multiple of column 1 to column 2 to eliminate u_1 and then add $-u_2$ multiple of column 1 to column 3 to eliminate u_2 ... so on and so forth and finally add $-u_{n-1}$ multiple of column 1 to column n to eliminate u_{n-1}. We can express this process compactly as elementary column operation by block: one add $-u^t$ multiple of the first column $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ to the column $\begin{pmatrix} u^t \\ B \end{pmatrix}$ to get $\begin{pmatrix} 0 \\ B \end{pmatrix}$ which is equivalent to the matrix multiplication $\begin{pmatrix} 1 & u^t \\ 0 & B \end{pmatrix}\begin{pmatrix} 1 & -u^t \\ 0 & I \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & B \end{pmatrix}$.

Since the determinant is invariant under adding multiples of other columns to one column, we have $\det\begin{pmatrix} 1 & u^t \\ 0 & B \end{pmatrix} = \det\begin{pmatrix} 1 & 0 \\ 0 & B \end{pmatrix}$. We define $D(B) = \det\begin{pmatrix} 1 & 0 \\ 0 & B \end{pmatrix}$. Clearly, this function satisfies the properties in Definition 7.12 since det does. Therefore, by the uniqueness of the determinant function, it must be $D(B) = \det(B)$.

We apply the previous result inductively and get

$$\begin{aligned} \det\begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \lambda_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} &= \lambda_1 \det\begin{pmatrix} 1 & * & \cdots & * \\ 0 & \lambda_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \\ =& \lambda_1 \det\begin{pmatrix} \lambda_2 & \cdots & * \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix} = \cdots = \lambda_1 \dots \lambda_n. \end{aligned}$$

□

Remark 8.15. A good way of computing the determinant is to use elementary row and column operations to reduce the matrix to the upper triangular form and apply the previous proposition.

Remark 8.16. In the proof above, we eliminate the block u^t as if it is a number that can be eliminated by the 1 in the same row. This is an efficient way to perform elementary row/column operations called elementary row/column operations by blocks. A general form of it is as follows.

Swap two block rows

$$\begin{pmatrix} A_{21} & A_{22} \\ A_{11} & A_{11} \end{pmatrix} = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix} \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}.$$

Multiply a block row by a square matrix Λ from the left:

$$\begin{pmatrix} \Lambda A_{11} & \Lambda A_{12} \\ A_{21} & A_{22} \end{pmatrix} = \begin{pmatrix} \Lambda & 0 \\ 0 & I \end{pmatrix} \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}.$$

Add a block multiple of one block row to another

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} + MA_{11} & A_{22} + MA_{12} \end{pmatrix} = \begin{pmatrix} I & 0 \\ M & I \end{pmatrix} \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$$

where M is a matrix of appropriate size (not necessarily invertible). This generalizes the usual row addition $R_i \leftarrow R_i + cR_j$ to the block setting.

The third one is most useful in computing determinants since the determinant is unchanged under such operations. The first two will differ by the determinant of the corresponding block elementary matrix.

Similarly, one can also perform elementary column operations by block. In particular adding a block multiple of one block column to another column is as follows

$$\begin{pmatrix} A_{11} & A_{12} + A_{11}M \\ A_{21} & A_{22} + A_{21}M \end{pmatrix} = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} I & M \\ 0 & I \end{pmatrix}$$

where M is a matrix of appropriate size (not necessarily invertible).

We note that when doing the third elementary column operations by blocks, one can only multiply M to the right of a block column while when doing elementary row operations by blocks, one can only multiply M to the left of a block row.

Definition 8.17. The (i, j)**-minor** of A, denoted by A_{ij} is the determinant of the $(n-1) \times (n-1)$ matrix obtained by deleting the i-th row and the j-th column of A. The **cofactor matrix** $\mathrm{Cof}(A)$ is the matrix with entries $\mathrm{Cof}(A)_{ij} = (-1)^{i+j} A_{ij}$.

Example 8.18. If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ then $\mathrm{Cof}(A) = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}$.

Theorem 8.19 (Laplace's expansion formula)**.** *Let $A \in M_{n\times n}(F)$. Then for any $j = 1, \ldots, n$*

$$\det(A) = \sum_{i=1}^{n} (-1)^{i+j} a_{ij} A_{ij} = \sum_{i=1}^{n} (-1)^{i+j} a_{ji} A_{ji}.$$

Proof. Let $e_1, \ldots, e_n$ be the standard basis of F^n. By multilinearity,

$$\begin{aligned}\det(A) &= \det(v_1 \, \ldots \sum_{i=1}^{n} a_{ij} e_i \ldots \, v_n) \\ &= \sum_{i=1}^{n} a_{ij} \det(v_1 \, \ldots \, e_i \, \ldots \, v_n)\end{aligned}$$

where e_i is in the j-th column. We first do adjacent column swaps to make e_i appear in the first column and then do adjacent row swaps to make the entry 1 in e_i appear in the first row. Then we made $i-1+j-1 = i+j-2$ adjacent swaps and hence the determinant change by a factor $(-1)^{i+j-2} = (-1)^{i+j}$. We apply Proposition 8.14 to see that $\det(v_1 \, \ldots \, e_i \, \ldots \, v_n) = (-1)^{i+j} A_{ij}$

The second equality follows from applying the formula to A^t and use Proposition 8.12. □

Definition 8.20. For $A = (v_1 \, \ldots \, v_n) \in M_{n\times n}(F)$ define $A_i(b) = (v_1 \, \ldots \, v_{i-1} \, b \, v_{i+1} \, \ldots \, v_n)$ as the matrix obtained by replacing the i-th column of A with the vector $b \in F^n$.

Lemma 8.21.

$$\det(A_i(v_j)) = \det(A)\delta_{ij}$$

Proof. If $j \neq i$, then $A_i(v_j)$ has two equal columns: v_j appears in both the j-th and the i-th column and $\det(A_i(v_j)) = 0$

If $j = i$, then $A_i(v_i) = A$, so $\det(A_i(v_i)) = \det(A)$. □

Corollary 8.22. $A(\mathrm{Cof}(A))^t = (\mathrm{Cof}(A))^t A = \det(A) I$. *If* $\det(A) \neq 0$ *then* $A^{-1} = \frac{1}{\det(A)}(\mathrm{Cof}(A))^t$.

Proof. By Theorem 8.19, $((\mathrm{Cof}(A))^t A)_{ij} = \sum_{k=1}^{n} (-1)^{i+k} A_{ki} a_{kj} = \det(A_i(v_j)) = \det(A)\delta_{ij}$. □

Example 8.23. If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ then $\det A = ad - bc$. If $ad - bc \neq 0$, then A is invertible and $A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

9 Lecture 9

9.1 Cramer's rule

Theorem 9.1 (Cramer's rule)**.** *Let $A \in M_{n\times n}(F)$ be invertible and $b \in F^n$. Then the solution $x = (x_1, x_2, \ldots, x_n)^t$ to $Ax = b$ is given by:*

$$x_i = \frac{\det(A_i(b))}{\det(A)}, \quad \textit{for } i = 1, 2, \ldots, n.$$

Proof. Let $x = (x_1, x_2, \ldots, x_n)^t$ be the unique solution to $Ax = b$. Then, $Ax = x_1v_1 + x_2v_2 + \cdots + x_nv_n = b$. Therefore $\det(A_i(b)) = \sum_{j=1}^n x_j \det(A_i(v_j)) = x_i \det(A)$ and hence $x_i = \frac{\det(A_i(b))}{\det(A)}$. □

Example 9.2. Consider the equation

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1 \\ a_{21}x_1 + a_{22}x_2 = b_2 \end{cases}$$

If $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ is invertible, then the unique solution to this equaiton is

$$x_1 = \frac{\det\begin{pmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{pmatrix}}{\det\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}} = \frac{b_1a_{22} - b_2a_{12}}{a_{11}a_{22} - a_{21}a_{12}}, \quad x_2 = \frac{\det\begin{pmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{pmatrix}}{\det\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}} = \frac{a_{11}b_2 - a_{21}b_1}{a_{11}a_{22} - a_{21}a_{12}}.$$

Proposition 9.3. *Let M be an $n \times n$ matrix of the form*

$$M = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix},$$

where $A \in M_{p\times p}(F)$, $B \in M_{p\times(n-p)}(F)$, $C \in M_{(n-p)\times(n-p)}(F)$ and 0 is a matrix of all zeros. Show that

$$\det(M) = \det(A)\det(C).$$

Proof. If C is singular, then $\operatorname{rank} C < n - p$. Then the last $n - p$ rows of $\begin{pmatrix} A & B \\ 0 & C \end{pmatrix}$ are linearly dependent. Hence M is singular. So $\det(M) = 0 = \det(A)\det(C)$.

If C is invertible, we consider the function $D(A) = \det(C)^{-1} \det \begin{pmatrix} A & B \\ 0 & C \end{pmatrix}$. Then $D(A)$ satisfies (i), (ii) in Definition 7.12 since det does. Moreover, by applying Proposition 8.14 repeatedly, we get

$$D(I) = \det(C)^{-1} \det \begin{pmatrix} I & B \\ 0 & C \end{pmatrix} = \det(C)^{-1}\det(C) = 1.$$

Thus by Theorem 8.1, we have $D(A) = \det(A)$. □

Remark 9.4. One can similarly show that

$$\det \begin{pmatrix} A_1 & * & \cdots & * \\ 0 & A_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A_n \end{pmatrix} = \det(A_1) \dots \det(A_n).$$

where A_i are square matrices whose diagonal aligns with the diagonal of the whole matrix. This should be compared with Proposition 8.14.

9.2 Polynomials

This section is an introduction to polynomial theory. We only need the conclusions. For a proof, see [1, Chapter 9]. For a proof of the fundamental theorem of algebra see https://en.wikipedia.org/wiki/Fundamental_theorem_of_algebra.

Definition 9.5. Let F be a field. The set of all polynomials in one variable x with coefficients in F is denoted by $F[x]$. Note that there is no restriction on the degree here. In other words, $F[x] = \cup_{n=1}^{\infty} \mathcal{P}_n(F)$. A general element of $F[x]$ has the form

$$f(x) = a_n x^n + \cdots + a_1 x + a_0,$$

where $a_0, \dots, a_n \in F$, and $n \in \mathbb{N}$.

If $f(x) \in F[x]$ is nonzero, its **degree**, denoted $\deg(f)$, is the highest power of x with nonzero coefficient. We define $\deg(0) = -\infty$ by convention.

A polynomial $f(x) \in F[x]$ is called **monic** if its leading coefficient is 1.

Theorem 9.6. *Given polynomials $f(x), g(x) \in F[x]$ with $g(x) \neq 0$, there exist unique polynomials $q(x), r(x) \in F[x]$ such that:*

$$f(x) = q(x)g(x) + r(x), \qquad \textit{with } \deg(r) < \deg(g).$$

*This is called the **Euclidean division**.*

Example 9.7. $x^3 - 1 = (x^2 + x + 1)(x - 1)$.

Definition 9.8. Let $f(x), g(x) \in F[x]$. We say that $g(x)$ **divides** $f(x)$, written $g(x) \mid f(x)$, if there exists $q(x) \in F[x]$ such that $f(x) = q(x)g(x)$.

Definition 9.9. Let $f(x) \in F[x]$, and let $\alpha \in F$. We say that α is a **root** (or **zero**) of f if $f(\alpha) = 0$.

Proposition 9.10. *$\alpha \in F$ is a root of $f(x)$ if and only if $(x - \alpha) \mid f(x)$ in $F[x]$.*

Definition 9.11. Let $p(x) \in F[x]$ be a nonzero polynomial of degree n. We say that $p(x)$ **splits in** F if it can be written as a product of linear factors, that is,

$$p(x) = a(x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_n),$$

for some $a \neq 0$ and $\lambda_1, \ldots, \lambda_n \in F$. One can also write it as

$$p(x) = a(x - \lambda_1)^{k_1}(x - \lambda_2)^{k_2} \cdots (x - \lambda_m)^{k_m}$$

where $\lambda_1, \ldots, \lambda_m$ are *distinct* roots of p and k_i is the multiplicity of λ_i.

Theorem 9.12 (Fundamental theorem of algebra)**.** *Let $p(x) \in \mathbb{C}[x]$. Then $p(x)$ splits in $\mathbb{C}$.*

Example 9.13. Let $p(x) = ax^2 + bx + c$ where $a, b, c \in \mathbb{R}$. Then $p(x) = a(x - \lambda_1)(x - \lambda_2)$ in $\mathbb{C}[x]$ where

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad \lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

here you can take the square root of any real number since if $b^2 - 4ac \geq 0$ then it's the usual square root, if $b^2 - 4ac < 0$, then $\sqrt{b^2 - 4ac} = \sqrt{4ac - b^2}\sqrt{-1} = \sqrt{4ac - b^2}i$.

For example $x^2 + 1$ does not splits in $\mathbb{R}$ since it has no roots in $\mathbb{R}$. However, $x^2 + 1 = (x - i)(x + i)$ splits in $\mathbb{C}$ and the roots are $\pm i$.

Also $x^3 - 1 = (x - 1)(x^2 + x + 1)$ does not splits in $\mathbb{R}$ since $x^2 + x + 1$ has no roots in $\mathbb{R}$. However, $x^3 - 1 = (x - 1)(x - \lambda_1)(x - \lambda_2)$ in $\mathbb{C}$ where $\lambda_1 = \frac{-1+\sqrt{3}i}{2}$ and $\lambda_2 = \frac{-1-\sqrt{3}i}{2}$. Note that $\lambda_1^2 = (\frac{-1+\sqrt{3}i}{2})^2 = \frac{1-2\sqrt{3}i-3}{4} = \frac{-1-\sqrt{3}i}{2} = \lambda_2$. So the above formula is also written as $x^3 - 1 = (x - 1)(x - \omega)(x - \omega^2)$ where $\omega = \lambda_1$.

Definition 9.14. Let $z = x + yi$ be a complex number $x, y \in \mathbb{R}$. The **complex conjugate** is the complex number $\bar{z} = x - yi$. x is called the **real part** of z and y is called the **imaginary part** of z. We write $x = \operatorname{Re} z$ and $y = \operatorname{Im} z$. (Note that the imaginary part Im should not be confused with the image im.)

We have the following properties for complex conjugates: $\overline{zw} = \bar{z}\bar{w}$ and $\overline{z + w} = \bar{z} + \bar{w}$.

Let $A = (a_{ij}) \in M_{m\times n}(\mathbb{C})$. We define the **complex conjugate matrix** $\bar{A} = (\overline{a_{ij}}) \in M_{m\times n}(\mathbb{C})$ and $\operatorname{Re} A = (\operatorname{Re} a_{ij}) \in M_{m\times n}(\mathbb{R})$ and $\operatorname{Im} A = (\operatorname{Im} a_{ij}) \in M_{m\times n}(\mathbb{R})$. By the properties above, it's not hard to show that $\overline{AB} = \bar{A}\bar{B}$ and $\overline{A + C} = \bar{A} + \bar{C}$ for $A, C \in M_{m\times n}(\mathbb{C})$, $B \in M_{n\times p}(\mathbb{C})$.

Example 9.15. If $A = \begin{pmatrix} 1-i & 1+i \\ 1 & 1 \end{pmatrix}$, then $\bar{A} = \begin{pmatrix} 1+i & 1-i \\ 1 & 1 \end{pmatrix}$, $\operatorname{Re} A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and $\operatorname{Im} A = \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix}$.

Lemma 9.16. *Suppose $p(x) \in \mathbb{R}[x]$ and we regard it as a polynomial in $\mathbb{C}[x]$. If $z \in \mathbb{C}$ is a root of p i.e. $p(z) = 0$, then $\bar{z}$ is also a root of p i.e. $p(\bar{z}) = 0$.*

Proof. Since $p(x) = a_n x^n + \cdots + a_0 \in \mathbb{R}[x]$, $\overline{a_i} = a_i$

$$p(\bar{z}) = a_n(\bar{z})^n + \cdots + a_1\bar{z} + a_0 = \overline{a_n z^n + \cdots + a_1 z + a_0} = \bar{0} = 0.$$

□

Proposition 9.17. *For any $p(x) \in F[x]$, there is a field extension E of F i.e. a field E containing F, such that $p(x)$ splits in E. The smallest such field E is called the **splitting field** of $p(x)$.*

9.3 Eigenvalue and eigenvectors

Let $T : V \to V$ be a linear transformation. We would like to find a basis such that $[T]_\beta^\beta$ is the simplest. The simplest possible matrix is the diagonal matrix $\mathrm{diag}(\lambda_1, \dots, \lambda_n)$. Although this is not always achievable, if we do have such a basis $\beta = (v_1, \dots, v_n)$, then the v_i satisfies the special property that $T(v_i) = \lambda_i v_i$ for $1 \le i \le n$.

Definition 9.18. We say that $\lambda \in F$ is an **eigenvalue** of T and $v \in V$ is an **eigenvector** corresponding to the eigenvalue λ if $v \neq 0$ and $Tv = \lambda v$.

A linear transformation $T : V \to V$ is **diagonalizable** over F if there is a basis β of V such that $[T]_\beta^\beta = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$.

The set of all eigenvalues of T is called the **spectrum** of T denoted by $\sigma(T)$.

In matrix language, $\lambda \in F$ is an **eigenvalue** of A and $v \in F^n$ is an **eigenvector** corresponding to the eigenvalue λ if $v \neq 0$ and $Av = \lambda v$.

$A \in M_{n\times n}(F)$ is **diagonalizable** over F if there is $P \in M_{n\times n}(F)$ invertible such that $P^{-1}AP = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$.

Remark 9.19. The word spectrum comes from quantum mechanics since the eigenvalues of the Hamiltonian operator are the energy of light emitted from a black body. So the study of eigenvalues and eigenvectors of a linear operator is also called spectral theory.

Proposition 9.20. *Let $A \in M_{n\times n}(F)$. Then λ is an eigenvalue and v is an eigenvector with eigenvalue λ if and only if $\det(\lambda I - A) = 0$ and $0 \neq v \in \ker(\lambda I - A)$.*

Proof. Since $Av = \lambda v$, we have $(\lambda I - A)v = 0$ i.e. $v \in \ker(\lambda I - A)$. Since $v \neq 0$, $\lambda I - A$ is not invertible. Hence $\det(\lambda I - A) = 0$. □

Definition 9.21. Let $A \in M_{n\times n}(F)$. The polynomial $p_A(x) = \det(xI - A) \in F[x]$ is the **characteristic polynomial** of A. λ is an eigenvalue of A if and only if λ is a root of $p_A(x)$.

The subspace $E(\lambda, A) = \ker(\lambda I - A)$ is called the **eigenspace** with eigenvalue λ.

The dimension $\dim E(\lambda, A) \ge 1$ is called the **geometric multiplicity** of λ.

The maximal integer k such that $(x-\lambda)^k \mid p_A(x)$ is called the **algebraic multiplicity** of λ.

The following four examples are all possible situations that one may encounter when finding the eigenvalues and eigenvectors of a matrix.

Example 9.22. This is an example where the matrix is diagonalizable.

$A = \begin{pmatrix} 7 & -10 \\ 5 & -8 \end{pmatrix} \in M_{n\times n}(\mathbb{R})$

$p_A(x) = \det(xI - A) = \begin{pmatrix} x-7 & 10 \\ -5 & x+8 \end{pmatrix} = (x-7)(x+8) + 50 = x^2 + x - 6 = (x-2)(x+3)$.

The eigenvalues of A are $\lambda_1 = 2$, $\lambda_2 = -3$ and both of algebraic multiplicity 1.

Solving $0 = (2I - A)v = \begin{pmatrix} -5 & 10 \\ -5 & 10 \end{pmatrix} v$ we have $v = c\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ for $c \in \mathbb{R}$.

Solving $0 = ((-3)I - A)v = \begin{pmatrix} -10 & 10 \\ -5 & 5 \end{pmatrix} v$ we have $v = c \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ for $c \in \mathbb{R}$.

Thus $\lambda_1 = 2$ is an eigenvalue with eigenvector $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\lambda_2 = -3$ is another eigenvalue with eigenvector $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$. The geometric multiplicities of λ_1, λ_2 are both 1. If we write $P = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ then $AP = P \begin{pmatrix} 2 & \\ & -3 \end{pmatrix}$ i.e. $P^{-1}AP = \operatorname{diag}(2, -3)$.

Example 9.23. This is an example where A is not diagonalizable since the field is too small to contain all eigenvalues of A.

$A = \begin{pmatrix} -1 & 2 \\ -1 & 1 \end{pmatrix} \in M_{2\times 2}(\mathbb{R})$

$p_A(x) = \det(xI - A) = \begin{pmatrix} x+1 & -2 \\ 1 & x-1 \end{pmatrix} = (x+1)(x-1) + 2 = x^2 + 1.$

There is no root of $p_A(x)$ in $\mathbb{R}$ which means A has no eigenvalues and eigenvectors over $\mathbb{R}$. In particular, A is not diagonalizable over $\mathbb{R}$.

Example 9.24. This is an example where we enlarge the field F to contain all eigenvalues of A and A becomes diagonalizable in the enlarged field.

$A = \begin{pmatrix} -1 & 2 \\ -1 & 1 \end{pmatrix} \in M_{2\times 2}(\mathbb{C})$

$p_A(x) = \det(xI - A) = \begin{pmatrix} x+1 & -2 \\ 1 & x-1 \end{pmatrix} = (x+1)(x-1) + 2 = x^2 + 1 = (x-i)(x+i).$

The eigenvalues of A in $\mathbb{C}$ are $\lambda_1 = i$, $\lambda_2 = -i$ both of algebraic multiplicity 1.

Solving $0 = (iI - A)v = \begin{pmatrix} i+1 & -2 \\ 1 & i-1 \end{pmatrix} v$ we have $v = c \begin{pmatrix} 1-i \\ 1 \end{pmatrix}$ for $c \in \mathbb{C}$.

Solving $0 = ((-i)I - A)v = \begin{pmatrix} -i+1 & -2 \\ 1 & -i-1 \end{pmatrix} v$ we have $v = c \begin{pmatrix} 1+i \\ 1 \end{pmatrix}$ for $c \in \mathbb{C}$.

Thus $\lambda_1 = i$ is an eigenvalue with eigenvector $\begin{pmatrix} 1-i \\ 1 \end{pmatrix}$ and $\lambda_2 = -i$ is another eigenvalue with eigenvector $\begin{pmatrix} 1+i \\ 1 \end{pmatrix}$. The geometric multiplicities of $\lambda_1 = i, \lambda_2 = -i$ are both 1. If we write $P = \begin{pmatrix} 1-i & 1+i \\ 1 & 1 \end{pmatrix}$ then $AP = P \begin{pmatrix} i & \\ & -i \end{pmatrix}$ i.e. $P^{-1}AP = \operatorname{diag}(i, -i)$.

Remark 9.25. Suppose we regard $A \in M_{n\times n}(\mathbb{R})$ as a matrix in $M_{n\times n}(\mathbb{C})$. If $\lambda \in \mathbb{C}$ is a complex eigenvalue with nonzero imaginary part, then $\bar{\lambda}$ is also an eigenvalue of A by Lemma 9.16. In fact if v is an eigenvector of A with eigenvalue λ, then $\bar{v}$ is an eigenvector of A with eigenvalue $\bar{\lambda}$. This is because if $Av = \lambda v$ then $A\bar{v} = \bar{A}\bar{v} = \overline{Av} = \overline{\lambda v} = \bar{\lambda}\bar{v}$. So in practice, we only need to compute half of the eigenvalues and eigenvectors.

Remark 9.26. In Example 9.23, A is not diagonalizable over $\mathbb{R}$. In Example 9.24, we enlarge $\mathbb{R}$ to $\mathbb{C}$ and A becomes diagonalizable. The price to pay is that the matrix P is a complex matrix, not a real matrix. In general, the issue of the field F not being large enough to contain all eigenvalues can be resolved by Proposition 9.17.

Example 9.27. This is an example where A is not diagonalizable since there are not enough eigenvectors.

$A = \begin{pmatrix} -1 & -4 \\ 1 & 3 \end{pmatrix}$

$p_A(x) = \det(xI - A) = \begin{pmatrix} x+1 & 4 \\ -1 & x-3 \end{pmatrix} = (x+1)(x-3)+4 = x^2-2x+1 = (x-1)^2.$

$\lambda = 1$ is the only eigenvalue of A. It has algebraic multiplicity 2.

Solving $0 = (I - A)v = \begin{pmatrix} 2 & 4 \\ -1 & -2 \end{pmatrix} v$ we have $v = c \begin{pmatrix} -2 \\ 1 \end{pmatrix}$ for $c \in \mathbb{R}$.

Thus $\lambda = 1$ is an eigenvalue with eigenvector $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$. Note that there is only 1 linearly independent eigenvector and the geometric multiplicity is 1.

Remark 9.28. In this example, $p_A(x)$ already splits in $\mathbb{R}[x]$, but there is not enough eigenvectors to diagonalize A i.e. geometric multiplicity $<$ algebraic multiplicity. The indiagonalizability is essential and cannot be overcome by choosing a larger field. To see that A is not diagonalizable from another perspective, suppose A is diagonalizable. Then $P^{-1}AP = \operatorname{diag}(\lambda_1, \lambda_2)$. In this case, $p_A(x) = (x-1)^2$ and we must have $\lambda_1 = \lambda_2 = 1$. That means $P^{-1}AP = I$ and $A = PP^{-1} = I$ which is not possible.

References

[1] Dummit, David Steven, and Richard M. Foote. Abstract algebra. Vol. 3. Hoboken: Wiley, 2004.

10 Lecture 10

10.1 Characteristic polynomial

Lemma 10.1. *Let $T : V \to V$ be a linear transformation, $v \in V$ an eigenvector of T with eigenvalue λ i.e. $Tv = \lambda v$. Let β, γ be two bases of V*

$$\begin{tikzcd}
& A[v]_\beta = \lambda[v]_\beta \\
Tv = \lambda v \arrow[ur, "A={[T]}_\beta^\beta"] \arrow[dr, "B={[T]}_\gamma^\gamma"'] & \\
& B[v]_\gamma = \lambda[v]_\gamma \arrow[uu, "\substack{B = P^{-1}AP \\ {[v]}_\beta = P{[v]}_\gamma}"']
\end{tikzcd}$$

We have $p_B(x) = p_A(x)$. We can define $p_T(x) = p_A(x)$ where $A = [T]_\beta^\beta$.

Proof. If $Tv = \lambda v$, then $[T]_\beta^\beta [v]_\beta = [T(v)]_\beta = \lambda[v]_\beta$.

$p_{PAP^{-1}}(x) = \det(xI - P^{-1}AP) = \det(P^{-1}(xI - A)P) = \det(xI - A) = p_A(x)$. □

Proposition 10.2. *The characteristic polynomial is of the form*

$$p_A(x) = x^n - c_1(A)x^{n-1} + \cdots + (-1)^n c_n(A)$$

where $c_k(A)$ is some function of A for example, $c_1(A) = \operatorname{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}$ and $c_n(A) = \det(A)$.

For any $P \in M_{n\times n}(F)$ invertible, we have $p_A(x) = p_{PAP^{-1}}(x)$. In particular $c_k(P^{-1}AP) = c_k(A)$.

If $p_A(x) = (x - \lambda_1)\cdots(x - \lambda_n)$ splits in F, then $c_k(A) = \sum_{i_1<\cdots<i_k} \lambda_{i_1}\cdots\lambda_{i_k}$.

Proof. Note that $(xI - A)_{ij} = x\delta_{ij} - a_{ij}$. By the Leibniz formula of the determinant (Theorem 8.1)

$$p_A(x) = \det(xI - A) = \sum_{\sigma\in S_n} \operatorname{sign}(\sigma)(x\delta_{\sigma(1)1} - a_{\sigma(1)1})\cdots(x\delta_{\sigma(n)n} - a_{\sigma(n)n}).$$

It is clear that $(-1)^n c_n(A) = p_A(0) = \det(-A) = (-1)^n \det(A)$. Therefore, $c_n(A) = \det(A)$.

To find the coefficient of x^n, we observe that the right hand side is the sum of products of n factors of degree ≤ 1. The only term that can contribute x^n is the one where there is an x in each factor. This is only possible when $\sigma(i) = i$ for all $1 \leq i \leq n$, i.e. $\sigma = \mathrm{id}$. Note that $\operatorname{sign}(\mathrm{id}) = 1$ and so this term is $(x - a_{11})\cdots(x - a_{nn})$. Thus we see that the coefficient of x^n is 1.

To find the coefficient of x^{n-1}, we observe that the term that contribute x^{n-1} should have an x in at least $n-1$ factors. This means $\sigma(i) = i$ for all but one $1 \leq i \leq n$. However, since $\sigma : \{1,\ldots,n\} \to \{1,\ldots,n\}$ is bijective, we must have $\sigma(i) = i$ for all

$1 \le i \le n$. The term that contributes x^{n-1} is $(x-a_{11})\cdots(x-a_{nn})$ again. The coefficient of x^{n-1} is $-a_{11} - \cdots - a_{nn} = -\operatorname{tr}(A)$.

Suppose $p_A(x) = (x-\lambda_1)\cdots(x-\lambda_n)$ We now expand the product. To obtain the coefficient of x^{n-k}, we must choose the term x from exactly $(n-k)$ of the factors, and choose the term $-\lambda_i$ from the remaining k factors. Any such choice contributes $(-1)^k\lambda_{i_1}\cdots\lambda_{i_k}$. Since each k-tuple of distinct indices $\{i_1,\dots,i_k\} \subset \{1,\dots,n\}$ occurs exactly once in the expansion, the coefficient of x^{n-k} is

$$(-1)^k \sum_{i_1<\cdots<i_k} \lambda_{i_1}\cdots\lambda_{i_k}.$$

Comparing coefficients with $x^n - c_1(A)x^{n-1} + \cdots + (-1)^n c_n(A)$ we get

$$c_k(A) = \sum_{i_1<\cdots<i_k} \lambda_{i_1}\cdots\lambda_{i_k}.$$

. □

Example 10.3. $A = \begin{pmatrix} 7 & -10 \\ 5 & -8 \end{pmatrix}$, $\operatorname{tr}(A) = 7-8 = -1$, $\det(A) = 7(-8) - 5(-10) = -6$. Thus $p_A(x) = x^2 + x - 6$. On the other hand, $P^{-1}AP = \operatorname{diag}(-3,2)$ and $\operatorname{tr}(A) = -3+2$, $\det(A) = (-3)(2) = -6$.

Remark 10.4. There is a formula for $c_k(A)$ in terms of the entries of A which is the sum of principal k-minors of A. For a definition of principal k-minors and a proof of the formula, see here.

Corollary 10.5. *$A \in M_{n\times n}(F)$ has at most n eigenvalues counting multiplicity. In other words, $(\lambda I - A)$ is invertible for all but at most n values of $\lambda \in F$.*

Proposition 10.6. *We have the following properties of the trace map.*
(i) $\operatorname{tr} : M_{n\times n}(F) \to F$ *is a linear function.*
(ii) Let $A, B \in M_{n\times n}(F)$. Then we have $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

Remark 10.7. (ii) is stronger than $\operatorname{tr}(P^{-1}AP) = \operatorname{tr}(A)$.

Remark 10.8. If $A, B, C \in M_{n\times n}(F)$, then by (ii) we have $\operatorname{tr}(ABC) = \operatorname{tr}(CAB) = \operatorname{tr}(BCA)$. However, in general $\operatorname{tr}(ABC) \neq \operatorname{tr}(BAC)$.

Proof. (i) This is clear from definition.

(ii) Let $A = (a_{ij})$ and $B = (b_{ij})$. By the definition of matrix multiplication,

$$(AB)_{ii} = \sum_{k=1}^{n} a_{ik}b_{ki}.$$

Therefore,

$$\operatorname{tr}(AB) = \sum_{i=1}^{n} (AB)_{ii} = \sum_{i=1}^{n}\sum_{k=1}^{n} a_{ik}b_{ki}.$$

We may interchange the order of summation:

$$\mathrm{tr}(AB) = \sum_{k=1}^{n}\sum_{i=1}^{n} a_{ik}b_{ki} = \sum_{k=1}^{n}\sum_{i=1}^{n} b_{ki}a_{ik}.$$

But this is exactly:

$$\mathrm{tr}(BA) = \sum_{k=1}^{n}(BA)_{kk} = \sum_{k=1}^{n}\sum_{i=1}^{n} b_{ki}a_{ik}.$$

Hence, $\mathrm{tr}(AB) = \mathrm{tr}(BA)$. □

10.2 Diagonalizability

Theorem 10.9. *Let $\lambda_1, \ldots, \lambda_m$ be distinct eigenvalues of $A \in M_{n\times n}(F)$. Let v_i be an eigenvector corresponding to λ_i for $1 \le i \le m$. Then $v_1, \ldots, v_m$ are linearly independent.*

In particular $E(\lambda_1, A) + \cdots + E(\lambda_m, A) = E(\lambda_1, A) \oplus \cdots \oplus E(\lambda_m, A)$ and if β_i is a basis of $E(\lambda_i, A)$, then $\beta = \beta_1 \cup \cdots \cup \beta_m$ is a basis of $E(\lambda_1, A) \oplus \cdots \oplus E(\lambda_m, A)$.

Proof. We prove by induction on m.

If $m = 1$, then $v_1 \neq 0$. Hence v_1 is linearly independent.

Suppose the conclusion is true for $m-1$, we now prove it for m. Let $a_1, \ldots, a_m \in F$ be such that

$$a_1v_1 + \cdots + a_mv_m = 0. \tag{2}$$

Applying T on both sides and using they are eigenvalues, we get

$$a_1\lambda_1v_1 + \cdots + a_m\lambda_mv_m = 0. \tag{3}$$

Multiplying λ_m on both sides of (2) we have

$$a_1\lambda_mv_1 + \cdots + a_m\lambda_mv_m = 0. \tag{4}$$

Subtracting (4) from (3), we get

$$a_1(\lambda_1 - \lambda_m)v_1 + \cdots + a_{m-1}(\lambda_{m-1} - \lambda_m)v_{m-1} = 0.$$

By induction hypothesis, $v_1, \ldots, v_{m-1}$ are linearly independent. Thus $a_i(\lambda_i - \lambda_m) = 0$ for $1 \le i \le m-1$. Since λ_i are distinct, we have $a_i = 0$ for $1 \le i \le m-1$. Plugging back into (2), we get $a_mv_m = 0$. Since $v_m \neq 0$, we have $a_m = 0$.

We now show that $E(\lambda_1, A)+\cdots+E(\lambda_m, A) = E(\lambda_1, A)\oplus\cdots\oplus E(\lambda_m, A)$. If this is not true, then by Proposition 3.5, there are $w_i \in E(\lambda_i, A)$ for $1 \le i \le m$ with some $w_i \neq 0$ such that $0 = w_1+\cdots+w_m$. Let $w_1, \ldots, w_r$ be all nonzero w_i's. Then $w_1+\cdots+w_r = 0$. This contradicts the fact that $w_1, \ldots, w_r$ are linearly independent.

Let $\beta_i = (v_1^{(i)}, \cdots, v_{k_i}^{(i)})$ be a basis of $E(\lambda_i, A)$. Then we would like to show that $\beta = (v_1^{(1)}, \cdots, v_{k_1}^{(1)}, \ldots, v_1^{(m)}, \cdots, v_{k_m}^{(m)})$ are linearly independent. Let $a_l^{(i)} \in F$ be such that

$$\sum_{i=1}^{m}\sum_{l=1}^{k_i} a_l^{(i)}v_l^{(i)} = 0.$$

Then with $w_i = \sum_{l=1}^{k_i} a_l^{(i)} v_l^{(i)}$, we have $w_1 + \cdots + w_m = 0$. Since $E(\lambda_1, A) \oplus \cdots \oplus E(\lambda_m, A)$ is a direct sum, $w_i = \sum_{l=1}^{k_i} a_l^{(i)} v_l^{(i)} = 0$. Since β_i is a basis, we have $a_l^{(i)} = 0$ for any $1 \leq i \leq m$, $1 \leq l \leq k_i$. □

Proposition 10.10. *Let $A \in M_{n\times n}(F)$ and $\lambda \in F$ be an eigenvalue of A. Then we have the geometric multiplicity of $\lambda \leq$ the algebraic multiplicity of λ.*

Proof. Since λ is an eigenvalue of A, we have the geometric multiplicity $m = \dim E(\lambda, A) \geq 1$. Let $v_1, \ldots, v_m \in F^n$ be a basis of $E(\lambda, A)$. We extend $v_1, \ldots, v_m$ to a basis $\beta = (v_1, \ldots, v_n)$ of $V = F^n$. Let P be the matrix whose columns are $v_1, \ldots, v_n$ since $v_i \in F^n$ are column vectors. (Or one can say P is the change of basis matrix from the standard basis to β.) We would like to find $P^{-1}AP$. Since $Av_i = \lambda v_i$ for $i = 1, \ldots, m$ and Av_j is unknown for $m+1 \leq j \leq n$, we have

$$P^{-1}AP = \begin{pmatrix} \lambda I_m & B \\ 0 & C \end{pmatrix}.$$

Then by Proposition 9.3

$$\begin{aligned} p_A(x) =& p_{P^{-1}AP}(x) = \det \begin{pmatrix} xI_m - \lambda I_m & -B \\ 0 & xI_{n-m} - C \end{pmatrix} \\ =& \det((x-\lambda)I_m) \det(xI_{n-m} - C) = (x-\lambda)^m p_C(x). \end{aligned}$$

In particular, $(x-\lambda)^m \mid p_A(x)$. By definition of algebraic multiplicity, $m \leq$ algebraic multiplicity of λ. □

Theorem 10.11. *Let $A \in M_{n\times n}(F)$. Then the following are equivalent.*

(i) A is diagonalizable over F

(ii) p_A splits over F and geometric multiplicity = algebraic multiplicity for each eigenvalue.

(iii) $n = \dim E(\lambda_1, A) + \cdots + \dim E(\lambda_m, A)$

(iv) $F^n = E(\lambda_1, A) \oplus \cdots \oplus E(\lambda_m, A)$

Proof. (i) $\Longrightarrow$ (ii) If A is diagonalizable, then $P^{-1}AP = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$. Then $p_A(x) = p_{P^{-1}AP}(x) = \det(xI - \mathrm{diag}(\lambda_1, \ldots, \lambda_n)) = (x-\lambda_1)\cdots(x-\lambda_n)$. For each λ_i, the algebraic multiplicity of λ_i is the number of times k_i that λ_i appear in $\mathrm{diag}(\lambda_1, \ldots, \lambda_n)$. The corresponding columns of P are linearly independent eigenvectors of eigenvalue λ_i. Thus the algebraic multiplicity k_i is equal to $\dim E(\lambda_i, A)$ which is the geometric multiplicity.

(ii) $\Longrightarrow$ (iii) Since $p_A(x)$ splits in F, we have

$$p_A(x) = (x-\lambda_1)^{k_1}(x-\lambda_2)^{k_2}\cdots(x-\lambda_m)^{k_m}$$

where λ_i are distinct roots of $p_A(x)$. Since geometric multiplicity = algebraic multiplicity, we have $\dim E(\lambda_i, A) = k_i$. Since $\deg p_A = n$, we have $n = k_1 + \cdots + k_m = \dim E(\lambda_1, A) + \cdots + \dim E(\lambda_m, A)$.

(iii) $\Longrightarrow$ (iv) By Theorem 10.9, (iii), Proposition 3.5 and Corollary 2.17, we have $F^n = E(\lambda_1, A) \oplus \cdots \oplus E(\lambda_m, A)$.

(iv) $\Longrightarrow$ (i) Let β_i be a basis of $E(\lambda_i, A)$. By Theorem 10.9, $\beta = \beta_1 \cup \cdots \cup \beta_m$ is a basis of $E(\lambda_1, A) \oplus \cdots \oplus E(\lambda_m, A) = F^n$. Thus we have a basis consisting of eigenvectors. $\square$

Corollary 10.12. *Let $A \in M_{n\times n}(F)$. If A has n distinct eigenvalues in F, then A is diagonalizable.*

Proof. For any eigenvalue λ_i, we have $\dim E(\lambda_i, A) \geq 1$. Then we have

$$\begin{aligned} n &\geq \dim(E(\lambda_1, A) + \cdots + E(\lambda_n, A)) \\ &= \dim(E(\lambda_1, A) \oplus \cdots \oplus E(\lambda_n, A)) \\ &= \dim E(\lambda_1, A) + \cdots + \dim E(\lambda_n, A) \geq n. \end{aligned}$$

Thus equality holds and we have A is diagonalizable. $\square$

10.3 Cayley-Hamilton theorem

Definition 10.13. Let $T : V \to V$ be a linear transformation on a vector space over F. Let $p(x) = a_n x^n + \cdots + a_0 \in F[x]$. Then we define

$$f(T) = a_n T^n + a_{n-1}T^{n-1} + \cdots + a_1 T + a_0 \mathrm{id}.$$

Note that $f(T) : V \to V$ is also a linear transformation. Similarly, for $A \in M_{n\times n}(F)$, we define $f(A) \in M_{n\times n}(F)$ as

$$f(A) = a_m A^m + a_{m-1}A^{m-1} + \cdots + a_1 A + a_0 I.$$

Proposition 10.14. *Let $A \in M_{n\times n}(F)$ be fixed. The map $F[x] \to M_{n\times n}(F)$ $f(x) \mapsto f(A)$ preserves polynomial addition and multiplication. That is for $f(x), g(x)$ in $F[x]$ we have $(f+g)(A) = f(A) + g(A)$ and $(fg)(A) = f(A)g(A)$.*

Proof. Suppose $f(x) = \sum_{j=0}^m a_j x^j$ and $g(x) = \sum_{k=0}^m b_k x^k$. Then

$$(f+g)(x) = \sum_{j=0}^m (a_j + b_j)x^j \quad (fg)(x) = \sum_{j=0}^m \sum_{k=0}^m a_j b_k x^{j+k}.$$

Thus

$$\begin{aligned} (f+g)(A) &= \sum_{j=0}^m (a_j + b_j)A^j = \sum_{j=0}^m a_j A^j + \sum_{j=0}^m b_j A^j = f(A) + g(A) \\ (fg)(A) &= \sum_{j=0}^m \sum_{k=0}^n a_j b_k A^{j+k} = \left(\sum_{j=0}^m a_j A^j\right)\left(\sum_{k=0}^n b_k A^k\right) = f(A)g(A). \end{aligned}$$

$\square$

Lemma 10.15. *If $B = P^{-1}AP$, then $f(B) = P^{-1}f(A)P$.*

Proof. $B^k = (P^{-1}AP)\cdots(P^{-1}AP) = P^{-1}A^kP$

$f(B)v = a_nB^n + \cdots + a_0I = a_nP^{-1}A^nP + \cdots + a_0I = P^{-1}(a_nA^n + \cdots + a_0I)P = P^{-1}f(A)P$. $\square$

11 Lecture 11

11.1 Cayley-Hamilton theorem

Proposition 11.1. *Let $A \in M_{n\times n}(F)$ and v be an eigenvector with eigenvalue λ.*

(i) For any $f(x) \in F[x]$, $f(A)(v) = f(\lambda)v$ thus $f(\lambda)$ is an eigenvalue of $f(A)$ and v is an eigenvector with eigenvalue $f(\lambda)$. In particular, $f(\sigma(A)) \subset \sigma(f(A))$.

(ii) If p_A splits in F, then $\sigma(f(A)) = f(\sigma(A))$.

Proof. (i) $A^k(v) = A^{k-1}(A(v)) = \lambda A^{k-1}(v) = \cdots = \lambda^k v$. Suppose $f(x) = a_n x^n + \cdots + a_0$. Then $f(A)v = (a_n A^n + \cdots + a_0 I)v = (a_n \lambda^n + \cdots + a_0)v = f(\lambda)v$.

(ii) Let $\mu \in \sigma(f(A))$, i.e. $f(A)v = \mu v$ for $0 \neq v \in F^n$. Consider the polynomial $f(x) - \mu$. Let E be a field extension of F such that $f(x) - \mu = (x - \lambda_1)\cdots(x - \lambda_m)$. Then $0 = (f(A) - \mu I)v = (A - \lambda_1 I)\cdots(A - \lambda_m I)v$. The matrix $(A - \lambda_1 I)\cdots(A - \lambda_m I)$ is singular since it maps a nonzero vector v to 0. So there must be $1 \leq i \leq m$ such that $A - \lambda_i I$ is singular. Thus λ_i is an eigenvalue of A. However, since $p_A(x)$ splits in F, $\lambda_i \in F$ and we have $f(\lambda_i) - \mu = 0$ that is $f(\lambda_i) = \mu$. □

Corollary 11.2. *Let $A \in M_{n\times n}(F)$ and $f(x) \in F[x]$. If $f(A) = 0$, then for any eigenvalue λ of A, we have $f(\lambda) = 0$.*

Example 11.3. If $A^2 = I$, then the eigenvalue of A is either 1 or -1.

If $A^2 = A$ then the eigenvalue of A is either 1 or 0.

If $A^m = 0$ then the eigenvalue of A is 0.

Example 11.4. If p_A does not splits in F, then there are $A \in M_{n\times n}(F)$ and $f \in F[x]$ such that $f(\sigma(A)) \subsetneq \sigma(f(A))$. Let $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \in M_{2\times 2}(\mathbb{R})$ and $f(x) = x^2$. Then $f(A) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$. We have $p_A(x) = x^2 + 1$ and $\sigma(A) = \emptyset$ and $\sigma(f(A)) = \{-1\}$. Hence $f(\sigma(A)) \subsetneq \sigma(f(A)) = \{-1\}$.

Theorem 11.5 (Cayley-Hamilton theorem)**.** *Let $A \in M_{n\times n}(F)$. Then $p_A(A) = 0$.*

Remark 11.6. Wrong but not very wrong proof: $p_A(A) = \det(AI - A) = \det 0 = 0$. This is wrong because plugging a matrix in $\det(xI - A)$ does not make sense and also $p_A(A)$ should be a matrix while in the above proof $\det 0$ is a scalar. However, with some efforts, one can make this argument to work. See Chapter 6, Theorem 5 of Linear algebra and its applications by Lax.

Proof when A is diagonalizable. Since A is diagonalizable, $P^{-1}AP = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$. Then $p_A(A) = P p_A(\mathrm{diag}(\lambda_1, \ldots, \lambda_n))P^{-1} = P\,\mathrm{diag}(p_A(\lambda_1), \ldots, p_A(\lambda_n))P^{-1} = 0$ since λ_i are roots of p_A. □

To deal with the case where A is not diagonalizable, we need the following theorem.

Theorem 11.7. *Let $A \in M_{n\times n}(F)$, p_A splits as in $F[x]$. Then there exists $P \in M_{n\times n}(F)$ invertible, such that*

$$P^{-1}AP = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \lambda_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A (with multiplicity).

Proof. We prove the statement by induction on n.

The base case $n = 1$ is trivial.

Assume the statement is true for all matrices of size $(n-1)\times(n-1)$ whose characteristic polynomial splits in F. Since p_A splits in F, A has at least one eigenvalue $\lambda_1 \in F$. Choose a nonzero eigenvector $v_1 \in F^n$ with $Av_1 = \lambda_1 v_1$. Extend $\{v_1\}$ to a basis $(v_1, v_2, \dots, v_n)$ of F^n and let $P_1 = (v_1 \ v_2 \ \cdots \ v_n)$ be the change-of-basis matrix. In this basis the linear map A has matrix

$$P_1^{-1}AP_1 = \begin{pmatrix} \lambda_1 & * \\ 0 & B \end{pmatrix},$$

where $B \in M_{(n-1)\times(n-1)}(F)$. By Proposition 8.14, $p_A(x) = (x-\lambda_1)p_B(x)$. Since $p_A(x)$ splits in F, we have $p_B(x)$ also splits in F. By the induction hypothesis there exists an $Q \in M_{(n-1)\times(n-1)}(F)$ invertible such that $Q^{-1}BQ$ is upper triangular with eigenvalues $\lambda_2, \dots, \lambda_n$ on the diagonal. Put

$$P = P_1 \begin{pmatrix} 1 & 0 \\ 0 & Q \end{pmatrix}.$$

Then a direct block multiplication gives

$$P^{-1}AP = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \lambda_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix},$$

which completes the induction and the proof. $\square$

Lemma 11.8. *If* $A = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \lambda_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$ *and* $B = \begin{pmatrix} \mu_1 & * & \cdots & * \\ 0 & \mu_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mu_n \end{pmatrix}$ *then*

$$A + B = \begin{pmatrix} \lambda_1+\mu_1 & * & \cdots & * \\ 0 & \lambda_2+\mu_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n+\mu_n \end{pmatrix}$$

and

$$AB = \begin{pmatrix} \lambda_1\mu_1 & * & \cdots & * \\ 0 & \lambda_2\mu_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n\mu_n \end{pmatrix}.$$

Note that $*$ may be different in different matrices.

In particular, we have for any integer $k \geq 1$ and $f(x) \in F[x]$

$$A^k = \begin{pmatrix} \lambda_1^k & * & \cdots & * \\ 0 & \lambda_2^k & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n^k \end{pmatrix}$$

and

$$f(A) = \begin{pmatrix} f(\lambda_1) & * & \cdots & * \\ 0 & f(\lambda_2) & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & f(\lambda_n) \end{pmatrix}.$$

Proof. Directly verify using matrix addition and multiplication. □

Lemma 11.9 (Linear transformation given by an upper triangular matrix)**.** *Let $T : V \to V$ be a linear transformation and $\beta = (v_1, \ldots, v_n)$ be a basis of V. If $A = [T]_\beta^\beta$ is an upper triangular matrix then $T(E_j) \subset E_j$ for all $j = 1, \ldots, n$ where $E_j =$* $\operatorname{span}(v_1, \ldots, v_j)$. *Moreover, if $a_{jj} = \lambda_j$, then $(T - \lambda_j \mathrm{id})(E_j) \subset E_{j-1}$.*

Proof. Since $A = [T]_\beta^\beta$ is upper triangular, for any $1 \leq j \leq n$, $T(v_j) = a_{1j}v_1 + \cdots + a_{jj}v_j \in E_j$. Thus $T(E_j) \subset E_j$.

Since $[T - \lambda_j \mathrm{id}]_\beta^\beta = A - \lambda_j I$ is upper triangular, we have $(T - \lambda_j \mathrm{id})(E_{j-1}) \subset E_{j-1}$. On the other hand, $(T - \lambda_j \mathrm{id})(v_j) = a_{1j}v_1 + \cdots + a_{jj}v_j - \lambda_j v_j = a_{1j}v_1 + \cdots + a_{(j-1)j}v_{j-1} \in E_{j-1}$. Since $(T - \lambda_j \mathrm{id})$ maps a basis of E_j into E_{j-1}, we have $(T - \lambda_j \mathrm{id})(E_j) \subset E_{j-1}$. □

Proof of Theorem 11.5. If $p_A(x) = (x - \lambda_1) \cdots (x - \lambda_n)$ splits, then by Theorem 11.7 there is $P \in M_{n \times n}(F)$ invertible such that

$$P^{-1}AP = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \lambda_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}.$$

Define the subspaces $E_j = \operatorname{span}(v_1, v_2, \ldots, v_j)$, where $v_1, v_2, \ldots, v_n$ are the columns of P.

Take an arbitrary vector $v \in F^n = E_n$. Then we have $p_A(A)v = (A - \lambda_1 I) \cdots (A - \lambda_n I)v$. Then we have

$$\begin{aligned}
&(A - \lambda_n I)v \in E_{n-1} \\
&(A - \lambda_{n-1} I)(A - \lambda_n I)v \in E_{n-2} \\
&\vdots \\
&(A - \lambda_2 I) \cdots (A - \lambda_n I)v \in E_1.
\end{aligned}$$

The last inclusion implies that $(A - \lambda_2 I) \cdots (A - \lambda_n I)v = cv_1$ for some $c \in F$. But $(A - \lambda_1 I)v_1 = 0$, and hence

$$0 = (A - \lambda_1 I)(cv_1) = (A - \lambda_1 I)(A - \lambda_2 I) \cdots (A - \lambda_n I)v.$$

Therefore, $p_A(A)v = 0$ for all $v \in F^n$, which means exactly that $p_A(A) = 0$.

If p_A does not splits in F, then by Proposition 9.17, there is a field extension E of F such that $p_A(x)$ splits in E. We now view A as a matrix over E. Then the previous argument shows $p_A(A) = 0$. Since p_A is the same polynomial and $p_A(A)$ is the same matrix regardless of whether you think of them as over E or F, we finish the proof. □

11.2 Minimal polynomial

Theorem 11.10. *There exists a unique polynomial such that*

(i) $m_A(x)$ is monic.

(ii) $m_A(A) = 0$

(iii) For any $f(x) \in F[x]$ such that $f(A) = 0$, we have $m_A(x) \mid f(x)$.

In fact, $m_A(x)$ is the monic nonzero polynomial with least degree such that $m_A(A) = 0$.

Proof. Consider the set $\mathcal{I}_A = \{f(x) \in F[x] : f(A) = 0\}$. By the Cayley-Hamilton theorem, the characteristic polynomial $p_A(x) = \det(xI - A) \in \mathcal{I}_A$, since $p_A(A) = 0$. So $\mathcal{I}_A \neq \{0\}$. Let m_A be a nonzero polynomial in $\mathcal{I}_A$ of least degree. We scale m_A such that it is monic.

Since $m_A \in \mathcal{I}_A$, we have $m_A(A) = 0$.

For any $f \in \mathcal{I}_A$ we have $f(A) = 0$. By Theorem 9.6, $f(x) = m_A(x)q(x) + r(x)$ where $\deg r < \deg m_A$. Since $0 = f(A) = m_A(A)q(A) + r(A) = r(A)$. We must have $r = 0$ since otherwise $r \neq 0$ and we have $r \in \mathcal{I}_A$. $\deg r < \deg m_A$. This contradicts m_A has least degree in $\mathcal{I}_A$. Hence $m_A(x) \mid f(x)$.

To show uniqueness, suppose that m_1, m_2 are two polynomials satisfying (i), (ii) and (iii). Then since $m_1(A) = 0$, $m_2(x) \mid m_1(x)$ by (iii) of $m_2(x)$. Since $m_2(A) = 0$, $m_1(x) \mid m_2(x)$ by (iii) of m_1. Thus $m_1 = cm_2$ for some $c \in F$. Since m_1 and m_2 are both monic, we have $m_1 = m_2$. □

Definition 11.11. The polynomial $m_A(x)$ is called the **minimal polynomial** of A.

Example 11.12. A $n \times n$ **Jordan block** with diagonal λ is a matrix of the form

$$J_n(\lambda) = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & 1 & \cdots & 0 & 0 \\ 0 & 0 & \lambda & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda & 1 \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{pmatrix} \in M_{n\times n}(F).$$

Then $p_{J_n(\lambda)}(x) = (x-\lambda)^n$ and by Propostion 11.13 $m_{J_n(\lambda)}(x) = (x-\lambda)^k$ for some $1 \le k \le n$. Let $N = J_n(\lambda) - \lambda I$. We have

$$\begin{aligned} &N(e_1) = 0, N(e_j) = e_{j-1} \text{ for any } j = 2, \dots, n \\ &N^2(e_1) = N^2(e_2) = 0, N^2(e_j) = e_{j-2} \text{ for any } j = 3, \dots, n \\ &\cdots \\ &N^{n-1}(e_1) = \cdots = N^{n-1}(e_{n-1}) = 0, N^{n-1}(e_n) = e_1 \\ &N^n(e_1) = \cdots = N^n(e_n) = 0. \end{aligned}$$

Thus $m_{J_n(\lambda)}(x) = (x-\lambda)^n$.

Proposition 11.13. *If the characteristic polynomial*

$$p_A(x) = (x-\lambda_1)^{k_1}(x-\lambda_2)^{k_2}\cdots(x-\lambda_m)^{k_m}$$

splits over F so the $\lambda_i \in F$ are the distinct eigenvalues, then the minimal polynomial $m_A(x)$ has the same distinct linear factors, i.e.

$$m_A(x) = (x-\lambda_1)^{l_1}(x-\lambda_2)^{l_2}\cdots(x-\lambda_m)^{l_m} \tag{5}$$

with $1 \le l_i \le k_i$. In other words, every factor $(x-\lambda_i)$ that appears in p_A also appears in m_A.

Proof. By Theorem 11.7, there exists $P \in M_{n\times n}(F)$ invertible such that

$$P^{-1}AP = \begin{pmatrix} \lambda_1 & * & \cdots & * \\ 0 & \lambda_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}.$$

By Lemma 11.8,

$$m_A(A) = P \begin{pmatrix} m_A(\lambda_1) & * & \cdots & * \\ 0 & m_A(\lambda_2) & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & m_A(\lambda_n) \end{pmatrix} P^{-1} = 0.$$

Thus $m_A(\lambda_i) = 0$ for all $1 \le i \le n$. By Proposition 9.10, $(x-\lambda_i) \mid m_A(x)$ for any eigenvalue λ_i of A. $\square$

Theorem 11.14. *Let $A \in M_{n\times n}(F)$. Then A is diagonalizable over F if and only if the minimal polynomial $m_A(x)$ splits over F as a product of distinct linear factors, i.e.*

$$m_A(x) = (x-\lambda_1)(x-\lambda_2)\cdots(x-\lambda_m)$$

where $\lambda_1, \ldots, \lambda_m$ are distinct. In other words, $l_1 = \cdots = l_m = 1$ in (5).

Proof. "$\Longrightarrow$" If A is diagonalizable over F then there exists an invertible P with $P^{-1}AP = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$, where each $\lambda_j \in F$. Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of A. We define $f(x) = (x-\lambda_1)\cdots(x-\lambda_m) \in F[x]$. Then we have $f(A) = Pf(\mathrm{diag}(\lambda_1, \ldots, \lambda_n))P^{-1} = 0$. Thus $m_A(x) \mid f(x)$. By Proposition 11.13, we have $f(x) \mid m_A(x)$. Thus $m_A(x) = f(x)$.

"$\Longleftarrow$" Conversely, suppose $m_A(x) = (x-\lambda_1)(x-\lambda_2)\cdots(x-\lambda_m)$ with distinct $\lambda_i \in F$. We define $p_i(x) = (x-\lambda_1)\cdots(x-\lambda_{i-1})(x-\lambda_{i+1})\cdots(x-\lambda_n)$. By Lagrange interpolation theorem (Homework 9 Problem 4),

$$1 = \sum_{i=1}^{m} c_i p_i(x), \quad c_i = \frac{1}{(\lambda_i-\lambda_1)\cdots(\lambda_i-\lambda_{i-1})(\lambda_i-\lambda_{i+1})\cdots(\lambda_i-\lambda_n)}.$$

Substituting A for x gives a decomposition of the identity matrix as a sum of matrices

$$I = \sum_{i=1}^{m} P_i, \qquad P_i := c_i p_i(A).$$

For any $v \in F^n$ we have $v = \sum_{i=1}^{m} P_i v$. For each $1 \le i \le m$,

$$(A-\lambda_i I)P_i v = c_i(A-\lambda_i I)p_i(A)v = c_i m_A(A)v = 0v = 0.$$

Thus $P_i v \in E(\lambda_i, A)$. Hence F^n decomposes as a direct sum of eigenspaces $F^n = E(\lambda_1, A) + \cdots + E(\lambda_m, A) = E(\lambda_1, A) \oplus \cdots \oplus E(\lambda_m, A)$. Therefore A is diagonalizable over F by Theorem 10.11. $\square$

Example 11.15. If $A^2 = A$ then A is diagonalizable with eigenvalue 1 or 0. This is because $f(x) = x^2 - x \in \mathcal{I}_A$. Then $m_A(x) \mid f(x)$. Since $f(x)$ splits as distinct linear factors, $m_A(x)$ splits as distinct linear factors. Similarly, if $A^2 = I$, then A is diagonalizable with eigenvalue 1 or -1.

12 Lecture 12

12.1 Commuting operators

Theorem 12.1. *If $A, B \in M_{n\times n}(F)$ are both diagonalizable. Then $AB = BA$ if and only if there exists $P \in M_{n\times n}(F)$ invertible such that $P^{-1}AP = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ and $P^{-1}BP = \operatorname{diag}(\mu_1, \ldots, \mu_n)$.*

Proof. $\Longleftarrow$: If $P^{-1}AP = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ and $P^{-1}BP = \operatorname{diag}(\mu_1, \ldots, \mu_n)$, then $AB = P \operatorname{diag}(\lambda_1\mu_1, \ldots, \lambda_n\mu_n)P^{-1} = BA$.

$\Longrightarrow$: Since A is diagonalizable over F, $F^n = E(\lambda_1, A) \oplus \cdots \oplus E(\lambda_m, A)$.

If $v \in E(\lambda_i, A)$ then $A(Bv) = B(Av) = B(\lambda_i v) = \lambda_i(Bv)$ so $Bv \in E(\lambda_i, A)$. Thus $B(E(\lambda_i, A)) \subset E(\lambda_i, A)$ for any $1 \le i \le m$. Thus if we let $Q \in M_{n\times n}(F)$ such that its columns are basis of $E(\lambda_i, A)$ for $1 \le i \le m$, then

$$Q^{-1}AQ = \operatorname{diag}(\lambda_1, \ldots, \lambda_n),\ Q^{-1}BQ = \begin{pmatrix} B_1 & 0 & \cdots & 0 \\ 0 & B_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & B_m \end{pmatrix}$$

where $B_i \in M_{k_i\times k_i}(F)$ for $k_i = \dim E(\lambda_i, A)$.

Since $m_B(B) = 0$, we have $m_{B_i}(x) \mid m_B(x)$. Since B is diagonalizable, $m_B(x)$ has no repeated factors. Hence $m_{B_i}(x)$ also has no repeated factors. Thus by Theorem 11.14 there exists a basis β_i of $E(\lambda_i, A)$ such that B_i is diagonalized.

By Theorem 10.9, $\beta = \beta_1 \cup \cdots \cup \beta_m$ is a basis of F^n. Since β_i is a basis of $E(\lambda_i, A)$, A is diagonalized under β. Since B_i is diagonalized under β_i, B is diagonalized under β. If P is the change of basis matrix then

$$P^{-1}AP = \operatorname{diag}(\lambda_1, \ldots, \lambda_n), \qquad P^{-1}BP = \operatorname{diag}(\mu_1, \ldots, \mu_n).$$

□

Remark 12.2. The theorem is saying that there is a basis consisting of vectors that are eigenvectors of both A and B. This is not saying that any eigenvector of A is an eigenvector of B. We take $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ as an example. Clearly $AB = BA$ and they are already in diagonalized under the standard basis e_1, e_2. However, since $A = I$, any nonzero vector is an eigenvectors of A but the eigenvectors of B are e_1 with eigenvalue 1 and e_2 with eigenvalue 2. Thus $e_1 + e_2$ is an eigenvector of A but not an eigenvector of B.

Remark 12.3. Theorem 12.1 also appears in quantum mechanics. If the operators corresponding to two physical quantities commute, then their values can be measured simultaneously. If they do not commute, the famous uncertainty principle states that their values cannot be simultaneously determined. In fact, the uncertainty principle has a qualitative version: we fix $\psi \in \mathbb{C}^n$. For any $A \in M_{n\times n}(\mathbb{C})$, we write

$\langle A \rangle = \bar{\psi}^t A \psi \in \mathbb{C}$. Let $A, B \in M_{n\times n}(\mathbb{C})$ with $\bar{A}^t = A$ and $\bar{B}^t = B$. Then we have $|(A - \langle A \rangle)^2 \psi|^2 |(B - \langle B \rangle)^2 \psi|^2 \geq |\frac{1}{2i}\langle [A, B] \rangle|^2$. The term $|(A - \langle A \rangle)^2 \psi|^2$ represents the variance of A in state ψ and the inequality is saying the product of variance is bounded below by the mean of the commutator so that if the commutator isn't 0, then lowering the variance of A will increase the variance of B.

12.2 A brief introduction to Jordan canonical form

In Lecture 9, we raised the question: given $T : V \to V$ a linear transformation, can we find a basis β such that $[T]_\beta^\beta$ is diagonal? We have seen in Example 9.27 and 11.12 that there are matrices that cannot be diagonalized. So we step back and ask what is the simplest form that $[T]_\beta^\beta$ can take? In Theorem 11.7, we see that we could make $[T]_\beta^\beta$ an upper triangular matrix. However, the issue with Theorem 11.7 is we have no information about the entries that are above the diagonal. The following theorem is the ultimate resolution to the question above.

Theorem 12.4. *Let $A \in M_{n\times n}(F)$ be such that $p_A(x)$ splits in F. Then there exists $P \in M_{n\times n}(F)$ invertible such that*

$$P^{-1}AP = \begin{pmatrix} J_{k_1}(\lambda_1) & 0 & \cdots & 0 \\ 0 & J_{k_2}(\lambda_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & J_{k_m}(\lambda_m) \end{pmatrix}$$

where $J_{k_i}(\lambda_i)$ is a Jordan block. This is called the ***Jordan canonical form*** *of A.*

Moreover, The Jordan canonical form for A unique up to a permutation of the Jordan blocks along the diagonal.

Remark 12.5. The λ_i here need not be distinct. E.g. we could have a Jordan canonical form like $\begin{pmatrix} \lambda & 1 & 0 & 0 \\ 0 & \lambda & 0 & 0 \\ 0 & 0 & \lambda & 1 \\ 0 & 0 & 0 & \lambda \end{pmatrix}$.

Remark 12.6. In Example 11.12, we see that a Jordan block is diagonalizable if and only if it is 1×1. In general, A is diagonalizable if and only if all its Jordan block are 1×1, i.e. its Jordan canonical form is diagonal.

Moreover, two matrices with different Jordan canonical form are not similar to each other. So we have:

Theorem 12.7. *A, B are similar if and only if their Jordan canonical form are the same up to permutation.*

The two theorems above are among the most difficult results to prove in linear algebra, and we unfortunately do not have time to cover their proofs in class. There are several different approaches in the literature, and I personally prefer the following two.

The first proof, from [1, Chapter 9], is completely elementary, and anyone with the background provided in these lecture notes should be able to follow it. The second proof, from [2, Chapter 12], is based on the structure theorem for finitely generated modules over a PID. I find the first proof more accessible for students encountering these ideas for the first time, while the second presents linear algebra from a more advanced, conceptual viewpoint and connects naturally with the broader theory underlying the Jordan canonical form.

In this note, we prove the simpler case for 2×2 matrices.

Theorem 12.8 (Jordan canonical form for 2×2 matrices)**.** *Let $A \in M_{2\times 2}(F)$ be such that $p_A(x)$ splits in F. Then there exists $P \in M_{2\times 2}(F)$ invertible such that either $P^{-1}AP = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$ or $P^{-1}AP = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$.*

Proof. If A has two distinct eigenvalues then A is diagonalizable by Corollary 10.12.

If A has one eigenvalue and $\dim E(\lambda, A) = 2$, then $A = \lambda I$ is diagonalizable.

If A has one eigenvalue and $\dim E(\lambda, A) = 1$, then by rank-nullity Theorem 4.14, $\operatorname{rank}(\lambda I - A) = 1$. Let $0 \neq v \in \operatorname{im}(A - \lambda I)$. Then $v = (A - \lambda I)w$ for $0 \neq w \in F^2$ i.e. $Aw = \lambda w + v$. The characteristic polynomial of A is $(x-\lambda)^2$. Then by Cayley-Hamilton theorem, $(A - \lambda I)^2 = 0$. Then we have $(A - \lambda I)v = (A - \lambda I)^2 w = 0$ i.e. $Av = \lambda v$. We claim that v, w are linearly independent. If $av + bw = 0$, then applying $(A - \lambda I)$ on both sides $0 = (A - \lambda I)(av + bw) = bv$. Since $v \neq 0$, we have $b = 0$. Thus $av = 0$ and hence $a = 0$. Thus v, w are linearly independent. We let $P = (v, w)$. Then P is invertible, and $P^{-1}AP = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$ □

The proof provides a good way to compute the Jordan canonical form for 2×2 matrices

Example 12.9. We recall Example 9.27 where $A = \begin{pmatrix} -1 & -4 \\ 1 & 3 \end{pmatrix}$ and $p_A(x) = (x-1)^2$. Then $\lambda = 1$ is the only eigenvalue of A. Since $A - I = \begin{pmatrix} -2 & -4 \\ 1 & 2 \end{pmatrix}$, $\dim\ker(A - I) = 1 < 2$. $\operatorname{im}(A - I) = \operatorname{span}\begin{pmatrix} -2 \\ 1 \end{pmatrix}$. Let $v = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$. Solving $(A - I)w = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$, we get $w = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. We set $P = \begin{pmatrix} -2 & 1 \\ 1 & 0 \end{pmatrix}$, then we have $P^{-1}AP = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

We have seen in Example 9.23 that a real matrix can have eigenvalues not in $\mathbb{R}$. Of course we can view it as a complex matrix and apply Theorem 12.8. However, we would like to know what is the simplest form of it under similarity by $P \in M_{2\times 2}(\mathbb{R})$ invertible.

Theorem 12.10 (Real canonical form for 2×2 matrices over $\mathbb{R}$)**.** *Let $A \in M_{2\times 2}(\mathbb{R})$. Then there exists $P \in M_{2\times 2}(\mathbb{R})$ invertible such that either $P^{-1}AP = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$ or $P^{-1}AP = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$ or $P^{-1}AP = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$ where $a \pm bi$ are the eigenvalues of A in $\mathbb{C}$.*

Proof. If $p_A(x)$ splits in $\mathbb{R}$, then the theorem follows from Theorem 12.8.

We deal with the case where A has no real eigenvalues. Then $p_A(x) = x^2 - \mathrm{tr}(A)x + \det(A)$ has complex roots $a \pm bi$ for $b \neq 0$. Let $v \in \mathbb{C}^2$ be an eigenvector with eigenvalue $a+bi$. Then $Av = (a+bi)v$. Let $v = v_1 + v_2 i$ where $v_1 = \mathrm{Re}\, v \in \mathbb{R}^2$ and $v_2 = \mathrm{Im}\, v \in \mathbb{R}^2$. We write $P = (v_1, v_2) \in M_{2\times 2}(\mathbb{R})$. Then $Av_1 + Av_2 i = A(v_1 + v_2 i) = (a+bi)(v_1 + v_2 i) = (av_1 - bv_2) + (av_2 + bv_1)i$. Thus $Av_1 = av_1 - bv_2$ and $Av_2 = av_2 + bv_1$. Then $P^{-1}AP = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$ provided P is invertible which can be seen as follows:

By Remark 9.25 , $v, \bar{v}$ are the eigenvector with distinct eigenvalues $a+bi$ and $a-bi$. Thus by Theorem 10.9 v and $\bar{v}$ are linearly independent. That v_1, v_2 are linearly independent follows from the following lemma.

Lemma 12.11. *Let $v \in \mathbb{C}^n$. If $v, \bar{v}$ are linearly independent, then* $\mathrm{Re}\, v$ *and* $\mathrm{Im}\, v$ *are linearly independent.*

Proof. We have $\mathrm{Re}\, v = \frac{1}{2}(v + \bar{v})$ and $\mathrm{Im}\, v = \frac{1}{2i}(v - \bar{v})$. Suppose $a\,\mathrm{Re}\, v + b\,\mathrm{Im}\, v = 0$ for $a, b \in \mathbb{C}$. Then $(\frac{1}{2}a + \frac{1}{2i}b)v + (\frac{1}{2}a - \frac{1}{2i}b)\bar{v} = 0$. Since $v, \bar{v}$ are linearly independent, we have

$$\begin{pmatrix} \frac{1}{2} & \frac{1}{2i} \\ \frac{1}{2} & -\frac{1}{2i} \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Since $\det \begin{pmatrix} \frac{1}{2} & \frac{1}{2i} \\ \frac{1}{2} & -\frac{1}{2i} \end{pmatrix} = -\frac{1}{2i} \neq 0$, we have $a = b = 0$. □

□

Example 12.12. We recall Example 9.24 where $A = \begin{pmatrix} -1 & 2 \\ -1 & 1 \end{pmatrix} \in M_{2\times 2}(\mathbb{C})$ and $p_A(x) = x^2 + 1 = (x-i)(x+i)$. We have $v_1 = \begin{pmatrix} 1-i \\ 1 \end{pmatrix}$ is an eigenvector with eigenvalue i. Then $\mathrm{Re}\, v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathrm{Im}\, v_1 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$. If $P = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}$ then $P^{-1}AP = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

12.3 Convergence of a sequence of matrices

Definition 12.13. A sequence of matrices $\{A_k = (a_{ij}^{(k)})\} \subset M_{m\times n}(\mathbb{C})$ converges to a matrix $A = (a_{ij}) \in M_{m\times n}(\mathbb{C})$ if $a_{ij}^{(k)} \to a_{ij}$ for any $i = 1, \ldots, m, j = 1, \ldots, n$ as $k \to \infty$. We write

$$\lim_{k\to\infty} A_k = A.$$

Definition 12.14. Let $A(t) = (a_{ij}(t)) \in M_{m\times n}(\mathbb{C})$ be a family of matrix depending on $t \in (a, b)$. Then we define the derivative at t to be

$$A'(t) = \lim_{h\to 0} \frac{A(t+h) - A(t)}{h}.$$

Lemma 12.15. *If $A(t) = (a_{ij}(t))$ where a_{ij} are differentiable at t then $A'(t) = (a'_{ij}(t))$.*

Proof. Since $a'_{ij}(t) = \lim_{h\to 0}\frac{a_{ij}(t+h)-a_{ij}(t)}{h}$ exists, we have $A'(t)$ exists and $A'(t) = (a'_{ij}(t))$. $\square$

12.4 Matrix function

Definition 12.16. Let $f(x) = \sum_{k=0}^{\infty} c_k x^k$ with $c_k \in \mathbb{C}$ be a function defined by a power series. We define

$$f(A) = \sum_{k=0}^{\infty} c_k A^k = \lim_{N\to\infty}\sum_{k=0}^{N} c_k A^k$$

for $A \in M_{n\times n}(\mathbb{C})$ if the right hand side converges.

Proposition 12.17. *Let $f(x) = \sum_{k=0}^{\infty} c_k x^k$ with $c_k \in \mathbb{C}$ be a function defined by a power series and $r = \frac{1}{\limsup_{k\to\infty}|c_k|^{1/k}}$ be the radius of convergence of the power series f.*

(i) If $f(A)$ exists, then $f(P^{-1}AP)$ also exists and $f(P^{-1}AP) = P^{-1}f(A)P$.

(ii) If $|\lambda| < r$ then $f(J_n(\lambda))$ converges and we have

$$f(J_n(\lambda)) = \begin{pmatrix} f(\lambda) & \frac{f'(\lambda)}{1!} & \frac{f''(\lambda)}{2!} & \cdots & \frac{f^{(n-2)}(\lambda)}{(n-2)!} & \frac{f^{(n-1)}(\lambda)}{(n-1)!} \\ 0 & f(\lambda) & \frac{f'(\lambda)}{1!} & \cdots & \frac{f^{(n-3)}(\lambda)}{(n-3)!} & \frac{f^{(n-2)}(\lambda)}{(n-2)!} \\ 0 & 0 & f(\lambda) & \cdots & \frac{f^{(n-4)}(\lambda)}{(n-4)!} & \frac{f^{(n-3)}(\lambda)}{(n-3)!} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & f(\lambda) & \frac{f'(\lambda)}{1!} \\ 0 & 0 & 0 & \cdots & 0 & f(\lambda) \end{pmatrix}.$$

(iii) Let $A \in M_{n\times n}(\mathbb{C})$ be such that $|\lambda| < r$ for each eigenvalue λ of A. Then $f(A) = \sum_{k=0}^{\infty} c_k A^k$ exists and if $P \in M_{n\times n}(\mathbb{C})$ is invertible such that

$$P^{-1}AP = \begin{pmatrix} J_{k_1}(\lambda_1) & 0 & \cdots & 0 \\ 0 & J_{k_2}(\lambda_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & J_{k_m}(\lambda_m) \end{pmatrix}$$

then

$$f(A) = P\begin{pmatrix} f(J_{k_1}(\lambda_1)) & 0 & \cdots & 0 \\ 0 & f(J_{k_2}(\lambda_2)) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & f(J_{k_m}(\lambda_m)) \end{pmatrix}P^{-1}$$

where $f(J_{k_i}(\lambda_i))$ is given by the formula in (ii).

Proof. (i) Let $f(x) = \sum_{k=0}^{\infty} c_k x^k$ and assume that the series $f(A) = \sum_{k=0}^{\infty} c_k A^k$ converges. Let $B = P^{-1}AP$ for some invertible matrix P. Then we have

$$\sum_{k=0}^{l} c_k B^k = \sum_{k=0}^{l} c_k (P^{-1}AP)^k = \sum_{k=0}^{l} c_k P^{-1}A^k P = P^{-1}(\sum_{k=0}^{l} c_k A^k)P.$$

Taking $l \to \infty$, we have $f(P^{-1}AP) = P^{-1}f(A)P$.

(ii) We write $J_n(\lambda) = \lambda I + N$, where $N = J_n(0)$ satisfies $N^n = 0$ as in Example 11.12. For any integer $k \geq 0$ the binomial expansion (finite because N is nilpotent) yields

$$\begin{aligned} J_n(\lambda)^k =&(\lambda I + N)^k = \sum_{j=0}^{n-1} \binom{k}{j} \lambda^{k-j} N^j = \sum_{j=0}^{n-1} \frac{k!}{j!(k-j)!} \lambda^{k-j} N^j \\ =&\sum_{j=0}^{n-1} \frac{1}{j!} k(k-1)\cdots(k-j+1)\lambda^{k-j} N^j. \end{aligned}$$

Then

$$\begin{aligned} f(J_n(\lambda)) =&\sum_{k=0}^{\infty} c_k J_n(\lambda)^k = \sum_{k=0}^{\infty} c_k \left(\sum_{j=0}^{n-1} \frac{1}{j!} k(k-1)\cdots(k-j+1)\lambda^{k-j} N^j \right) \\ =&\sum_{j=0}^{n-1} \frac{1}{j!} N^j \left(\sum_{k=0}^{\infty} c_k k(k-1)\cdots(k-j+1)\lambda^{k-j} \right). \end{aligned}$$

Since $|\lambda| < r$, we can differentiate f term-by-term and get $f^{(j)}(\lambda) = \sum_{k=0}^{\infty} c_k k(k-1)\cdots(k-j+1)\lambda^{k-j}$ where the series converges uniformly in λ. Thus $f(J_n(\lambda)) = \sum_{j=0}^{n-1} \frac{f^{(j)}(\lambda)}{j!} N^j$. Since N^j is a matrix with 1's on the j-th superdiagonal and 0 elsewhere we get the result.

(iii) By the Jordan canonical form Theorem 12.4, there exists an invertible $P \in M_{n\times n}(\mathbb{C})$ such that

$$P^{-1}AP = \begin{pmatrix} J_{k_1}(\lambda_1) & 0 & \cdots & 0 \\ 0 & J_{k_2}(\lambda_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & J_{k_m}(\lambda_m) \end{pmatrix}.$$

Since $|\lambda_i| < r$ for all $1 \leq i \leq m$, by (ii) $f(J_{k_i}(\lambda_i))$ converges. By (i) we have $f(A)$ converges and

$$f(A) = P \begin{pmatrix} f(J_{k_1}(\lambda_1)) & 0 & \cdots & 0 \\ 0 & f(J_{k_2}(\lambda_2)) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & f(J_{k_m}(\lambda_m)) \end{pmatrix} P^{-1}.$$

□

Corollary 12.18. *For any $A \in M_{n\times n}(\mathbb{C})$, $\exp(A) = e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!}$ exists.*

Proof. The radius of convergence of $f(x) = e^x = \sum_{j=0}^{\infty} \frac{x^j}{j!}$ is ∞. □

Example 12.19. $\exp(J_n(\lambda)) = \begin{pmatrix} e^\lambda & \frac{e^\lambda}{1!} & \frac{e^\lambda}{2!} & \cdots & \frac{e^\lambda}{(n-2)!} & \frac{e^\lambda}{(n-1)!} \\ 0 & e^\lambda & \frac{e^\lambda}{1!} & \cdots & \frac{e^\lambda}{(n-3)!} & \frac{e^\lambda}{(n-2)!} \\ 0 & 0 & e^\lambda & \cdots & \frac{e^\lambda}{(n-4)!} & \frac{e^\lambda}{(n-3)!} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & e^\lambda & \frac{e^\lambda}{1!} \\ 0 & 0 & 0 & \cdots & 0 & e^\lambda \end{pmatrix}$

References

[1] Linear algebra done wrong, by Sergei Treil 2025,

[2] Dummit, David Steven, and Richard M. Foote. Abstract algebra. Vol. 3. Hoboken: Wiley, 2004.

13 Lecture 13

13.1 Matrix exponential

Theorem 13.1. *(i)* $e^0 = I$.

(ii) $\det(e^A) = e^{\mathrm{tr}(A)}$.

(iii) If $AB = BA$, *then* $e^{A+B} = e^A e^B$. *In particular* $e^A e^{-A} = I$.

(iv) $\frac{d}{dt} e^{tA} = Ae^{tA} = e^{tA}A$.

Proof. (i) By definition $e^0 = \sum_{k=0}^{\infty} \frac{0^k}{k!} = I + \sum_{k=1}^{\infty} 0 = I$.

(ii) $e^A = P \begin{pmatrix} \exp(J_{k_1}(\lambda_1)) & 0 & \cdots & 0 \\ 0 & \exp(J_{k_2}(\lambda_2)) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \exp(J_{k_m}(\lambda_m)) \end{pmatrix} P^{-1}$. Then

$$\det(e^A) = e^{k_1\lambda_1} \cdots e^{k_m\lambda_m} = e^{k_1\lambda_1 + \cdots + k_m\lambda_m} = e^{\mathrm{tr}(A)}.$$

(iii) Since $AB = BA$, we have for each integer $k \geq 0$

$$(A+B)^k = \sum_{j=0}^{k} \binom{k}{j} A^j B^{k-j}.$$

Hence

$$e^A e^B = \Big(\sum_{m=0}^{\infty} \frac{A^m}{m!}\Big)\Big(\sum_{n=0}^{\infty} \frac{B^n}{n!}\Big) = \sum_{k=0}^{\infty}\sum_{j=0}^{k} \frac{A^j B^{k-j}}{j!(k-j)!} = \sum_{k=0}^{\infty} \frac{(A+B)^k}{k!} = e^{A+B}.$$

(iv) Let $A \in M_{n\times n}(\mathbb{C})$. We have

$$e^{tA} = \sum_{k=0}^{\infty} \frac{t^k A^k}{k!}.$$

We differentiate term by term to get

$$\frac{d}{dt} e^{tA} = \sum_{k=1}^{\infty} \frac{t^{k-1}A^k}{(k-1)!} = A \sum_{k=0}^{\infty} \frac{t^k A^k}{k!} = Ae^{tA}.$$

Similarly, e^{tA} commutes with A, so

$$\frac{d}{dt} e^{tA} = e^{tA} A.$$

$\square$

Remark 13.2. In proving (ii) and (iii) we can reorder the double infinite sum and differentiate the series term by term since e^A, e^B converge absolutely and e^{tA} converges absolutely and uniformly for t in a compact interval. To fully justify this we need the idea of the matrix norm. With that, the proof is exactly the same as the proof when A, B are number. We omit the details and refer to p.89 of Lax's book for an introduction to the matrix norm and to Theorem 2.6.5 and Corollary 6.2.13 of this book for a proof when A, B are numbers.

Example 13.3. We compute $\exp\begin{pmatrix} a & b \\ -b & a \end{pmatrix}$. Let $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Then $\begin{pmatrix} a & b \\ -b & a \end{pmatrix} = aI + bJ$. Note that $J^2 = -I$, $J^3 = -J$ and $J^4 = I$. We have $\exp(bJ) = \sum_{k=0}^{\infty} \frac{(bJ)^k}{k!} = I + bJ - \frac{1}{2!}b^2 I - \frac{1}{3!}b^3 J^3 + \cdots = (1 - \frac{b^2}{2!} + \frac{b^4}{4!} + \cdots)I + (b - \frac{b^3}{3!} + \frac{b^5}{5!} + \cdots)J = (\cos b)I + (\sin b)J = \begin{pmatrix} \cos b & \sin b \\ -\sin b & \cos b \end{pmatrix}$. Thus $\exp(aI + bJ) = e^a \begin{pmatrix} \cos b & \sin b \\ -\sin b & \cos b \end{pmatrix}$. In general, $f\begin{pmatrix} a & b \\ -b & a \end{pmatrix} = \frac{f(a+ib)+f(a-ib)}{2} I + \frac{f(a+ib)-f(a-ib)}{2i} J$.

13.2 Application: solving linear ODEs

Corollary 13.4 (Solution to first order linear ordinary differential equation)**.** *Let $A \in M_{n\times n}(\mathbb{R})$. Let $x(t) \in \mathbb{R}^n$ be a one parameter family of vectors statisfying $x' = Ax$. Then $x(t) = e^{tA}x(0)$.*

Proof. $\frac{d}{dt}(e^{-tA}x(t)) = -Ae^{-tA}x(t) + e^{-tA}Ax(t) = 0$. Thus $e^{-tA}x(t) = x(0)$ that is $x(t) = e^{tA}x(0)$. Here, we used $(A(t)x(t))' = A'(t)x(t) + A(t)x'(t)$ and if $A'(t) \equiv 0$ then $A(t)$ is constant. ☐

Example 13.5. We would like to solve the second order linear ODE: for $p, q \in \mathbb{R}$, $f : \mathbb{R} \to \mathbb{R}$

$$f''(t) + pf'(t) + qf(t) = 0, \quad f(0) = f_0,\ f'(0) = f_1.$$

Let $x(t) = \begin{pmatrix} f(t) \\ f'(t) \end{pmatrix}$. Then

$$x'(t) = \begin{pmatrix} f'(t) \\ f''(t) \end{pmatrix} = \begin{pmatrix} f'(t) \\ -qf(t) - pf'(t) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -q & -p \end{pmatrix} \begin{pmatrix} f(t) \\ f'(t) \end{pmatrix} = Ax(t).$$

By Theorem 13.4, $x(t) = e^{tA}x(0)$.

We have $p_A(x) = \det\begin{pmatrix} x & -1 \\ q & x+p \end{pmatrix} = x^2 + px + q$.

If $p^2 - 4q > 0$, then $p_A(x) = (x - \lambda_1)(x - \lambda_2)$ for $\lambda_1 \neq \lambda_2 \in \mathbb{R}$. There is $P \in M_{2\times 2}(\mathbb{R})$ $P^{-1}AP = \begin{pmatrix} \lambda_1 & \\ & \lambda_2 \end{pmatrix}$. Then

$$e^{tA} = P \begin{pmatrix} e^{\lambda_1 t} & \\ & e^{\lambda_2 t} \end{pmatrix} P^{-1}.$$

Since finding $f(t)$ is all we want, there is a way to avoid computing P and P^{-1} explicitly as follows: $f(t) = x_1(t)$ is of the form $c_1e^{t\lambda_1} + c_2e^{t\lambda_2}$. We then use $f(0) = f_0$ and $f'(0) = f_1$ to determine c_1 and c_2 and obtain the solution.

If $p^2 - 4q = 0$, then $p_A(x) = (x-\lambda)^2$ where $\lambda = -\frac{p}{2}$. In this case, $A + \frac{p}{2}I = \begin{pmatrix} \frac{p}{2} & 1 \\ -\frac{p^2}{4} & -\frac{p}{2} \end{pmatrix}$. Since $\operatorname{rank}(A+\frac{p}{2}I) = 1$, the Jordan canonical form is $P^{-1}AP = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$. Then

$$e^{tA} = P \begin{pmatrix} e^{\lambda t} & te^{\lambda t} \\ 0 & e^{\lambda t} \end{pmatrix} P^{-1}.$$

Again we do not need to compute P and P^{-1} explicitly but $f(t) = c_1e^{\lambda t} + c_2te^{\lambda t}$ and use $f(0) = f_0$ and $f'(0) = f_1$ to determine c_1, c_2.

If $p^2 - 4q < 0$, then $p_A(x) = (x-a-bi)(x-a+bi)$ has two complex roots. In this case, the real canonical form is $P^{-1}AP = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$. Thus

$$e^{tA} = Pe^{at} \begin{pmatrix} \cos(bt) & \sin(bt) \\ -\sin(bt) & \cos(bt) \end{pmatrix} P^{-1}.$$

Then $f(t) = c_1e^{at}\cos(bt) + c_2e^{at}\sin(bt)$. Then we use $f(0) = f_0$ and $f'(0) = f_1$ to determine c_1, c_2.